

The Morgan Stanley AI Guidebook: Navigating the Age of Inference

Collaborative work across 7 teams details our views on 8 key GenAI debates, why we expect investor dollars to flow toward enablers/software and where we are seeing AI adopter signal (tech, comm services, financials, healthcare). We also detail next AI debates to consider for '27 and beyond.

8 GenAI Debates as We Enter the Age of Inference: Today's collaborative guidebook across 7 Morgan Stanley teams details our views and frameworks on the myriad of GenAI debates swirling the economy including: **1)** hyperscaler capex trajectory; **2)** GW compute capacity coming online; **3)** GenAI ROIC; **4)** the \$50-\$60tn knowledge work and consumer TAMS to be digitized; **5)** early GenAI adoption signal across the economy; **6)** the importance of open-weights models; **7)** AI credit financing; and **8)** regulatory concerns and other bottlenecks to monitor. ***We hope this guidebook is helpful in detailing our expansive views and please do let us know if interested in any of our interactive models across the space.***

Thematically, Where Do We See Investment Dollars Flowing Next? From an equity perspective, we continue to believe we are in the early days of a fund flow rotation following the typical multi-year tech cycle... as flows move marginally out of hardware/semis/memory and into the enabler and software layers. While we don't necessarily expect the hardware layer stocks to categorically fall materially (as they are not expensive and this investment cycle is still going to be multi-year), we expect larger revisions and multiple expansion opportunity further up the stack as capex growth slows (which we expect in '28) and revenue/EBIT beats grow up the stack. This makes us **constructive AMZN, META, GOOGL, MSFT and other GenAI enablers.**

What About the Adopters? We also expect outperformance from early adopters across the economy. Our latest survey work shows that the **technology, comm services, financials, healthcare and industrials sectors** are furthest along in adopting and quantifying benefits from GenAI.

Also Remain Constructive Bottlenecks: Time to market on power remains a bottleneck, which also **leaves us constructive onsite generation names like BE, INIO and SEI and powered shell names like WULF, HUT, CIFR, and RIOT.** On the data center regulatory front, while we see state/local data center political scrutiny as largely manageable (essentially leading to higher costs of power for hyperscalers that are already in our out-year forecasts), we are highly focused on the build up to the '28 election and political risk to data center expansion from the executive branch if the Democratic Party wins.

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For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

4 Next Areas of Research and Key '27 Debates: As we have learned the past 3 years, this space and technology are evolving quickly. As such, we also remain focused on the next debates to potentially flare up as we enter the age of inference including: **1)** broader workforce reduction and re-training across the economy; **2)** further political scrutiny around datacenter buildouts and GenAI security; **3)** the relationship between hyperscalers and frontier labs; and **4)** the open-weights models vs the frontier labs and efforts to capture the trillions of dollars of inference spend ahead.

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The Morgan Stanley AI Guidebook: Navigating the Age of Inference

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Executive Summary

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8 Key GenAI Debates As We Enter the Age of Inference

- 1) **Forward Hyperscaler Capex:** We see data center capex growing 60% (\$550bn) in '27 but see a deceleration in capex growth going forward, with ~12% growth in '28 (\$170bn). Expect this (and accelerating hyperscaler/software revenue) to lead to investor fund rotation.
 - 2) **Forward Compute Capacity Growth:** We see total compute capacity growing ~4X from 35 GW in '25 to ~145 GW in '28. GOOGL will add the most capacity, and we see ASICs growing in the mix, rising from 34% of incremental capacity in '25 to 66% in '28.
 - 3) **GenAI ROIC:** We paint paths to 25-50% GenAI ROIC across hyperscalers and models with APIs. The highest returns accrue to the model layer running on top of owned infrastructure (META, GOOGL, SPCX).
 - 4) **The \$50-\$60tn Global Knowledge Work & Consumer TAMs:** We see a \$20-\$30tn opportunity in the global knowledge work spend to be digitized through GenAI tools. We also see ~\$30tn of consumer spend (across retail, travel, autonomous, advertising, etc) to be digitized. We also analyze hyperscaler and Facebook adoption curves as analogies to consider... speaking to the trillion-dollar opportunities ahead.
 - 5) **GenAI Adoption Signal:** We focus on diffusion across the economy. So far, we see most signal within the tech sector, followed by financials, healthcare and industrials.
 - 6) **Open-Weights Models:** Leading cost to serve will matter through the age of inference. We think open-weights and lower-cost models are critical to driving GenAI adoption across the economy. These will pressure leading labs to innovate to capture share of inference spend, while also reinforcing the value of the hyperscaler concierge layer (AWS Bedrock, MSFT Foundry, GOOGL Vertex) that optimizes compute efficiency by task.
 - 7) **The AI Credit Financing Market:** Despite ~\$450bn of AI-related debt issuance globally year-to-date, we think there's still significant capacity to absorb additional issuance, especially from high-quality hyperscalers. Rather than constraining issuance, the market adjustment is likely to come through wider spreads. Hyperscaler '27/'28 OCF is also expected to be 7-8X larger than required debt.
 - 8) **Regulatory Concerns and Bottlenecks:** We view state and local pushback to data centers as largely manageable, although remain focused on the buildup to the '28 election as a political risk to exacerbating this bottleneck. Time to market on power will remain a significant bottleneck, which leaves us constructive on power suppliers.
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Source: Company data, Morgan Stanley Research

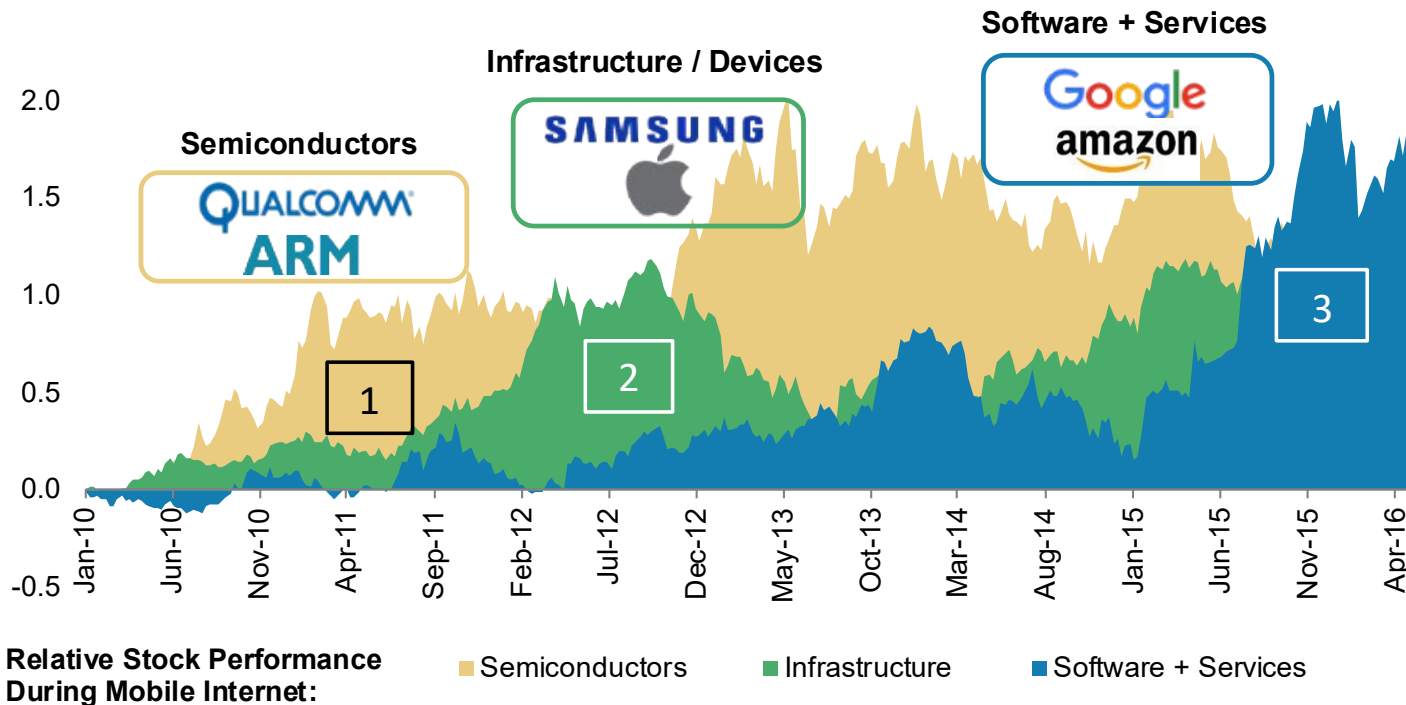
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How Can We Map AI to Past Tech Cycles?

- In the Mobile Era, we saw clear outperformance of Semis names (like Qualcomm/ARM), followed by Infrastructure/Devices (Samsung/Apple), and then Software & Services (Google/Amazon).
- We see a similar trend playing out in the AI cycle, with the infrastructure and software layer enabled by the hyperscalers via products like AWS Bedrock, MSFT Foundry, and Google's Vertex API.

Monetization in Mobile Era: Indexed Relative Stock Performance



Source: Company data, Morgan Stanley Research estimates

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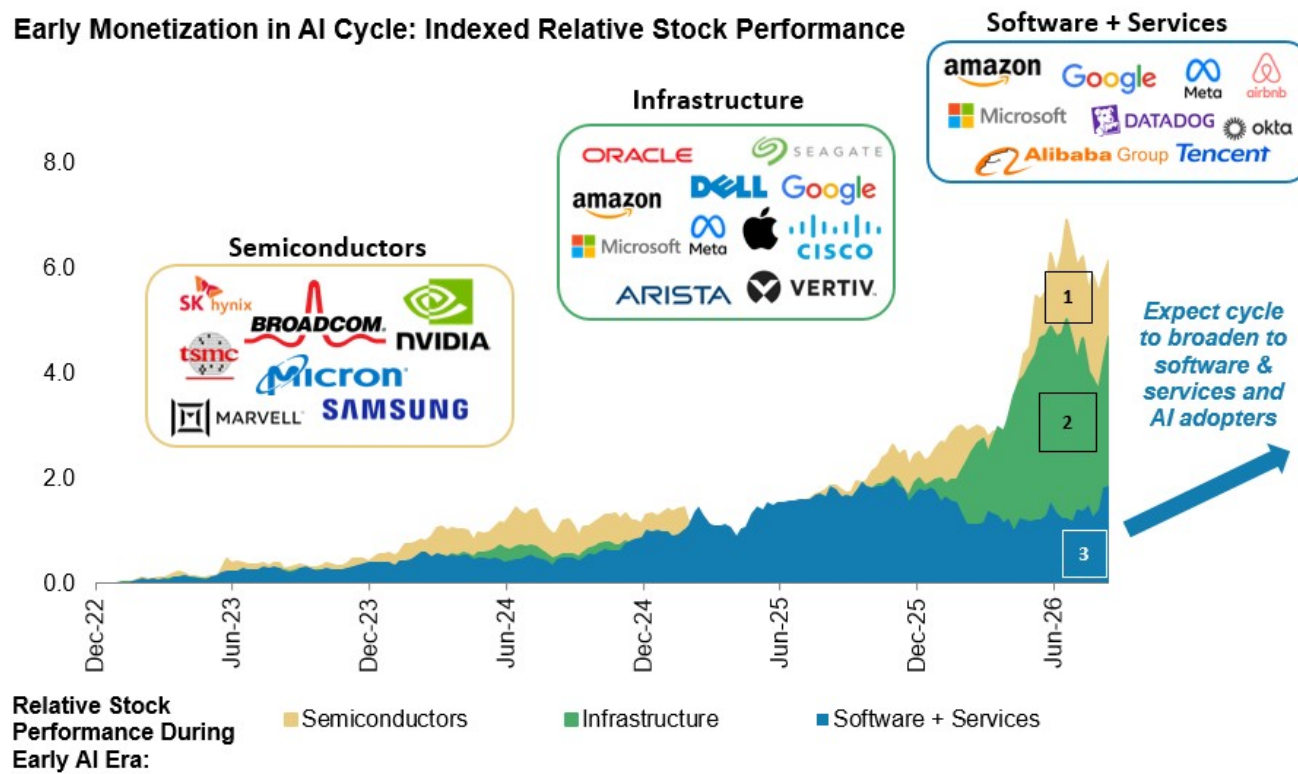
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Semis and Infrastructure Have Outperformed So Far... While Software + Services Appear Well Positioned for the Next Leg

- **Semis and Infrastructure** have outperformed thus far, and at an accelerated rate relative to past tech cycles.
- **Software and Services** layer appears to be primed for the next leg of outperformance as broader AI adoption diffuses throughout the global economy.

Early Monetization in AI Cycle: Indexed Relative Stock Performance



Source: Company data, Morgan Stanley Research estimates

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Next Areas of Research and Key '27 Debates

The Build-Out Debates

- **Power:** Will further power constraints impact the trajectory of data center opening, cost or ROIC?
- **Further Constraints:** Will physical (labor? materials?) constraints and/or expansion of regulatory headwinds (in particular heading into the mid-terms and '28 election cycle) impact the trajectory of data center cost, opening, or ROIC?

Adoption Debates

- **Workforce Reduction:** Which companies or industries will be most aggressive first in what could be material workforce reductions as GenAI benefits diffuse across the economy?
- **Workforce Re-alignment:** The US workforce will need some skill re-training through this transition... how will this be funded and how could this impact the political discussion around GenAI and pace of GenAI roll outs?
- **Adoption Drivers:** Which industries/AI tools will drive adoption, spend and diffusion across the economy?
- **Hyperscaler Cost to Serve:** How will hyperscaler “cost to serve” improve through custom silicon and other advances to drive higher token throughput?
- **Hyperscalers vs Labs:** Will the labs be largely on-prem or served through the hyperscalers?
- **Open-Weights vs Labs:** There will be trillions of dollars spent on inference over the next decade... how will competitive dynamics and offerings between open-weights and frontier labs evolve to capture these dollars?

Source: Company data, Morgan Stanley Research

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Hyperscale Capex Spend Going Forward

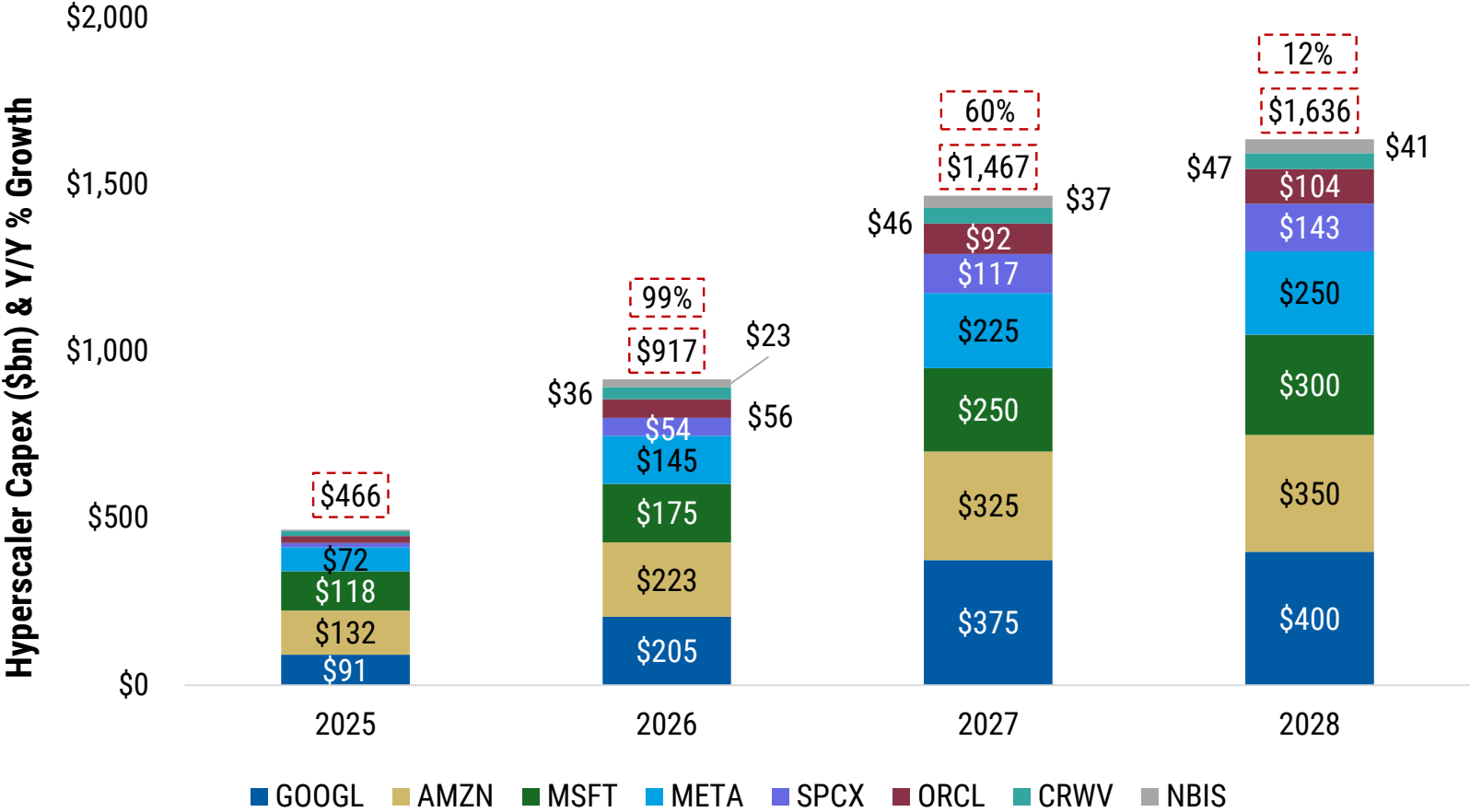
Our View: We see hyperscale capex growing ~\$500bn y/y (60%) in '27 reaching \$1.5tn. But looking into '28, we expect a slowing in spend growth for multiple reasons.

Stock Implications: This slowing in spend along with scaling revenue and improving cash flow dynamics are important to long-term investor flow dynamics marginally out of hardware/semis/memory, into the enabler/software layers. This should **benefit hyperscalers and GenAI enablers at the infrastructure and software layers (AMZN, GOOGL, MSFT, META)**

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We See Hyperscaler Capex Reaching ~\$1.5tr/~1.6tr in '27/'28...



Source: Company data, Morgan Stanley Research estimates; Note: AMZN = Total Company Capex excluding proceeds from property and equipment sales and incentives. Note ORCL capex is projected using consensus estimates.

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Why We See Capex Growth Slowing in '28 and Where We Could Be Wrong

Why We Think it Slows

- **Bottom-Up Chip/Rack Work:** We build our capex on a bottom-up basis based on chip and rack counts/estimated shipments and cost/chip... and there are limits on the numbers of chips that are built/available to purchase.
- **Outyear Pre Builds are Material... and Have Limits:** We estimate a substantial portion (in some cases 30-40%) of '26 capex is being spent on long-dated land, buildings, equipment, power for data centers to open in '27/'28/'29. Hyperscalers are building further out for **strategic reasons** (being able to light up compute as it comes available) and **political reasons** (getting ahead of potential further political backlash against data center builds if the Executive branch changes parties). While forward building will continue, there are limits as to how far out these pre-builds will go.
- **Physical Limits:** We see physical limits (labor, materials, power, etc.) also holding back further substantial growth off of already large numbers and rapid pace of build.
- **META '28 Downward Capex Potential:** The extent to which META chooses to leverage hyperscalers (AWS/Azure) to scale its family of models and API (as it did in the past with Llama) would lower long-term compute needs and potentially put downward pressure on '28 capex.

Where We Could Be Wrong?

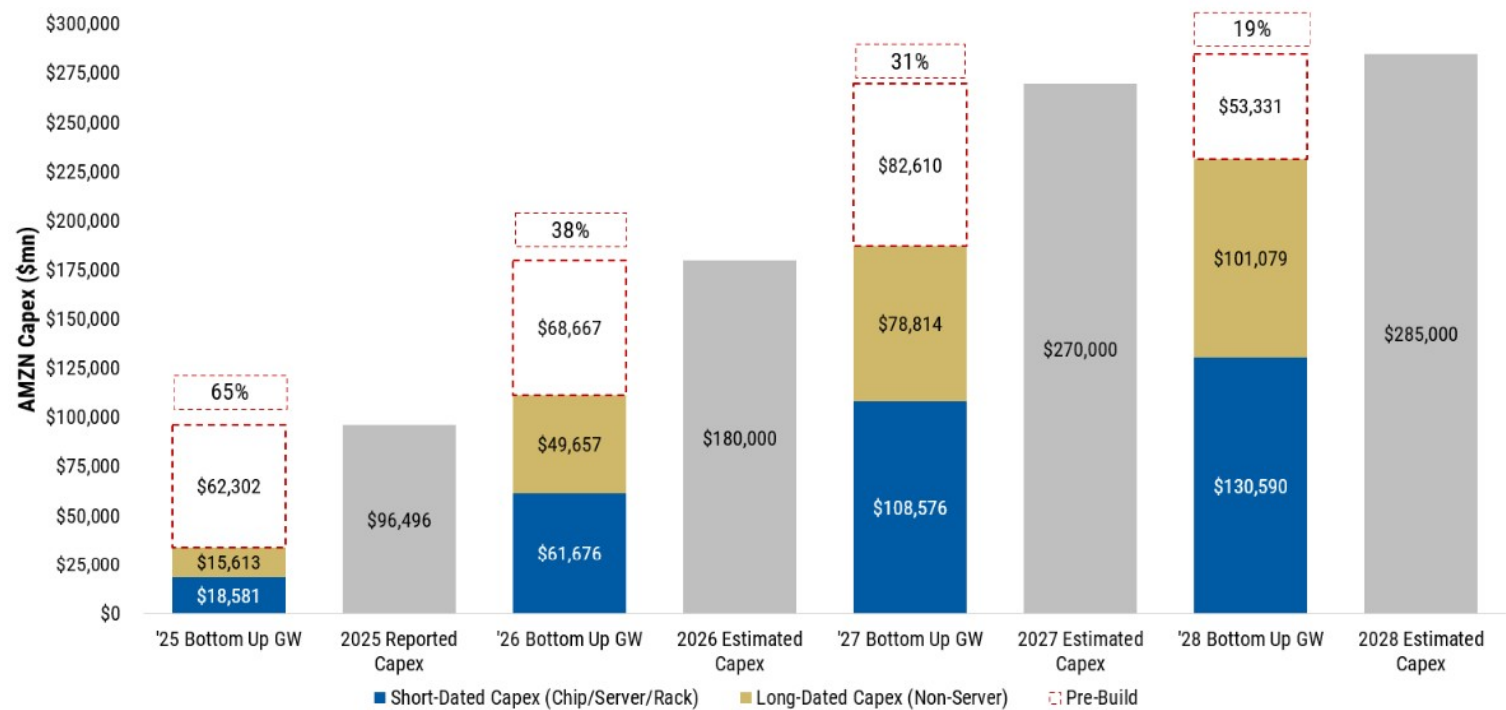
- **Component Inflation:** Component inflation (memory, cost of energy, etc.) remains high and could continue to creep higher. Next generations of chips are also more expensive... though with better performance and (we believe) better revenue opportunities.
- **Larger-than-expected chip production in '29 and beyond:** Could lead to more prebuild in '27/'28.

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AMZN Datacenter Capex Breakdown

- We see AMZN data center capex reaching \$180bn in '26, growing 50% in '27 and 6% in '28.
- ~38% of '26 capex is dedicated to “pre-build” for capacity to open in forward years.
- We see this prebuild growing 20% in '27 (rising by \$14bn), before contracting by 35% (or \$30bn) in '28 as the data center footprint scales.



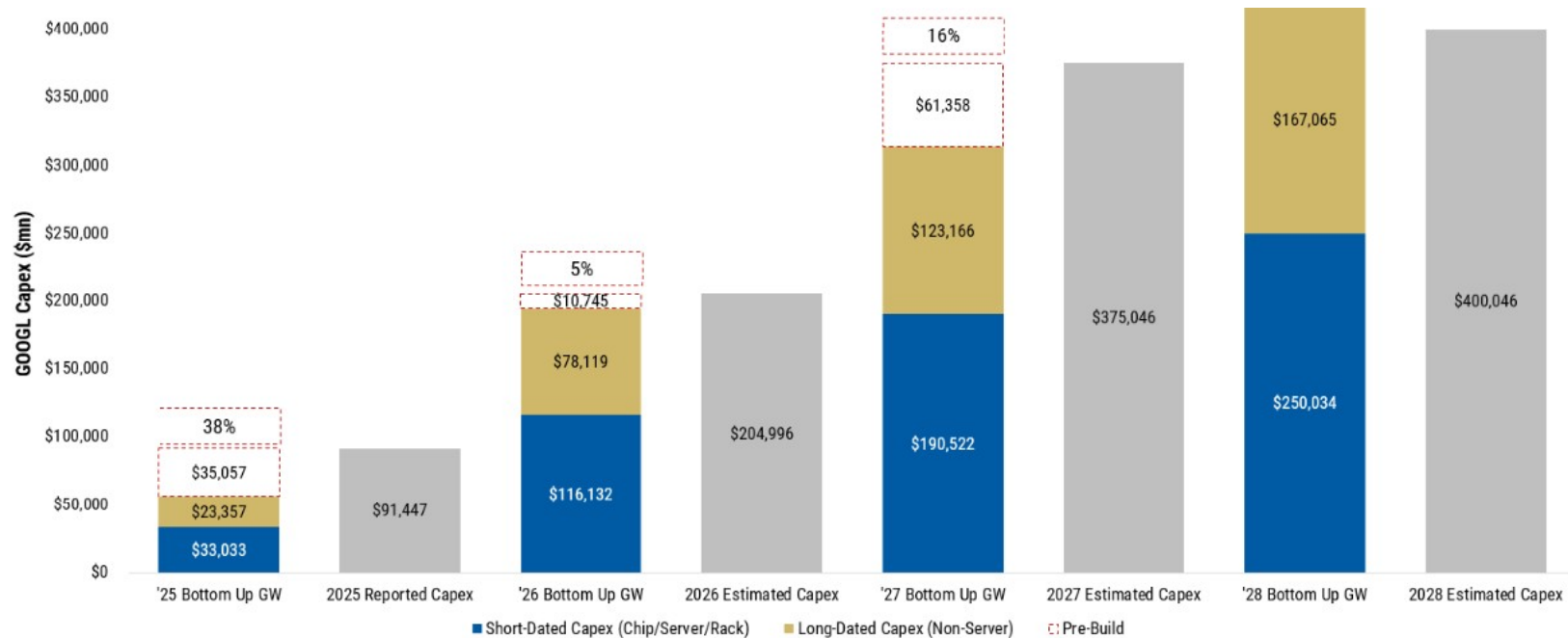
Source: Company data, Morgan Stanley Research estimates

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GOOGL Datacenter Capex Breakdown

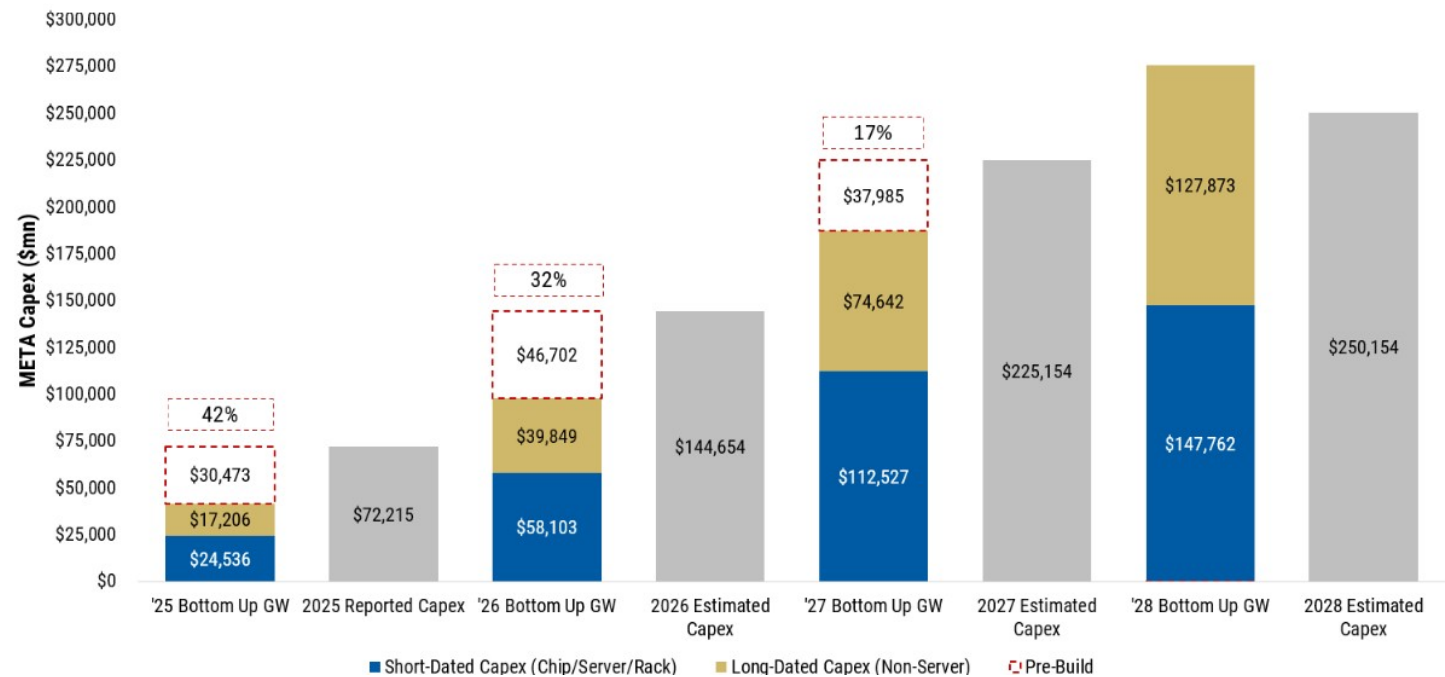
- We see GOOGL data center capex reaching \$205bn in '26, growing 83% in '27 and 7% in '28.
- Only ~5% of '26 capex is dedicated to “pre-build” ahead of capacity to open in forward years, speaking to the need to build more in '27, which is why we assume the “pre-build” grows ~6X in '27, reaching \$61bn. Like AMZN, expect prebuild needs to shrink in '28.



Source: Company data, Morgan Stanley Research estimates

META Datacenter Capex Breakdown

- We see META data center capex reaching \$145bn in '26, growing 55% in '27 and 11% in '28.
- ~32% of '26 capex is dedicated to “pre-build” ahead of capacity to open in forward years.
- We see this pre-build contracting 19% in '27 (or \$9bn) and see minimal pre-build needs in '28 as footprint scales.
- **META '28 Downward Capex Potential:** The extent to which META chooses to leverage hyperscalers (AWS/Azure) to scale its family of models and API (as it did in the past with Llama) would lower long-term compute needs and potentially put downward pressure on '28 capex. So too will the fact that META has \$279bn of leases yet to be commenced (much of which is for powered shells), which will bias downwards the long-dated portion of capex.



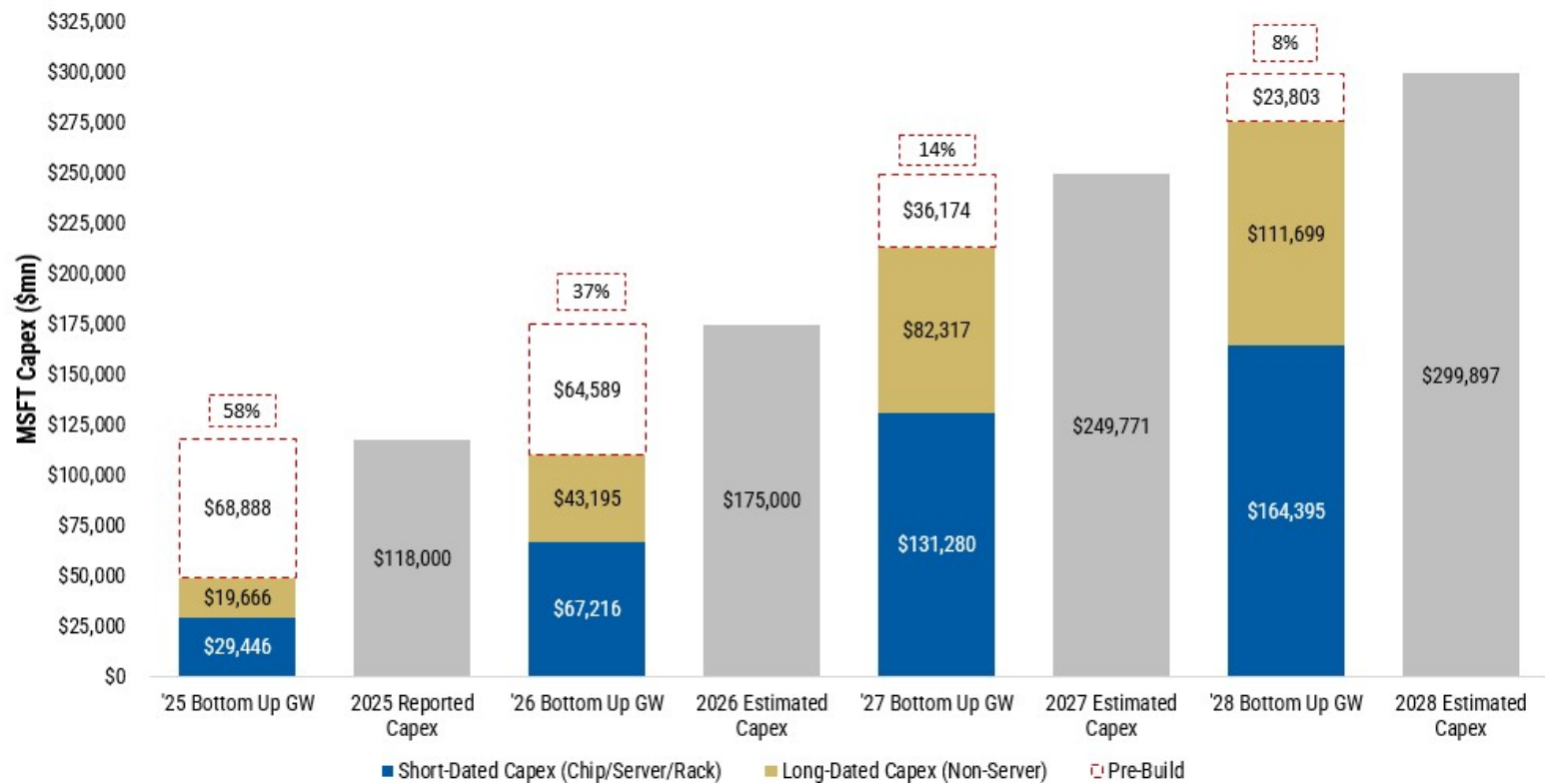
Source: Company data, Morgan Stanley Research estimates

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MSFT Datacenter Capex Breakdown

- We see MSFT data center capex reaching \$175bn in '26, growing 43% in '27 and 20% in '28.
- ~37% of '26 capex is dedicated to “pre-build” ahead of capacity to open in forward years.
- We see this pre-build slowing to 14% in '27 (or \$36bn) and 8% in '28 (or \$24bn).



Source: Company data, Morgan Stanley Research estimates

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How Much Compute Capacity Do We See Coming on?

Our View: We see the capex spend leading to ~80 GW of capacity coming online in '27/'28 (38 GW/44 GW). For perspective, this implies we reach ~145 GW of total compute capacity in '28, up ~4X from '25.

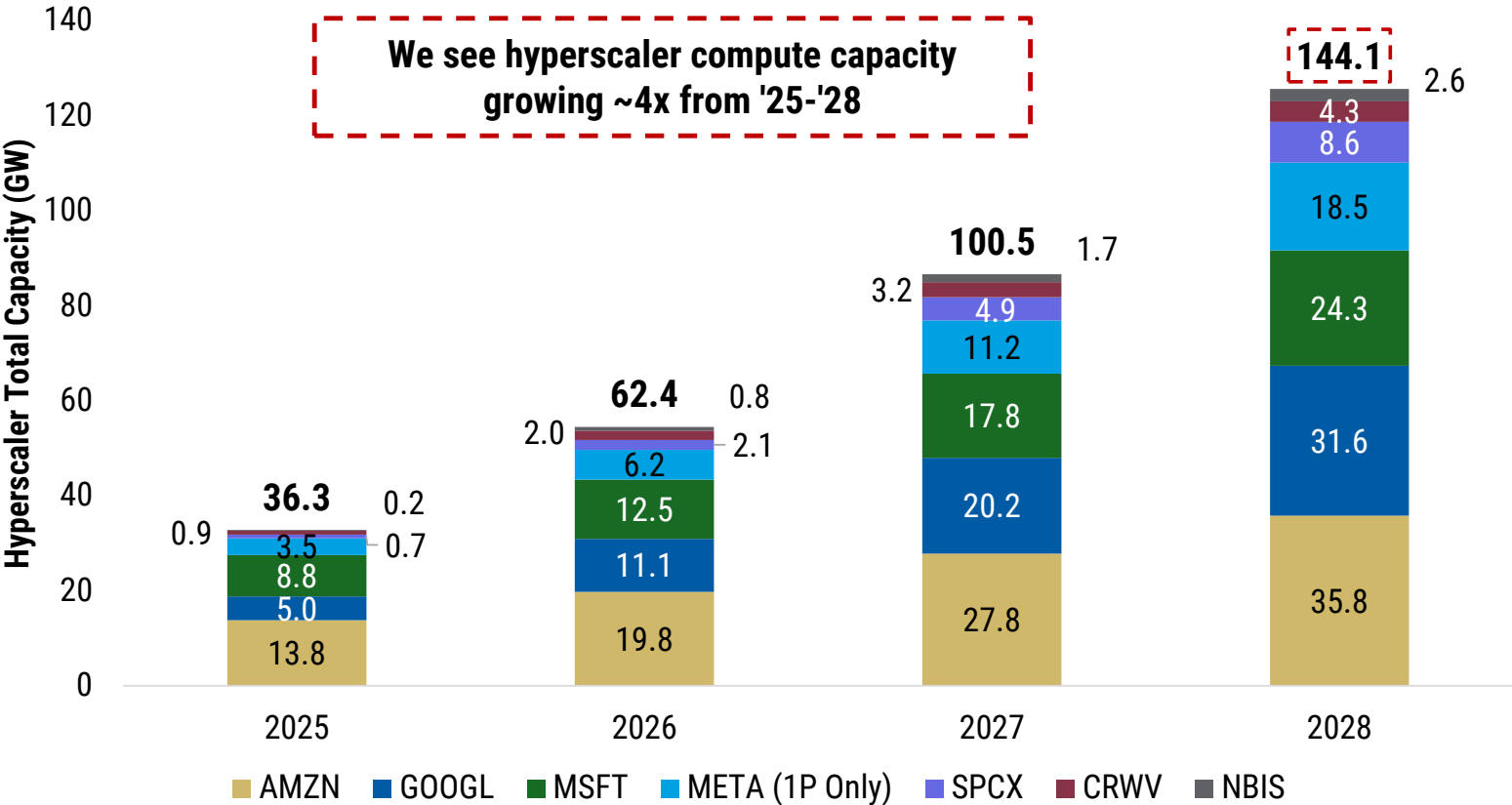
We see custom ASICs rising from 34% of incremental capacity in '25 to 66% of incremental capacity in '28, driven primarily by Trainium and TPU, of which we think a combined 12 GW will come online in '28 (vs. 13 GW for NVDA).

Stock Implications: Compute capacity remains a scarce resource that will be monetized as it comes online (new GenAI enabled tools, agentic workflows, etc). This means **hyperscalers, labs and enablers (AMZN, GOOGL, MSFT), and companies benefiting from diffusion** across the economy benefit.

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We See a Path to ~145GW of Compute Capacity Online by '28...



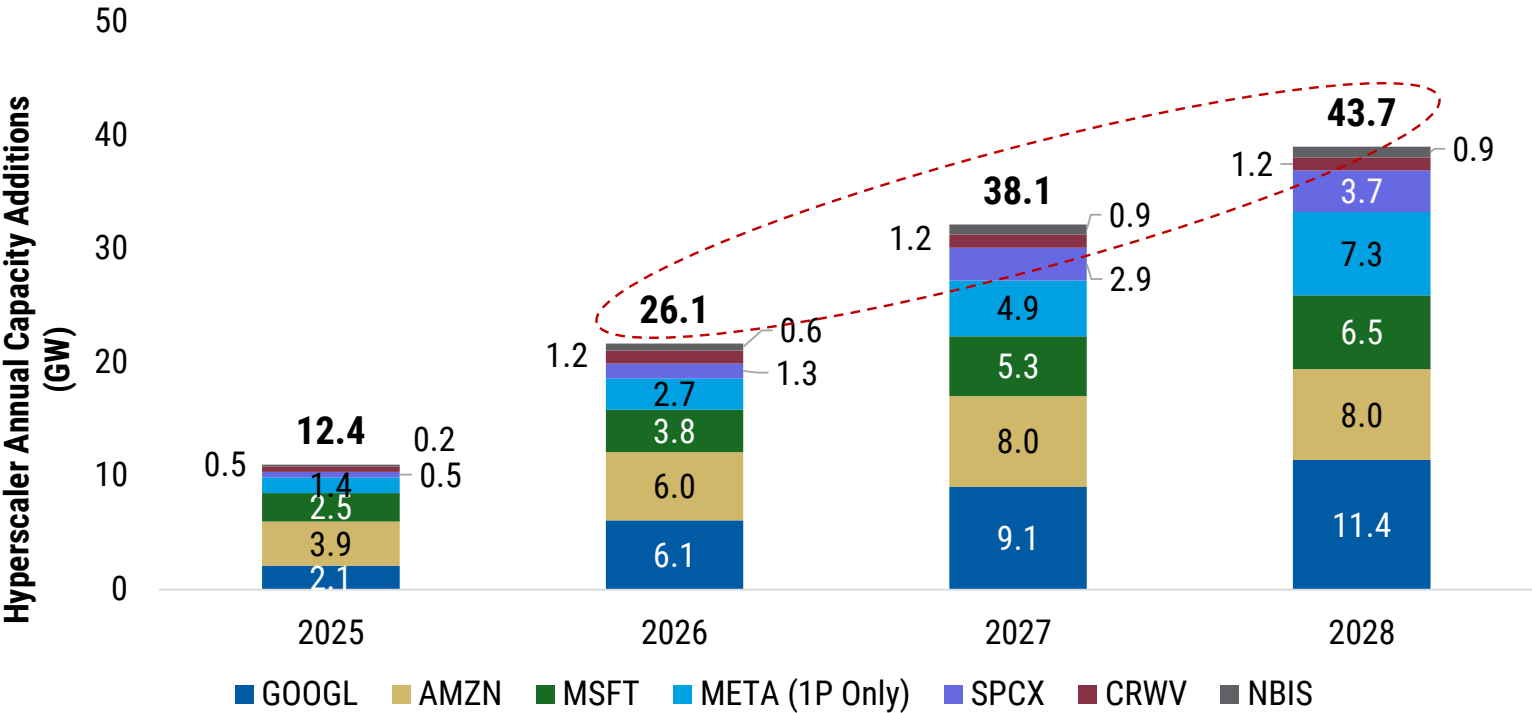
Source: Company data, Morgan Stanley Research estimates

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... With ~38/44 GW of Capacity Coming Online in '27/28

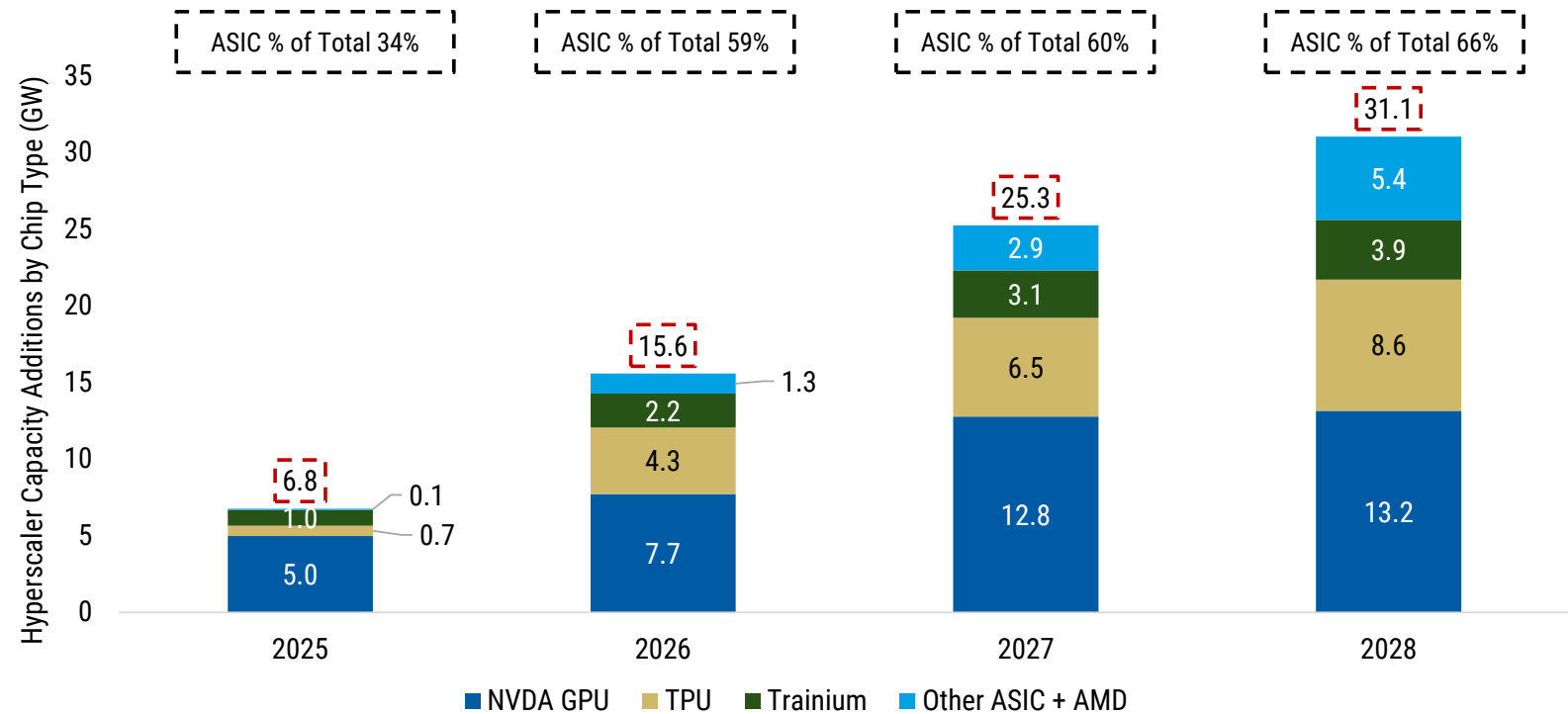
- Notably, we estimate **GOOGL adds the most capacity in '27/'28 (~9 GW/~11 GW)**... followed by AMZN, MSFT and META. This speaks to GOOGL's surging demand to: **1)** train leading frontier models (Gemini); **2)** enable GCP growth; and **3)** deploy new GenAI-enabled offerings across its core Search/YouTube offerings.



Source: Company data, Morgan Stanley Research

ASICs Incremental Capacity Share to Grow Over Next Few Years

- Focusing on the hyperscalers (AMZN, GOOGL, META and MSFT), we see total compute capacity reaching ~145 GW by '28 (up from ~35 in '25).
- We see custom ASICS growing from 34% of incremental compute (~2 GW) to 66% of incremental compute by '28 (~17 GW). GOOGL's TPU and AMZN's Trainium lead this shift... with GOOGL TPU bringing on ~15 GW in '27/'28.



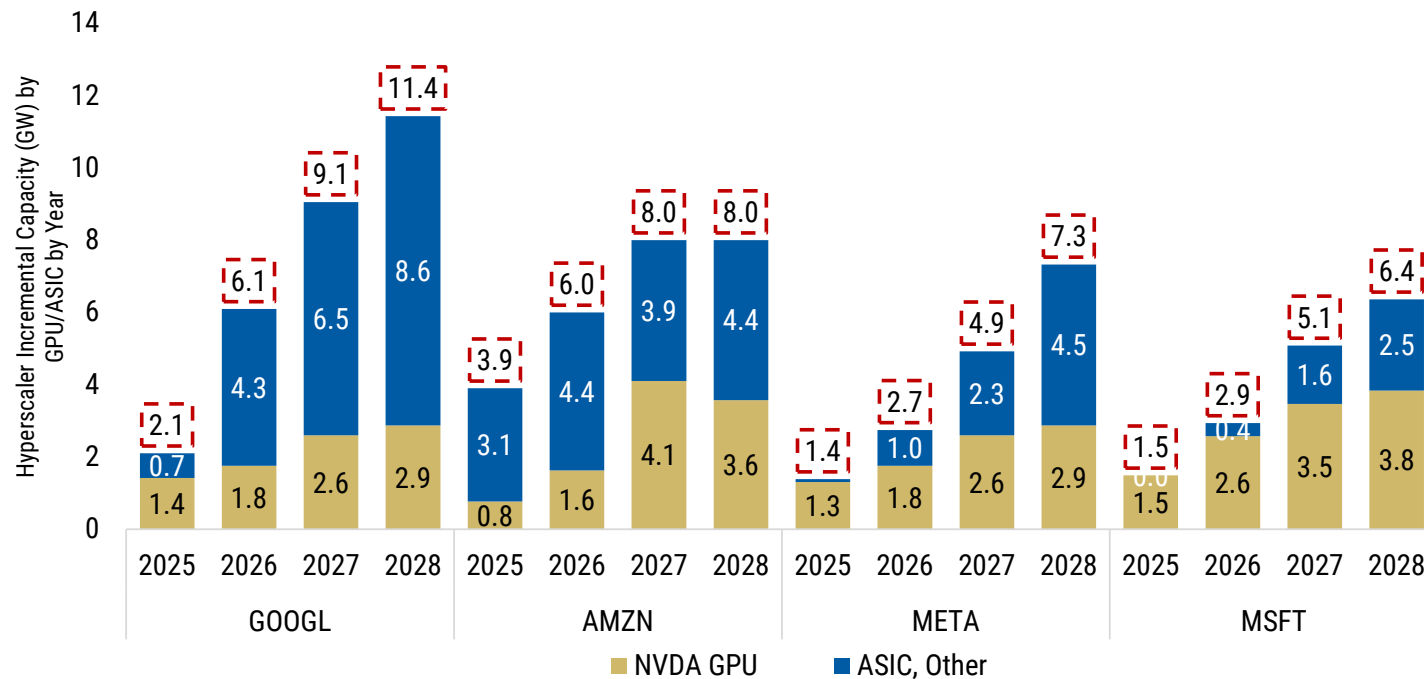
Source: Company data, Morgan Stanley Research estimates

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GOOGL and AMZN Are Driving the Custom Silicon Mix Shift

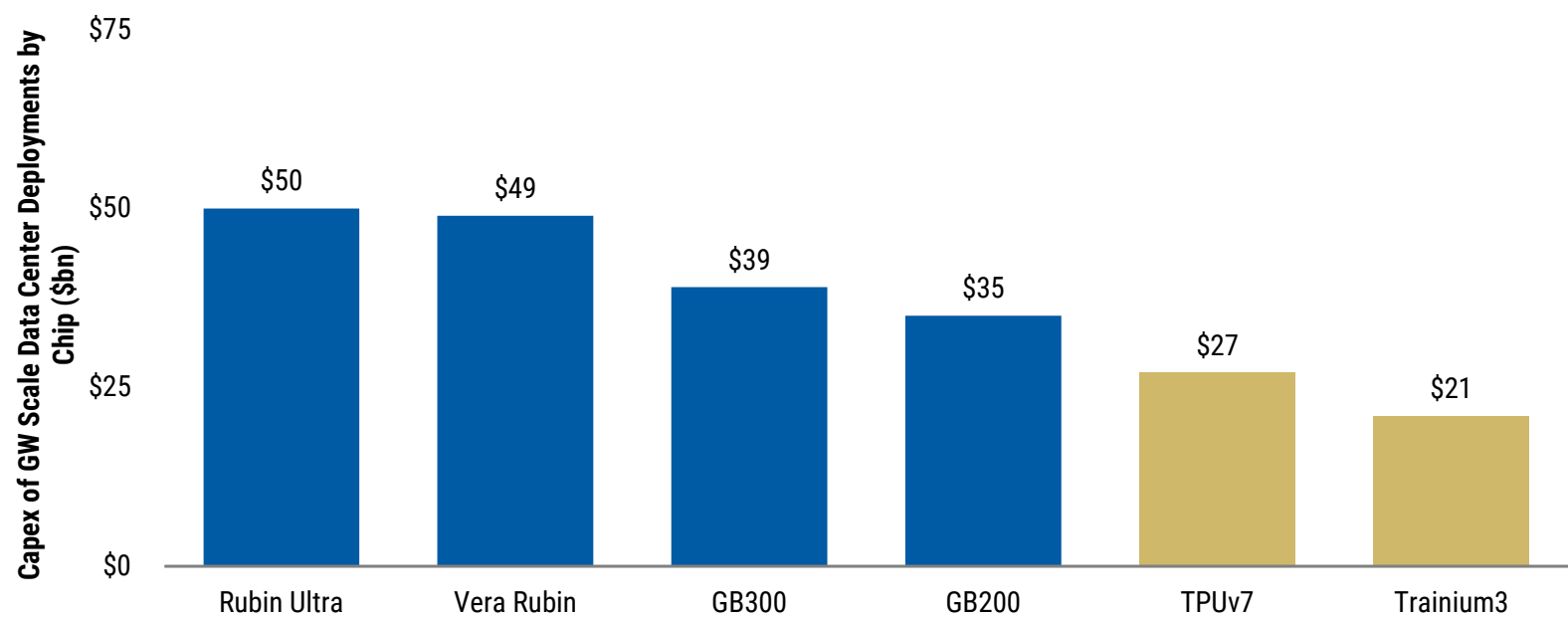
- **GOOGL:** Adding more ASIC than GPU capacity in '26 and beyond.
 - We estimate ~50% of capacity added in '26/'27 is for Google Cloud.
- **AMZN:** Ramping NVDA capacity in '26/'27 to meet AWS demands, while also forward selling Trn3/Trn4 1-2 years out.
 - We believe the substantial majority of capacity is for AWS.
- **META and MSFT:** ASIC and AMD deployments ramp over time, with >50% of incremental '27 capacity (internally owned) still NVDA.



Source: Company data, Morgan Stanley Research estimates

Our GW Capacity Forecast Is Informed by Our Bottom-up Cost per GW Estimates

- Our capex forecasts are also driven by updates to our bottom-up cost per GW which account for: **1)** increasing rack costs, as we align our rack BoMs with our Asia team’s [latest estimates](#), which account for memory inflation across GB200/GB300/VR200 racks; and **2)** increasing costs outside of the racks (powered shell, labor construction costs, etc.), where we have received better data points on the all-in non-rack costs.



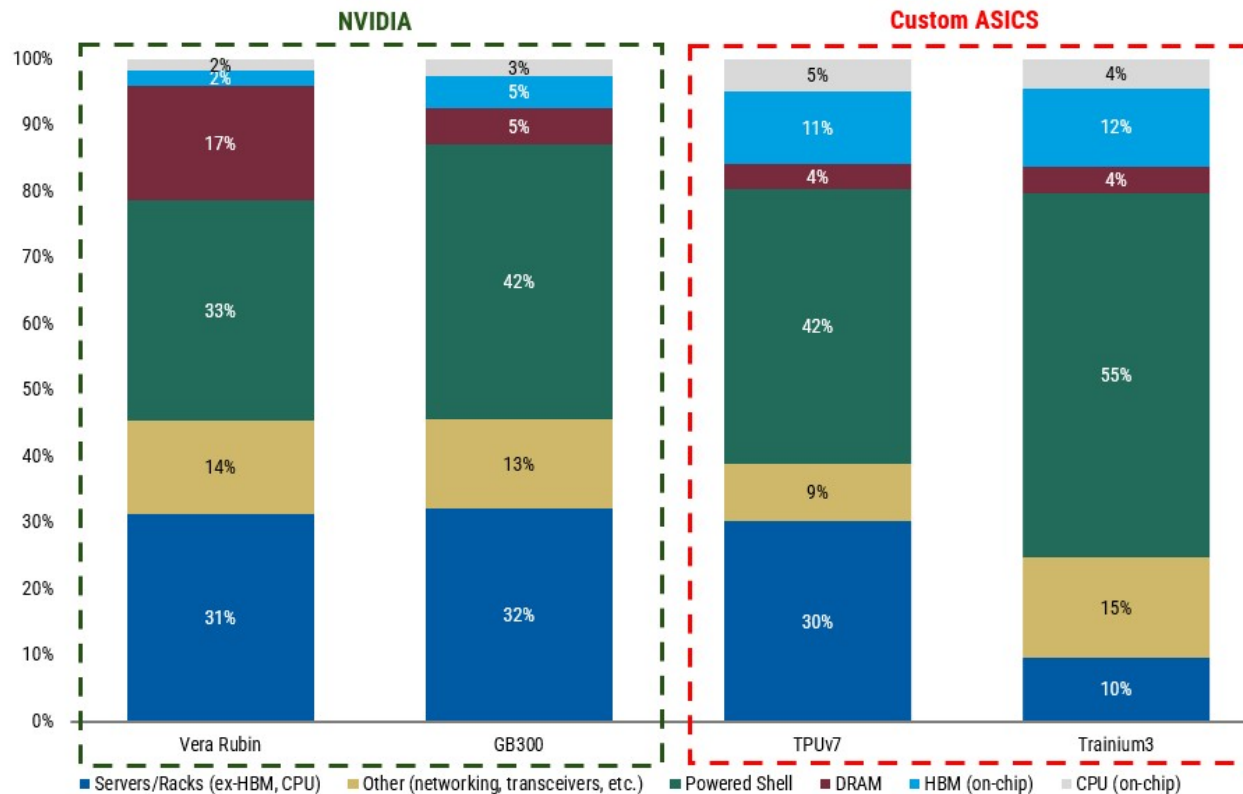
Source: Company data, Morgan Stanley Research estimates

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Powered Shell and Racks Still Biggest Driver of GW Investment, but Skew Towards Memory Grows as Component Prices Rise

- Powered Shell/Accelerator Racks lead cost categories at ~40%/~30% today for NVDA GB300 architectures, but we see memory inflation growing DRAM and HBM contribution meaningfully with new chip generations.



Source: Company data, Morgan Stanley Research estimates

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What Is the ROIC on GenAI Capex?

Our View: We detail three frameworks paving paths to 25-50% return on invested capital, with the highest returns accruing to the model layer on top of owned infrastructure... but the hyperscaler model is still set to generate ~30% ROIC.

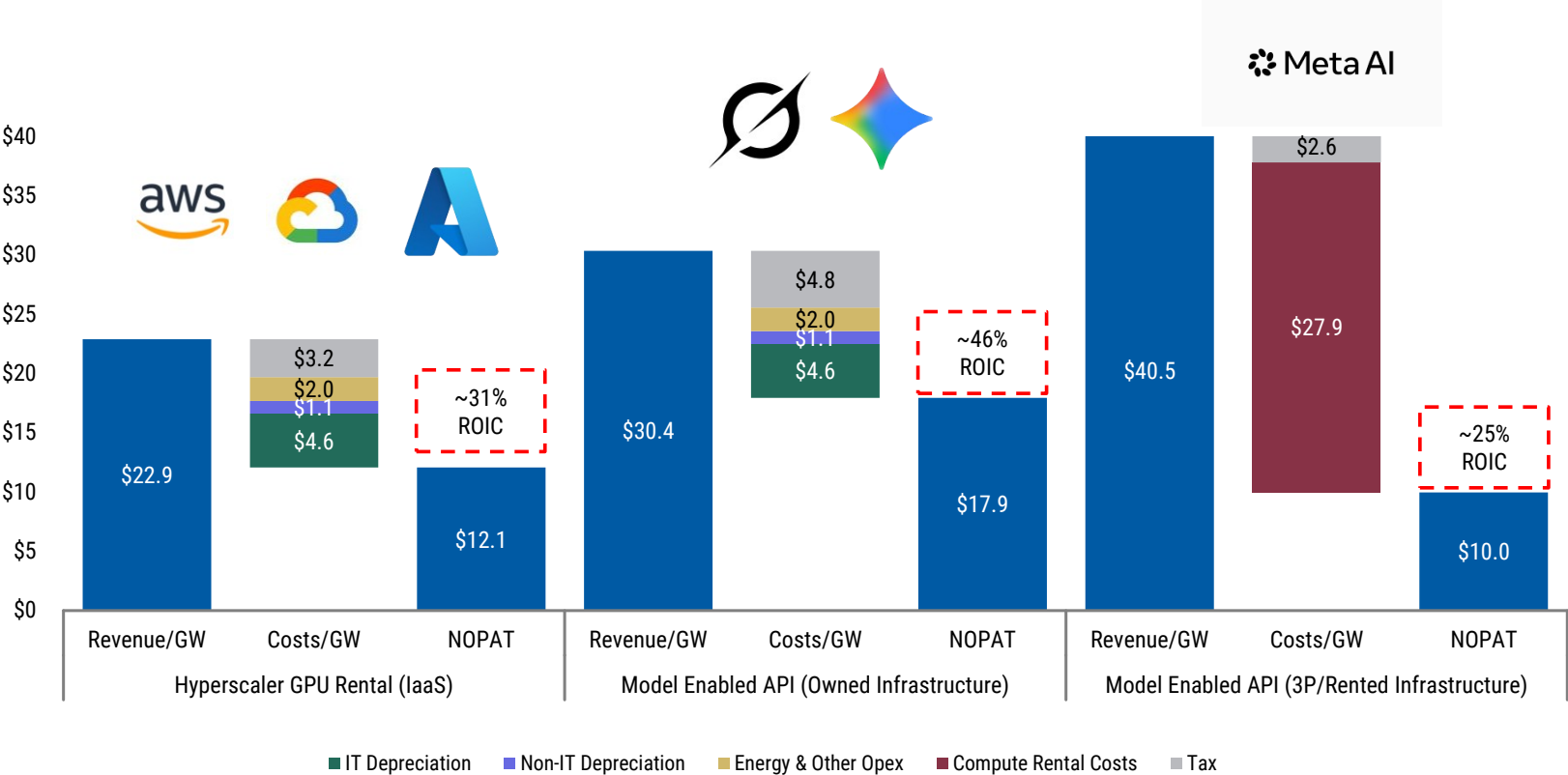
Stock Implications: Enablers and fully integrated model providers benefit here by showcasing return on spend (**GOOGL/AMZN/MSFT**)... with **META/SPCX** having more to prove from a product perspective but fully part of these beneficiaries.

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We Present 3 Frameworks Detailing GenAI ROIC of 25-50%

We estimate ROIC across 3 different monetization vectors including: **1) hyperscaler** renting out GPU/compute capacity; **2) model builders** serving their API through their own, internal infrastructure; and **3) model builders** serving their API through 3P infrastructure.



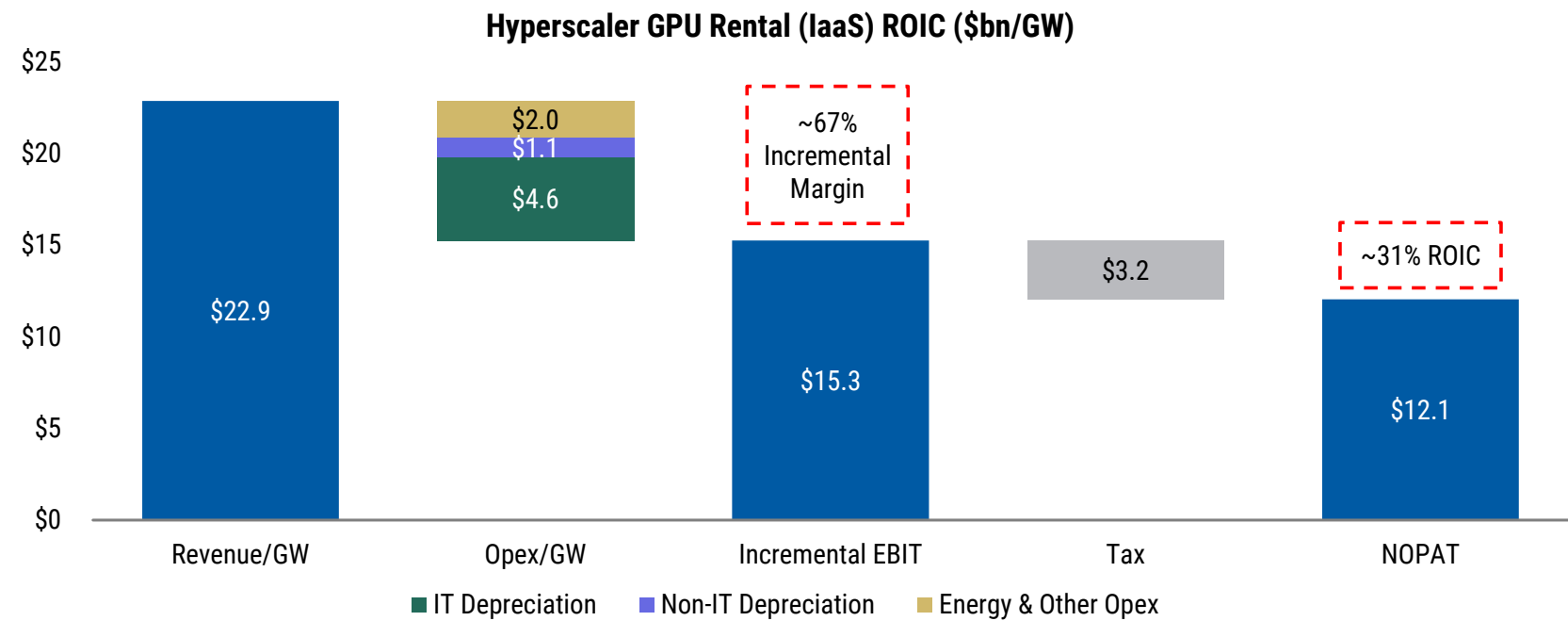
Source: Company data, Morgan Stanley Research estimates

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Hyperscalers: \$23bn in Rev/GW and 31% ROIC...

- We estimate hyperscalers would be able to monetize rented GB300 capacity at ~\$23bn/GW.
- \$8bn/GW of operating costs (which includes \$4.6bn of IT depreciation) leave ~\$15bn/GW of incremental EBIT at 67% incremental EBIT margins.
- Post-tax, this equates to ~\$12bn/GW of NOPAT or ~30% ROIC.



Source: Company data, Morgan Stanley Research estimates

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... Driven Primarily by GPU Rental Pricing, Capacity Utilization, and Capex/GW

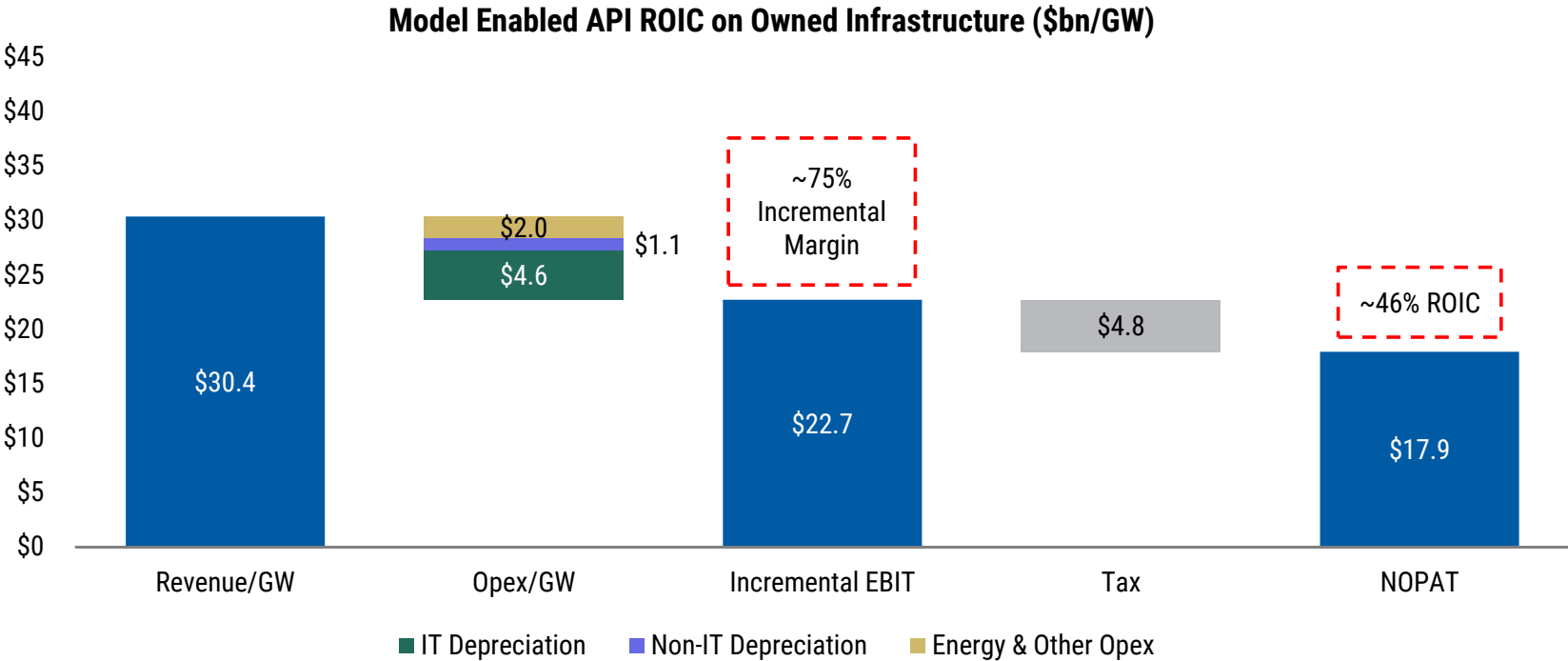
- We estimate ~30% ROIC on the GPU IaaS business for a GB300 GW, with ~70% incremental EBIT margins.
- Note that aggregate revenue/GW across the hyperscalers will still be lower than ~\$23 shown below given the majority of the compute fleet will be older chip generations (and traditional CPU workloads). The GB300s provide a useful framework for the long-term ROIC as hyperscalers deploy more performant capacity (which will command better pricing).

ROIC of Hyperscaler GPU Rentals (IaaS) on 1GW GB300 DC (\$bn)						Notes:
Revenue:						
Annual Hours	8,760	8,760	8,760	8,760	8,760	24 hrs. x 365 days
(X) GB300 Chips in 1GW	410,256	410,256	410,256	410,256	410,256	MS est of GB300 chips in 1GW datacenter
(=) GPU Hours (billions)	3.6	3.6	3.6	3.6	3.6	
(X) Utilization (%)	75%	75%	75%	75%	75%	MS est. of utilization
(X) Est. Price/Hour	\$7.0	\$8.0	\$8.5	\$9.0	\$10.0	MS est: GPU rental px/hr (range supported by industry/company data)
(=) Revenue/GW	\$18.9	\$21.6	\$22.9	\$24.3	\$27.0	
Costs:						
IT Depreciation Cost	\$5	\$5	\$5	\$5	\$5	IT capex divided by IT useful life
(+) Non-IT Depreciation Cost	\$1	\$1	\$1	\$1	\$1	Non-IT capex divided by Non-IT useful life
(+) Energy + Other Opex	\$2	\$2	\$2	\$2	\$2	Energy, labor, maintenance, etc, industry/company data
(=) Total Opex/GW	\$7.6	\$7.6	\$7.6	\$7.6	\$7.6	
EBIT & NOPAT:						
Incremental EBIT/GW	\$11.2	\$13.9	\$15.3	\$16.6	\$19.3	
Incremental EBIT Margins	60%	65%	67%	69%	72%	
(X) Tax Rate	21%	21%	21%	21%	21%	
(=) NOPAT	\$8.9	\$11.0	\$12.1	\$13.1	\$15.3	
% ROIC (NOPAT / Total Capex)	23%	28%	31%	34%	39%	
Capex for 1GW, GB300 Datacenter						
IT Capex	\$23	\$23	\$23	\$23	\$23	MS est. IT capex per 1 GW DC of GB300s
Useful Life	5	5	5	5	5	MS est. servers + network useful life
Non IT Capex	\$16	\$16	\$16	\$16	\$16	MS est. Non IT capex per 1 GW DC of GB300s
Useful Life	15	15	15	15	15	MS est. outside the rack useful life
Total Capex (IT + Non IT)	\$39	\$39	\$39	\$39	\$39	

Source: Company data, Morgan Stanley Research estimates

Model-Enabled API: \$30bn of Revenue/GW and 46% ROIC...

- We estimate model builders that serve APIs on their own infrastructure would be able to monetize capacity at ~\$30bn/GW.
- \$8bn/GW of operating costs (which includes \$4.6bn of IT depreciation) leave ~\$23bn/GW of incremental EBIT at 75% incremental EBIT margins.
- Post-tax, this equates to ~\$18bn/GW of NOPAT or ~46% ROIC.



Source: Company data, Morgan Stanley Research estimates

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... Driven Primarily by Capacity Allocation to Inference/Training, Token Throughput, and Token Pricing

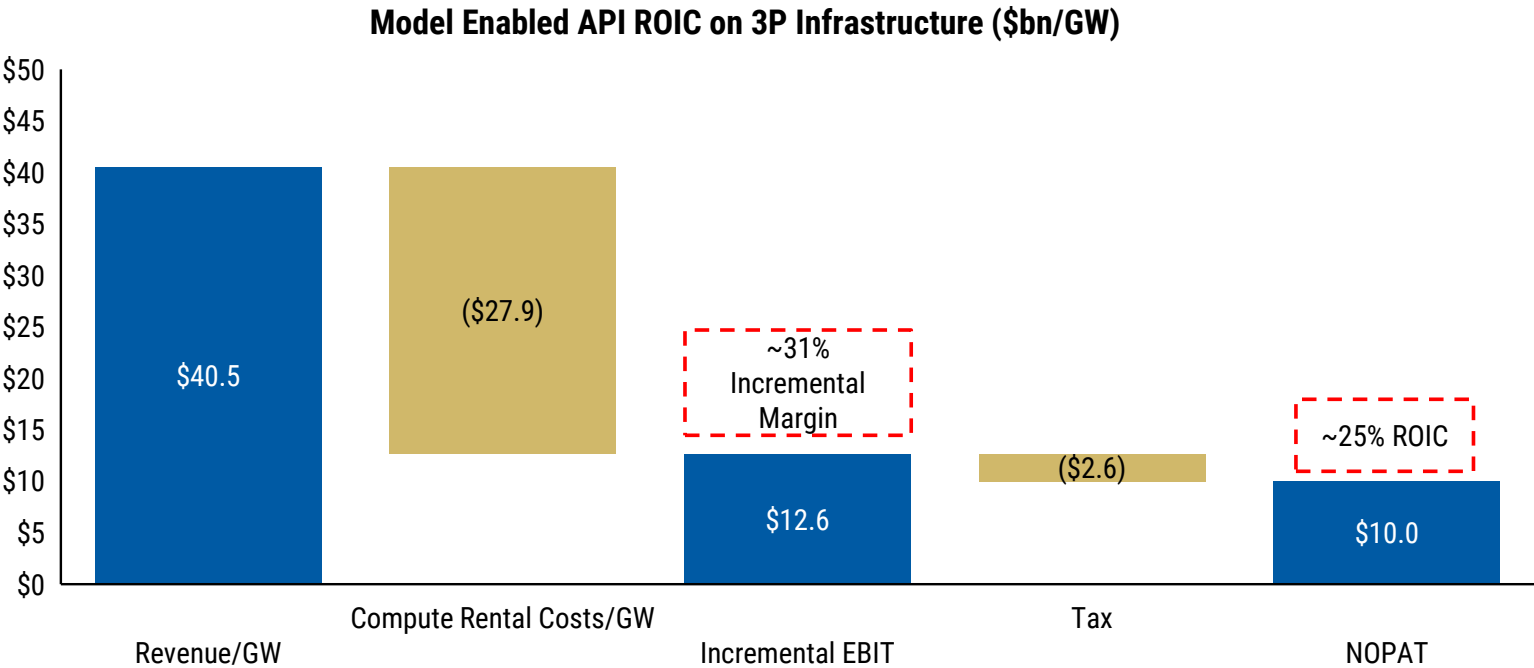
- We estimate ~45% ROIC on APIs run on internally owned infrastructure, with ~75% incremental EBIT margins.
- As part of our sensitivity analysis, we consider that some percentage of GB300 capacity will be allocated towards model training and will therefore not be used for revenue generating inference.
- In addition, we sensitize token throughput and pricing per million tokens to arrive at \$30bn/GW in base case revenue.

ROIC of Model Enabled API on 1GW of Owned Infrastructure (\$bn)						Notes:
Revenue:						
GB300 Chips in 1GW	410,256	410,256	410,256	410,256	410,256	MS est. of GB300 chips in 1GW datacenter
(X) % of Capacity Used for Inference	50%	60%	65%	70%	80%	Illustrative variable on % of capacity used for inference vs. training
(=) Revenue Generating Chips	205,128	246,154	266,667	287,179	328,205	
(X) Tokens/Second/GPU	3,500	3,000	2,750	2,500	2,000	MS est. range, Semianalysis Inference X, NVDA GTC
(X) Seconds per Year (k)	31,536	31,536	31,536	31,536	31,536	
(X) Utilization	75%	75%	75%	75%	75%	MS est. of uptime
(=) Tokens per Year (tn)	16,981	17,466	17,345	16,981	15,525	
(X) Token Pricing per Million	\$1.00	\$1.50	\$1.75	\$2.00	\$2.50	MS est. range. Muse Spark 1.1 blended pricing \$1.58 (8K/1K I/O)
(=) Revenue/GW	\$17	\$26	\$30	\$34	\$39	
Costs:						
IT Depreciation Cost	\$5	\$5	\$5	\$5	\$5	IT capex divided by IT useful life
(+) Non-IT Depreciation Cost	\$1	\$1	\$1	\$1	\$1	Non-IT capex divided by Non-IT useful life
(+) Energy + Other Opex	\$2	\$2	\$2	\$2	\$2	Energy, labor, maintenance, etc, industry/company data
(=) Total Opex/GW	\$7.6	\$7.6	\$7.6	\$7.6	\$7.6	
EBIT & NOPAT:						
Incremental EBIT/GW	\$9.3	\$18.6	\$22.7	\$26.3	\$31.2	
Incremental EBIT Margins	55%	71%	75%	78%	80%	
(X) Tax Rate	21%	21%	21%	21%	21%	
(=) NOPAT	\$7.4	\$14.7	\$17.9	\$20.8	\$24.6	
% ROIC (NOPAT / Total Capex)	19%	38%	46%	53%	63%	
Capex for 1GW, GB300 Datacenter						
IT Capex	\$23	\$23	\$23	\$23	\$23	MS est. IT capex per 1 GW DC of GB300s
Useful Life	5	5	5	5	5	MS est. servers + network useful life
Non IT Capex	\$16	\$16	\$16	\$16	\$16	MS est. Non IT capex per 1 GW DC of GB300s
Useful Life	15	15	15	15	15	MS est. outside the rack useful life
Total Capex (IT + Non IT)	\$39	\$39	\$39	\$39	\$39	

Source: Company data, Morgan Stanley Research estimates

Model-Enabled API on 3P Capacity: ~\$40bn of Revenue/GW and 25% ROIC...

- We estimate model builders that serve APIs on external infrastructure would be able to monetize capacity at ~\$41bn/GW.
- \$28bn/GW of rental costs leaves ~\$13bn/GW of incremental EBIT.
- Post-tax, this equates to ~\$10bn/GW of NOPAT.



Source: Company data, Morgan Stanley Research estimates

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... Driven Primarily by GPU Rental Pricing, Capacity Allocation to Inference/Training, Token Throughput, and Token Pricing

- We estimate ~25% NOPAT Margin on APIs run on 3P infrastructure, with ~30% incremental EBIT margins.
- We consider that some percentage of GB300 capacity will be allocated towards model training and will therefore not be used for revenue generating inference.
- In addition, we sensitize token throughput and pricing per million tokens to arrive at \$30bn/GW in base case revenue.

NOPAT Margin of Model Enabled API on 1GW of 3P Infrastructure (\$bn)						Notes:
Revenue:						
GB300 Chips in 1GW	410,256	410,256	410,256	410,256	410,256	MS est. of GB300 chips in 1GW datacenter
(X) % of Capacity Used for Inference	50%	60%	65%	70%	80%	Illustrative variable on % of capacity used for inference vs. training
(=) Revenue Generating Chips	205,128	246,154	266,667	287,179	328,205	
(X) Tokens/Second/GPU	3,500	3,000	2,750	2,500	2,000	MS est. range, Semianalysis Inference X, NVDA GTC
(X) Seconds per Year (k)	31,536	31,536	31,536	31,536	31,536	
(=) Tokens per Year (qdn)	23	23	23	23	21	
(X) Token Pricing per Million	\$1.00	\$1.50	\$1.75	\$2.00	\$2.50	MS est. range. Muse Spark 1.1 blended pricing \$1.58 (8K/1K I/O)
(=) Revenue/GW	\$23	\$35	\$40	\$45	\$52	
Costs:						
GPU Rental Price/Hour	\$7.0	\$7.5	\$7.8	\$8.0	\$8.5	MS est: GPU rental px/hr (supported by industry/company data)
(X) Hours/Year	8,760	8,760	8,760	8,760	8,760	
(X) GB300 Chips in 1GW	410,256	410,256	410,256	410,256	410,256	MS est: GB300 chips in 1GW datacenter
(=) Cost of Compute/GW (\$bn)	\$25	\$27	\$28	\$29	\$31	
EBIT & NOPAT:						
Incremental EBIT/GW	(\$2.5)	\$8.0	\$12.6	\$16.5	\$21.2	
Incremental EBIT Margins	(11%)	23%	31%	37%	41%	
(X) Tax Rate	21%	21%	21%	21%	21%	
(=) NOPAT	(\$2.0)	\$6.3	\$10.0	\$13.1	\$16.8	
% Margin (NOPAT / Revenue)	-9%	18%	25%	29%	32%	

Source: Company data, Morgan Stanley Research estimates

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How Big Are the Global GenAI TAMs Across Knowledge Work and Consumer Spend?

Our View: \$50-\$60tn: We identify a \$20-\$30tn opportunity in the global knowledge work spend to be digitized and augmented through GenAI tools. In addition, we see ~\$30tn of consumer spend (across retail, travel, autonomous, advertising, etc.) to be digitized over the next decade.

We also analyze hyperscaler and Facebook adoption curves as analogies to consider... speaking to the trillion-dollar opportunities ahead.

Stock Implications: Enterprise AI adoption benefits the hyperscalers and software enablers, while consumer AI adoption benefits enabling platforms.

We See a \$20-\$30tn Global Knowledge Work TAM to Be Augmented and Further Digitized

Global Knowledge Work TAM (\$bn)					
Avg. Labor Comp	Est. Knowledge Work Mix of 3.6bn global employment				
		15%	20%	25%	30%
	\$20,000	\$10,797	\$14,396	\$17,995	\$21,594
	\$25,000	\$13,497	\$17,995	\$22,494	\$26,993
	\$30,000	\$16,196	\$21,594	\$26,993	\$32,392
	\$35,000	\$18,895	\$25,194	\$31,492	\$37,790

Global Knowledge Work TAM	(abs/\$ in bn)	Note:
Global Employment	3.6	Int. Labor Org. "ILO" -modelled estimate of all working-age persons worldwide
Est. Knowledge Workers	0.9	ILO-modelled estimate for managers, professionals, technicians & assoc. professionals, clerical support workers
Implied Knowledge Work Mix (%)	25%	
Knowledge Workers (X) Avg. Comp	\$25,200	Per ILO 2024-2025 wage report, top 25% (knowledge work proxy) of wage workers globally (proxy for knowledge work) earn equivalent of ~\$2,100/month
(=) Knowledge Work TAM	\$22,557	~\$22.5 trillion of knowledge work TAM

Source: International Labor Organization, Morgan Stanley Research estimates



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With Use Cases Spanning the Economy

Healthcare	Financials	Advertising & Digital Content	Retail & Freight	Energy & Industrials	Customer Service	Business Intelligence
Smart Implants (Ex: Hip & Knee)	AI Thematic Fund Offerings	Marketing Efficiency	3D Shopping & Product Try On	AI Tools for Pipelines	Customer Analytics	Developer Tools & Assisted Code
Medical Imaging	New Trade Pricing Tools	Enhanced Ad Targeting	Sidewalk Robots for Deliveries	Geological Models	Insurance Claims and Records	AI-Related Consulting
Pathology Detection	Generate Client Portfolio Reviews	Recommendation Engine	Supply Chain Efficiency	AI-Enabled Microgrid Tools	Transcribe and Summarize	Integrated Smart Home Interface
Personalized Medicine	Automated Trade Prices, Execution	Higher Ad Conversion	Inventory Management	Energy Usage Analytics	Client Onboarding	Automate Back Office Tasks
Identify Biomarkers	AI Portfolio Management	Ad Auction Dynamics	Demand Prediction	Energy Trading Enhancements	Shopping AI Agents	Fraud Prevention and Protection
New Drug Development	Increase Trading Velocity	Content Creation	Truck Routing Capabilities	Labor Scheduling		Cybersecurity Capabilities
Analysis of Genomic Data	Fraud Detection/Prevention	Customized Travel Itineraries	Higher Fleet Utilization	Reservoir Optimization		Risk Assessment & Management
Physician Point of Service Tools		Video Game Development	Autonomous Fleet Networks	Leak Detection & Track Emissions		Accelerated Data Analysis
Predictive Health Insights		Campaign Creation Tools		Nuclear Energy Projects		Predictive Modeling

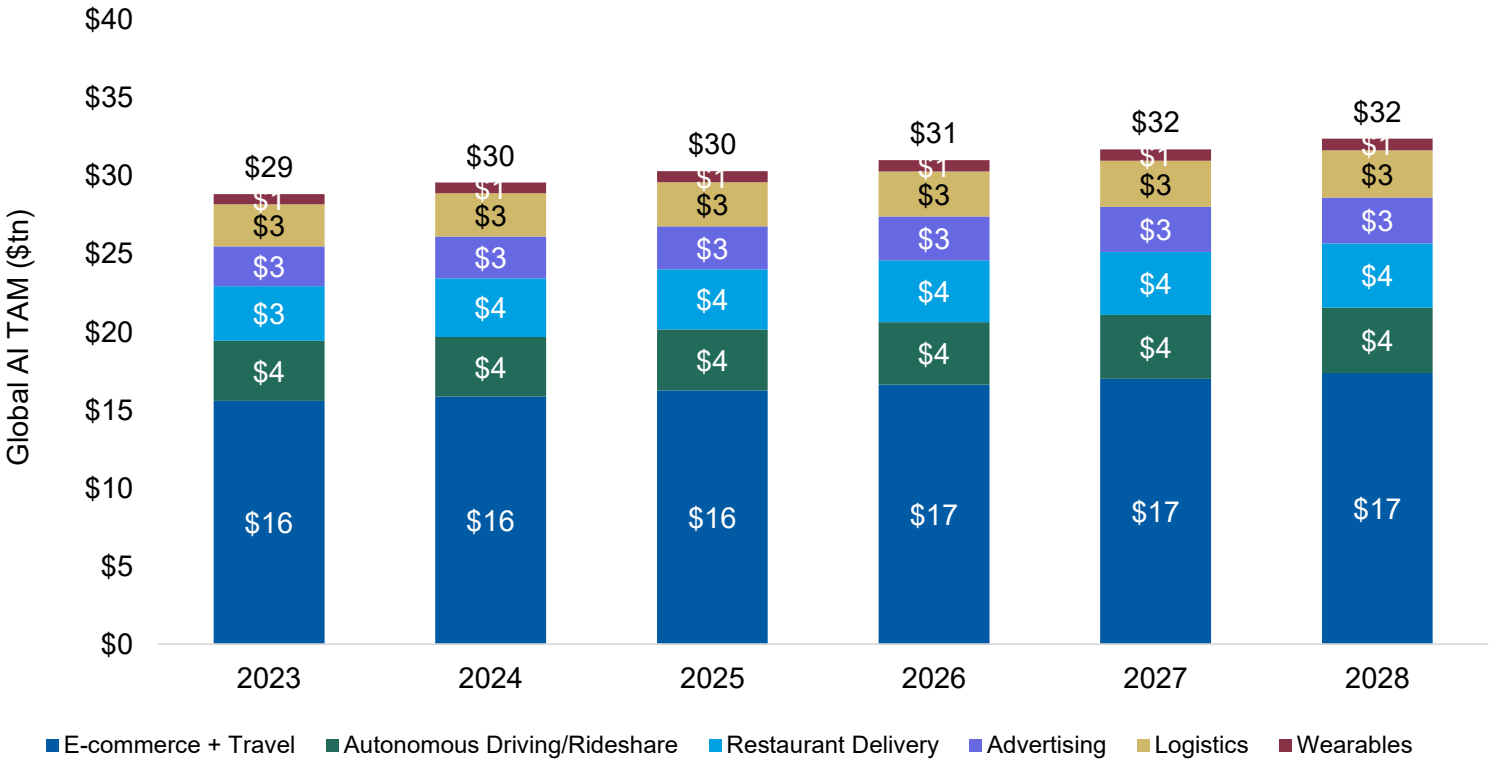
Source: Company data, Morgan Stanley Research estimates.

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We Identify ~\$30tn of Consumer Spend to Be Further Digitized...

- We see a ~\$30tn consumer TAM across Retail + Travel (~\$16tn), Rideshare/Autonomous (\$4tn), Food Delivery (~\$4tn) and Advertising (~\$3tn).
- As consumer behavior and spend is digitized, these dollars will move online.



Source: Company data, PCE, BEA, Euromonitor, Morgan Stanley Research estimates

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... With Personal Agents and Agentic Shoppers a Substantial Potential Unlock Alone...

Higher Consumer Utility Leads to Further Digitization of Consumers' Wallets



Source: Morgan Stanley Research



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... And Next Generation Use Cases and Sources of Utility to Emerge Too

Next Gen Wearables

Meta Ray-Ban Smart Glasses

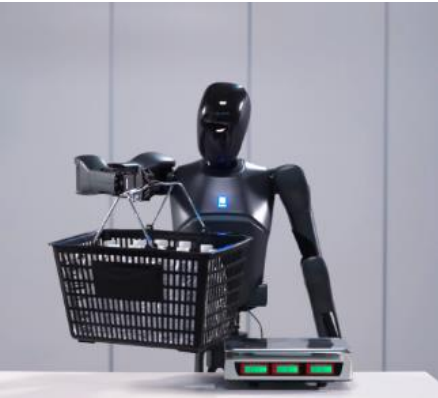


Meta Orion Glasses



Industrial Robotics

PUDU D7 Semi-Humanoid



Amazon Warehouse Robots



Autonomous Driving

Waymo



Tesla Robotaxi



DoorDash/Wing Drone Delivery



Source: Meta, Pudu Robotics, Waymo, Tesla, Morgan Stanley Research



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And Remember... on Average, Wall Street Underestimates the Opportunity... by Almost 40% Based on History

Compute Cycle	Initial Forecast	Actual	% Underestimation
 PC	225mn <i>PC Users in 2000</i>	354mn <i>PC Users in 2000</i>	36%
 Internet	152mn <i>Internet Users in 2000</i>	361mn <i>Internet Users in 2000</i>	58%
 Mobile	657mn <i>Smartphones Shipments in 2013</i>	1,019mn <i>Smartphones Shipments in 2013</i>	36%
 Cloud	\$90.2bn <i>Top 3 Cloud Provider Revenue in 2020</i>	\$115.6bn <i>Top 3 Cloud Provider Revenue in 2020</i>	22%

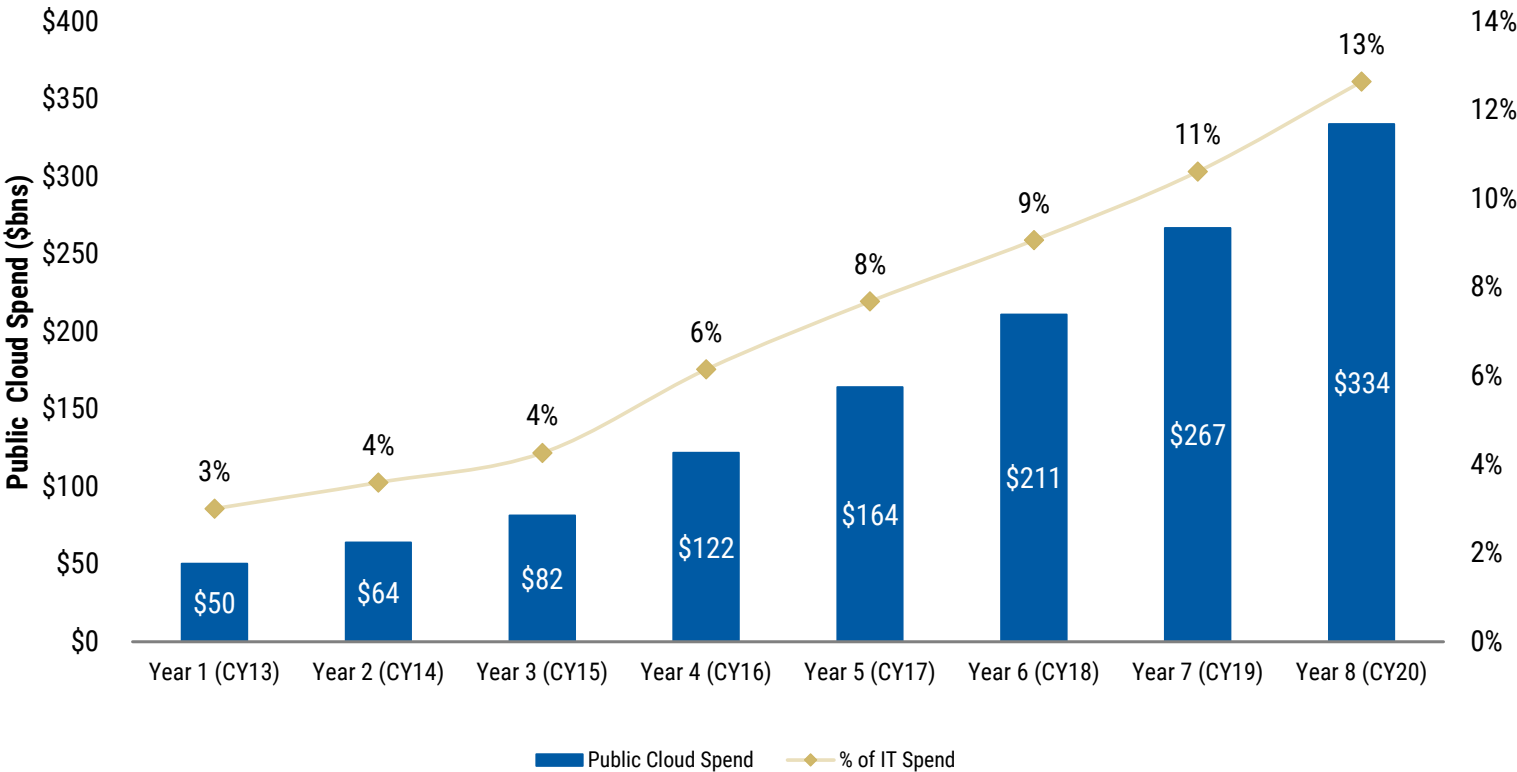
Average Underestimation: 38%

Source: Morgan Stanley Research

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What About Adoption Analogies? On the Enterprise Side, Consider that Public Cloud Spend Reached 4% of IT Budgets in Year 2 (2014)



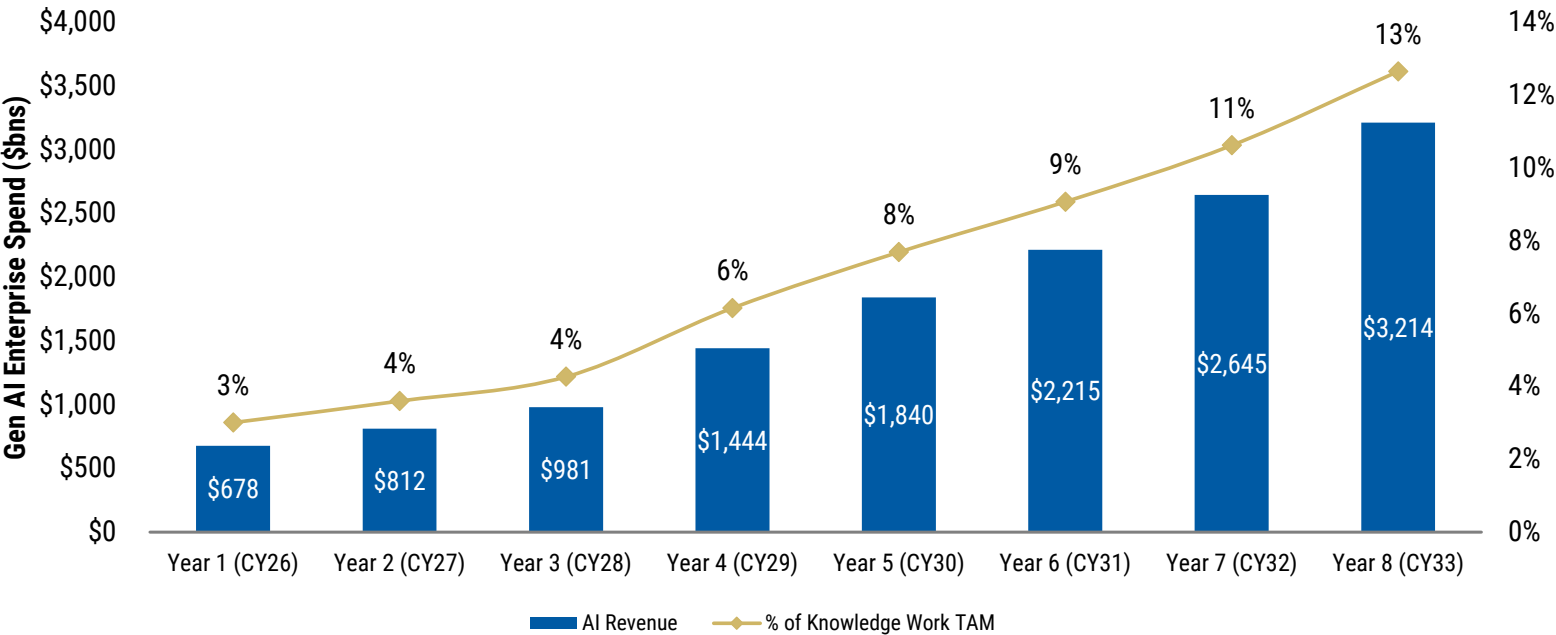
Source: Company data, Morgan Stanley Research estimates, IDC

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GenAI Should Be Faster for Multiple Reasons... but Even 4% Penetration in Year 2 (Conservatively Saying its '27 Here) Could Imply ~\$800bn of GenAI Enterprise Spend

- Using the public cloud adoption cycle as an analogy for future AI adoption, we see ~\$800bn of Enterprise AI spend by 2027 or about 4% penetration of the broader Knowledge Work TAM.
- However, **AI should diffuse faster than public cloud** for a few reasons including: **1)** AI doesn't require wholesale infrastructure migration; **2)** shorter time-to-value from impactful productivity gains; and **3)** digital is known and enterprises face a more substantial competitive opportunity cost if they fall behind peers on AI adoption.

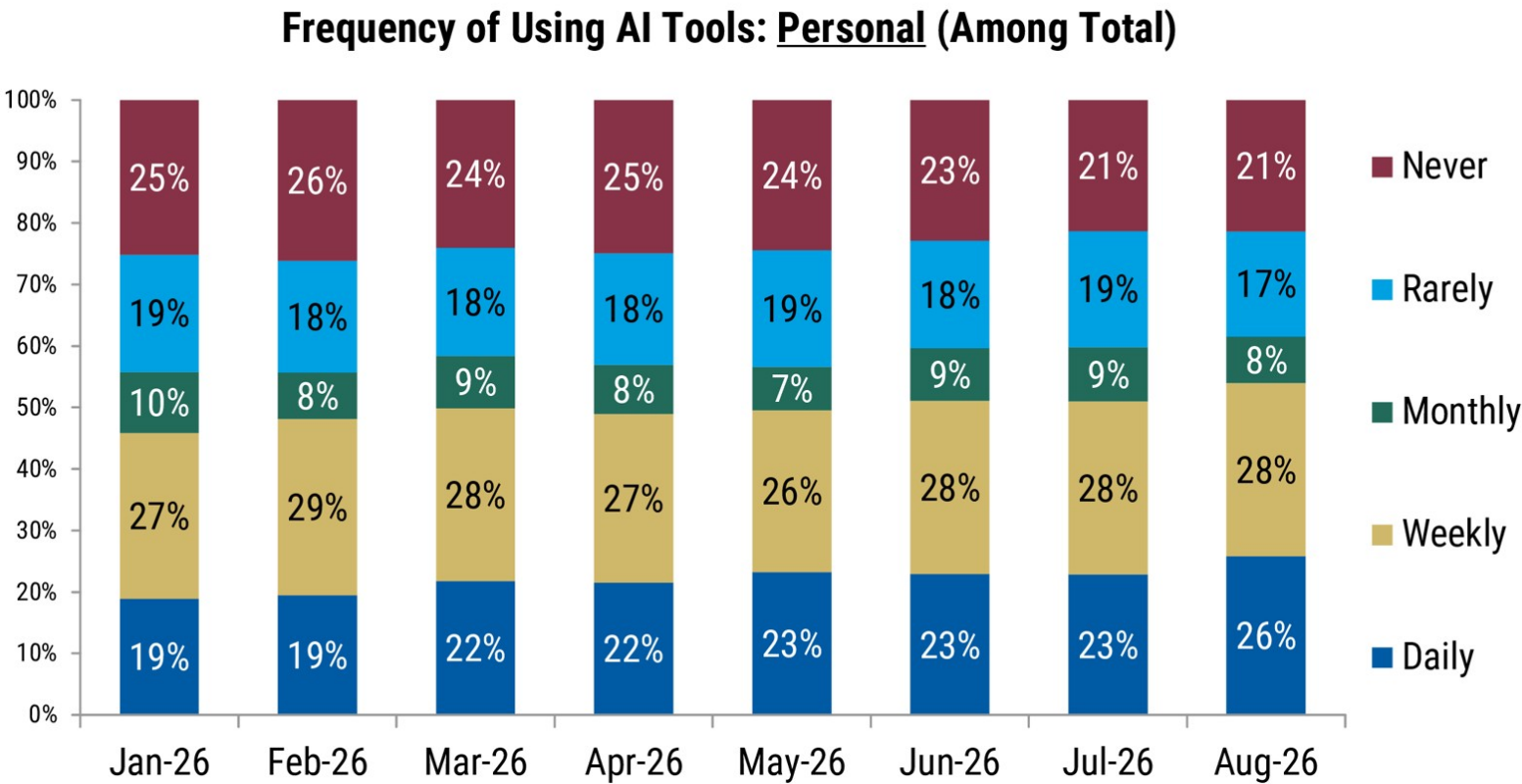


Source: Company data, Morgan Stanley Research estimates. Note: Exhibit above assumes Knowledge Work TAM starts at \$22.6tn in 2027 and grows at 2% per year, thereafter.

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On the Consumer Front, AI Tools Are Already Used Daily by ~25% of the US Population...



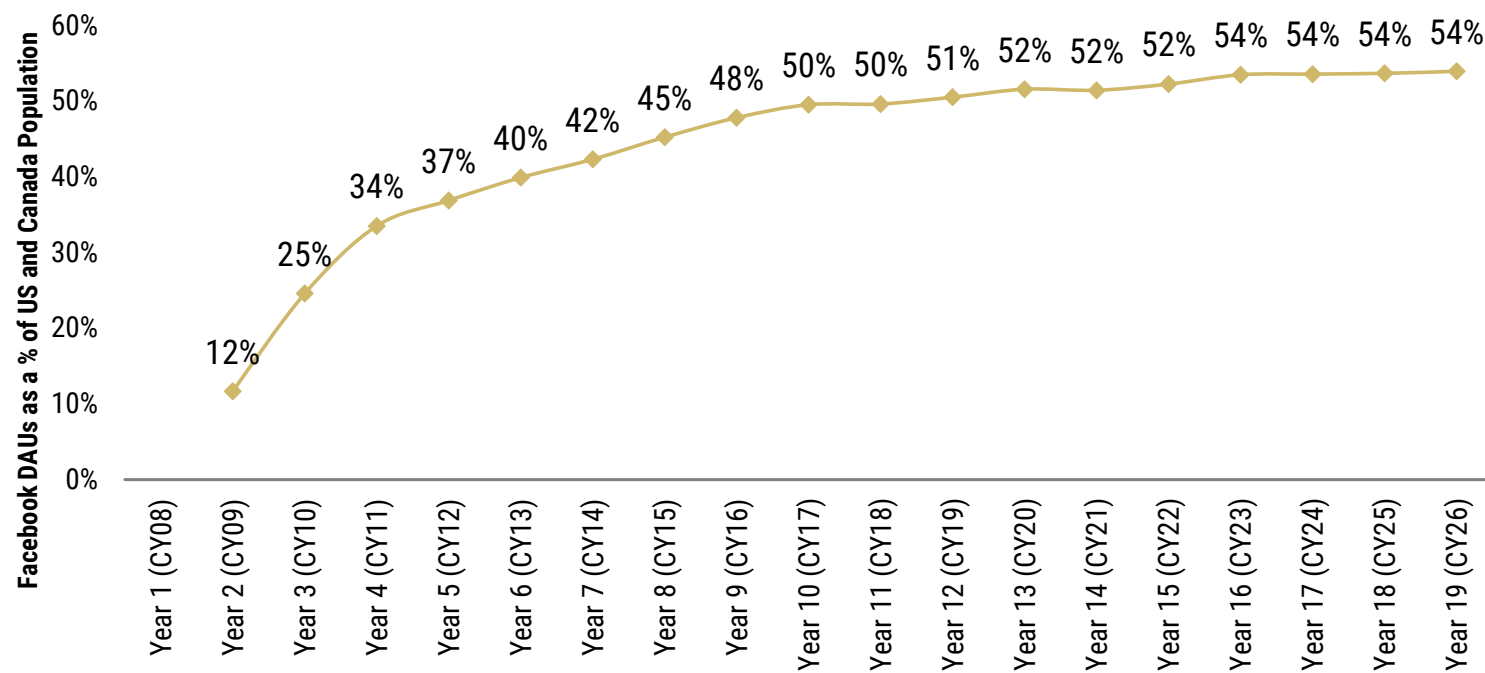
Source: Company data, Morgan Stanley Research estimates, Alphawise

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... And Expect Engaging High-Utility AI Products to Drive More Adoption Across the Economy... Consider Facebook Adoption

- Early in the mobile era, it was products like Facebook from META that drove adoption of internet/mobile technologies, as **penetration of the US/Canada population reached ~45% just 8 years after the release of the product... and ~55% penetration today.**

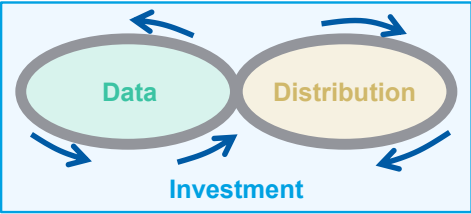


Source: Company data, Morgan Stanley Research estimates, AlphaWise, United Nations

And in this Case, We See Existing Leading Digital Platforms with Distribution, Data and Investment Leading this Adoption

Our View

- Value of leading first-party data sets/distribution has never been higher, favoring **large platforms, which should be investing aggressively... and early adopters.**
- We also think the application layer (and domain-specific uses) is likely to drive most value creation.

Data, Distribution, Investment		Mega Cap Examples		
		  3bn+ users on 5 apps, +2.5bn MAUs on AI Overviews, +1bn MAUs on AI Mode; 5tn annual search queries, Gmail, shopper & inventory data (50bn listings & 2bn updates to shopping graph per hour)	  3.5bn+ daily active people across Family of Apps; Interest and engagement data. More than 10mn advertisers globally	  ~200mn+ Prime Members, ~30bn units sold annually, ~200mn+ Prime Video viewers, millions of AWS customers; ~2mn+ sellers
3 Differentiators				
Data	Scaled and unique first party data sets			
Distribution	Channels to release products, drive adoption and generate new data	Reach audience across full product offering (Gmail, Search, Cloud, etc.)	Access to users across social and messaging channels (Instagram, WhatsApp, etc.)	Access to global shopper base and millions of AWS customers
Investment	Willingness and ability to invest in AI opportunities	\$205bn/\$375bn in '26E/'27E Infrastructure Capex	\$145bn/\$225bn in '26E/'27E Infrastructure Capex	\$140bn/\$170bn in '26E/'27E Infrastructure Capex

Source: Company data, Morgan Stanley Research

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Adoption: Where Are We Seeing Early GenAI Adoption Signal?

Our View: ~25% of S&P 500 companies are now quantifying GenAI benefits as of 2Q:26 (up from 14% 1 year ago).

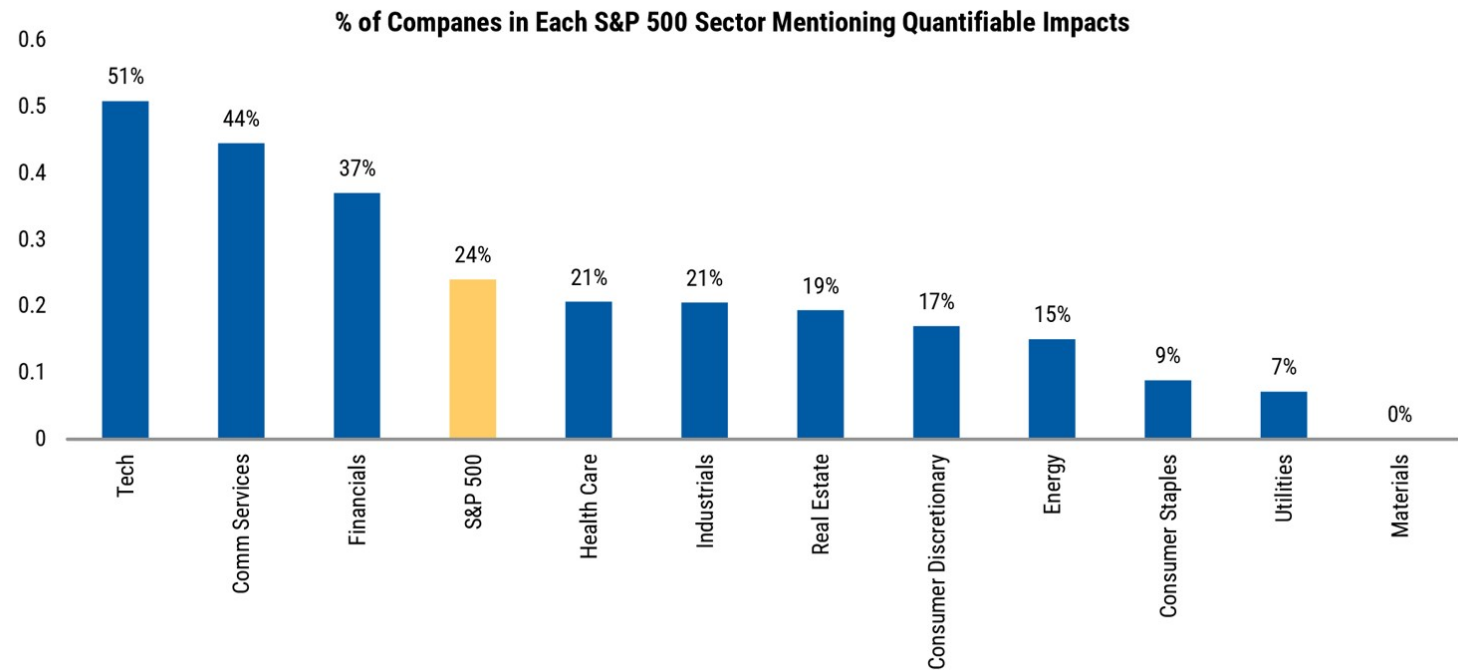
Stock Implications: **Model companies and hyperscalers** benefit from driving adoption but expect early enablers across the economy to see outsized financial benefits and potential multiple expansion. For now, **Technology, Comm Services, Financials and Healthcare** seem furthest along.

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AI Diffusion Has Already Begun Across S&P 500 Companies... with Tech, Financials Healthcare and Industrials Earliest Adopters

- ~25% of S&P 500 companies are now quantifying GenAI benefits as of 2Q:26 (up from 14% 1 year ago).
- Technology, Comm Services, Financials and Healthcare lead sector-level adoption rates across the S&P 500.
- On a YoY basis, Tech saw the largest increase in mentions (51% today vs 28% a year ago), followed by Financials (37% today vs. 8% a year ago), and Industrials (21% today vs 19% a year ago).



Source: Company data, Morgan Stanley Research estimates. Note Data on company commentary is as of July 8, 2026

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Within Tech, We See a Broad-based Signal of Top-line and Efficiency Gains...

Company	Commentary	Benefit Category
AMZN	"Advertisers using Ads agent targeting see 8% lower cost per impression and 6% cost per acquisition" "Customers who use Alexa for Shopping spend an average of 40% more per order than those who don't, and customers who've tried Alexa are signing up for Prime at nearly 25% higher rates."	Top-Line and Efficiency
EBAY	"For buyers, our agentic search pilot reinforced the value of natural language understanding and helping customers uncover more of the products they'll love within eBay's 2.6 billion listings."	Top-Line
SHOP	"Both AI-driven traffic and also orders to Shopify stores tripled year-over-year in the second quarter. New buyer orders are coming in at nearly twice the rate of other channels." "We saw that AI search was starting to disproportionately benefit the long-tail in 2025 and that trend has continued with 75% of AI attributed orders in the second quarter coming from outside our top 100 categories in Q2. " "Buyer shopping journeys are being compressed as half of all AI referred sessions are landing directly on a product description page, that is 2.5 times more than what we see with traditional search. " "Agentic transactions carry the exact same economics as an online store transaction. There's no new fees, there's no separate pricing, but more Agentic GMV, it means more Shopify revenue, and that's the model, and it's been working really well for almost two decades."	Top-Line
RVLV	"[AI is bringing] an enhancement to our on-site search algorithms that will soon allow consumers to upload photo images to discover similar items from our assortment. For example, a customer could upload a photo of a celebrity wearing a cute dress and our AI-driven innovation will show the exact item if available, along with similar items on REVOLVE."	Top-Line

Company	Commentary	Benefit Category
UBER	"We are seeing at the moment near 100% adoption with our engineers on AI-based coding tools... we are seeing doubling in the code output for engineers... At a minimum, it has allowed us to moderate the pace with which we are adding headcount there." "Three-quarter of our rides on Uber happen via a personalized destination suggestion...that's another small example of what AI can do." "The size of [AI built carts] is often twice the size of non-AI built carts."	Top-Line and Efficiency
CART	"Orders placed with our AI assistant are, on average, larger than our typical basket. That's especially notable given our industry-leading average order value of \$115."	Top-Line

Source: Company data, Morgan Stanley Research estimates

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... And Online Advertising Leaders Continue to Show Improvements...

Company	Commentary	Benefit Category
META	"We deployed the first generative model into our ads retrieval system and saw notable improvements in ad performance. Early pilots using LLMs to better understand user preferences drove a 1% increase in app event conversions on Instagram." "In Q2, we also advanced our user understanding models to analyze ads and organic activity and simultaneously improve both user experience and advertiser performance. Combined with our GEM model for ads ranking and sequence learning, these advancements generated an 8.3% increase in ad clicks and a 15.7% uplift in conversions on Facebook." "Adoption of our Gen AI ad creative tools continues to scale, with over 9 million small businesses using at least one AI creative tool. Image generation, which now lets advertisers produce more creatives at scale from existing content, including a new ability to create images from video assets, saw adoption more than double this quarter."	Top-Line and Efficiency
GOOGL	"Our agentic development platform Antigravity... has more than 2.4 million weekly active users. Antigravity is a powerful tool for users and enterprises, and it's completely accelerated how we build internally. As just one example, a team in Chrome is now on track to accelerate delivery by eight times, compressing a two-year timeline into three months through model-driven refactoring." "Our Ads Customer Support teams use Gemini-powered agentic solutions that now autonomously address 75% of support queries, freeing them to solve our customers' most complex challenges." "83% of our Sales team uses Gemini-assisted tools weekly, driving up a 20% higher win rate when using customized pitch narratives." "Those who adopt our AI-powered campaigns like AI Max or PMax see an average of 15% more conversions or value on Search at a similar ROAS." "Since expanding AI Mode globally last October, we've surpassed one billion monthly active users."	Top-Line and Efficiency
PINS	"With open models, we are achieving cost per transaction at less than 8% of the cost of comparable closed proprietary models." "In the second half, we're expanding this model-routing infrastructure to even more internal use cases... In July, weekly pull requests per engineer increased 45% versus last year, while incident rates and service uptime remained relatively consistent."	Efficiency
SNAP	"Our internal AI code reviewer now reviews 90% of pull requests across Snap and has saved an estimated 30,000 hours of code-review time. " "Our AI-powered support agent answers approximately 3.9 million questions from Snapchatters each month and has reduced support ticket volume by approximately 62% since the start of the year." "In advertising, AI increased first-pass image-review automation from 40% in Q2 2025 to nearly 90% in Q2 2026, resulting in faster approvals for advertisers, stronger content safety, and lower operating costs."	Efficiency
RDDT	"Reddit Max launched a private beta earlier this year, and we're seeing strong adoption and performance for mid- and lower-funnel advertisers. The number of advertisers using Max grew over 60% from Q1, while Max revenue grew over 150% during the same period."	Top-Line

Source: Company data, Morgan Stanley Research estimates

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... As Do Ad Tech and Real Estate Players...

Company	Commentary	Benefit Category
APP	"Using AI Technologies to accelerate the rate of those tests too... And the rate of testing has for sure gone up in our company over the last couple of years as LLMs have gotten a lot more sophisticated writing code. "	Efficiency
U	In the second quarter alone, the Vector team successfully executed over 20 major updates, advancing the capabilities and intelligence of our AI prediction platform, bringing new runtime data online and transforming how we identify, understand and value gamer attention. This incredible rate of change enabled Unity to optimize real-time bidding precision for our advertisers to a degree that was formerly impossible." "As I've noted before on the call and I think we've discussed before, AI is going to make the creation of interactive entertainment much easier over time. And so, you're going to see a massive expansion of TAM for the interactive creation marketplace. " "Unity AI product went into open beta in May... It's an integrated agent that's tuned specifically for Unity game development. It's still very early, just still in beta, but the results have been really encouraging." "AI that is driving this not only drives more content creation, in that, we think it will – it's not only driving more innovation, but it's driving easier and more use of our platform. As that happens, we will earn money downstream from services, from live operating services, from Vector, from other things. That is the flywheel we are trying to explain and why we're so excited about this."	Top-Line and Efficiency
CRTD	"Traffic from ChatGPT converts at approximately 1.5 times to 2 times the rate of traditional referral traffic, while roughly 80% of paid traffic is new to the brand." "Agape Diamonds used Criteo Go to expand beyond traditional retargeting, combining AI-powered customer acquisition with performance campaigns across the open web and social channels. This approach increased return on ad spend and conversion rate by 20%, while increasing average cart value by 13%."	Top-Line and Efficiency
MNTN	"We've had over 37,000 QuickFrame AI signups in Q2, bringing us to over 73,000 signups year-to-date."	Top-Line
Company	Commentary	Benefit Category
Z	"We are actively expanding what AI mode can do, adding skills and evaluations that serve buyers, sellers, renters, and homeowners. The opportunity in front of us is to deepen our engagement with our already broad audience across every stage in ways that weren't possible before." "We're really pleased with the ZHL continued improvement is coming from AI initiatives we put in place there to manufacture loans more efficiently. " "AI mode is in early stages, but these are the signals we want to continue to see as we iterate. Consumers who use AI mode spend more than three times as long on Zillow, view more than twice as many homes, run nearly three times as many searches, and contact an agent at nearly three times the rate of consumers who don't use AI mode."	Top-Line and Efficiency
COMP	"In July, we demoed our AI Assistant more broadly, which is an integrated assistant that helps agents orchestrate more than 90 of our platform tools simply through natural language prompts." "We see opportunity to continue lowering the cost to serve per transaction through greater utilization of AI and deploying other best practices." "We are using AI to reduce OpEx. As you would expect, in Q2, we began deploying FDEs or forward deployed engineers within business functions including transaction management, legal and growth to build automated AI workflows directly into day-to-day operations. To date, the team has identified a mix of roughly \$8 million in savings and cost avoidance opportunities. Additionally, across the technology organization, 50% to 60% of all new code is now produced by AI."	Efficiency

Source: Company data, Morgan Stanley Research estimates

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And Financials, Industrials and Healthcare Quantifying Benefits as Well

Sector	Company	Commentary	Benefit Category
Financials	BLK	"Our systematic equity investors are leveraging decades of proprietary data, over 1,000 alpha signals, and our AI-driven research engine to deliver returns." "AUM has doubled in just the last two years from \$200 billion to now \$400 billion."	Top-Line
	JPM	"We've already reduced headcount by 30% to 40% in certain specific areas." "You don't uniquely benefit from AI."	Efficiency
Industrials	HON	"So to me, AI is an integral part of our offering set. Honeywell believes that it's how the automation industry will move towards autonomy." "We are going to inch from about \$900 million to a little over \$1 billion of ARR in that segment this year."	Top-Line
	WM	"One example is our WM Smart Truck platform, which now generates more than \$300 million of annual run rate operating EBITDA through service upgrades, optimized routing, and lower operating costs." "Importantly, we are still in the early innings of capturing the full value of these capabilities."	Efficiency
Healthcare	IQV	"Our own agentic roadmap has continued to make great progress. In fact, we now have 294 agents deployed across 90 use cases." "Four of the top 10 pharma companies have already contracted with us to co-develop AI solutions, and 19 of the top 20 pharma companies have already deployed IQVIA AI solutions in their workflow."	Top-Line
	UNH	"AI is a very core initiative for us. We are really thinking of it in terms of reimagining our entire enterprise, virtually everything that we do, we see it basically as the operating infrastructure of the future."	Efficiency

Source: Company data, Morgan Stanley Research estimates

Morgan Stanley

What Are the Likely Impacts of Open-Weights Models?

Our View: Leading cost to serve is critical through the age of inference. Open-weights models do not change underlying demand for compute. In fact, we think lower-cost open-weights and open-source models are critical to adoption across the economy. Expect these models to also showcase the value of the distribution layer (hyperscalers).

Stock Implications: We see the proliferation of open-weights models furthering Jevon's Paradox, lowering average token prices and thus expanding demands for tokens and enabling inference to scale across the economy.

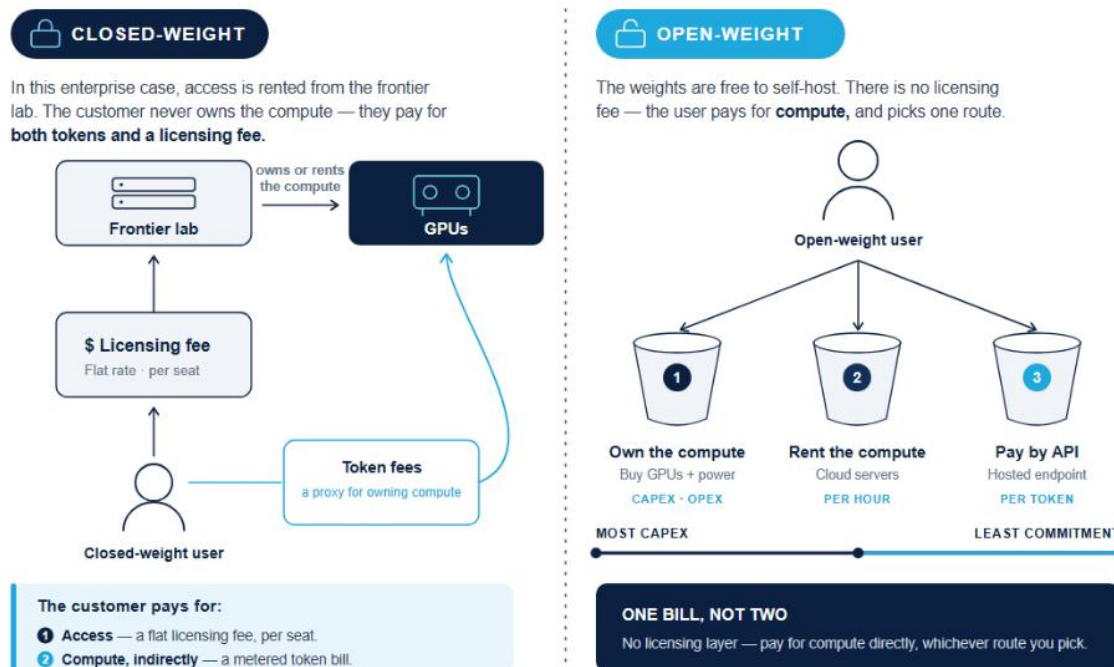
While it puts pressure on the models to maintain differentiation and competitive pricing, **it re-enforces the value of hyperscale “concierge” services like AWS’s Bedrock, MSFT’s Foundry and GOOGL’s Vertex API.**

Morgan Stanley

MORGAN STANLEY RESEARCH

What Are Open-Weights Models?

- **Open-Weights models** give users more control over model weighting and are: **1)** free to download by any end user; **2)** able to be fine-tuned/re-trained on a user's proprietary data; and **3)** deployable wherever the user chooses (on-prem or in the cloud).
- As compared with closed-weight models (like those offered by frontier AI labs), open-weights models add a greater breadth of distribution routes, tend to be lower cost, and enable more customization... **all of which are important to driving adoption and tech diffusion across the economy.**



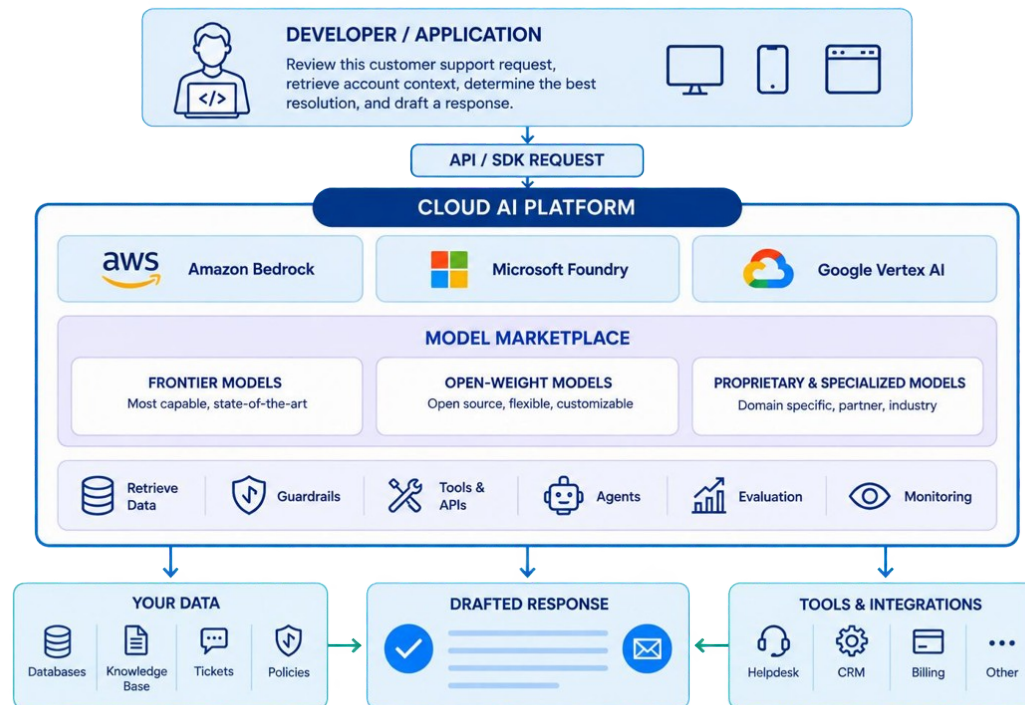
Source: Company data, Morgan Stanley Research

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Hyperscalers Monetize Open-Weights as They Own Capacity and Provide a “Concierge Layer,” Choosing the Workload-Appropriate Model and Maximizing Adopter ROIC

- Hyperscalers aggregate closed and open-weights models into platforms like **AWS Bedrock**, **MSFT Foundry**, and **GOOGL Vertex AI**, charging for API/inference consumption while abstracting away model hosting and infrastructure complexity.
- Open-weights models may have little or no direct licensing cost, but they still drive spend on **GPU compute, fine-tuning, data/embedding pipelines, among other value-added services** – allowing hyperscalers to monetize whichever model a customer chooses.



Source: Company data, Morgan Stanley Research

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How Do Open-Weights Impact Hyperscaler ROIC?

- While Open-weights models put pricing pressure on the model layer with lower price-per-token, **our API ROIC analysis already assumes token pricing well below the largest frontier models** (and even below some open-weight models).
- Even with deeper proliferation of open-weights models, the value of hyperscaler compute remains high, and **our analysis still shows ~20-40% ROIC** on 1GW of GB300 capacity.
- Also re-enforces importance of token throughput improvements across the system.

ROIC of Hyperscaler GPU Rentals (IaaS) on 1GW GB300 DC (\$bn)						Notes:
Revenue:						
Annual Hours	8,760	8,760	8,760	8,760	8,760	24 hrs. x 365 days
(X) GB300 Chips in 1GW	410,256	410,256	410,256	410,256	410,256	MS est of GB300 chips in 1GW datacenter
(=) GPU Hours (billions)	3.6	3.6	3.6	3.6	3.6	
(X) Utilization (%)	75%	75%	75%	75%	75%	MS est. of utilization
(X) Est. Price/Hour	\$7.0	\$8.0	\$8.5	\$9.0	\$10.0	MS est: GPU rental px/hr (range supported by industry/company data)
(=) Revenue/GW	\$18.9	\$21.6	\$22.9	\$24.3	\$27.0	
Costs:						
IT Depreciation Cost	\$5	\$5	\$5	\$5	\$5	IT capex divided by IT useful life
(+) Non-IT Depreciation Cost	\$1	\$1	\$1	\$1	\$1	Non-IT capex divided by Non-IT useful life
(+) Energy + Other Opex	\$2	\$2	\$2	\$2	\$2	Energy, labor, maintenance, etc, industry/company data
(=) Total Opex/GW	\$7.6	\$7.6	\$7.6	\$7.6	\$7.6	
EBIT & NOPAT:						
Incremental EBIT/GW	\$11.2	\$13.9	\$15.3	\$16.6	\$19.3	
Incremental EBIT Margins	60%	65%	67%	69%	72%	
(X) Tax Rate	21%	21%	21%	21%	21%	
(=) NOPAT	\$8.9	\$11.0	\$12.1	\$13.1	\$15.3	
% ROIC (NOPAT / Total Capex)	23%	28%	31%	34%	39%	

Source: Company data, Morgan Stanley Research estimates

Morgan Stanley

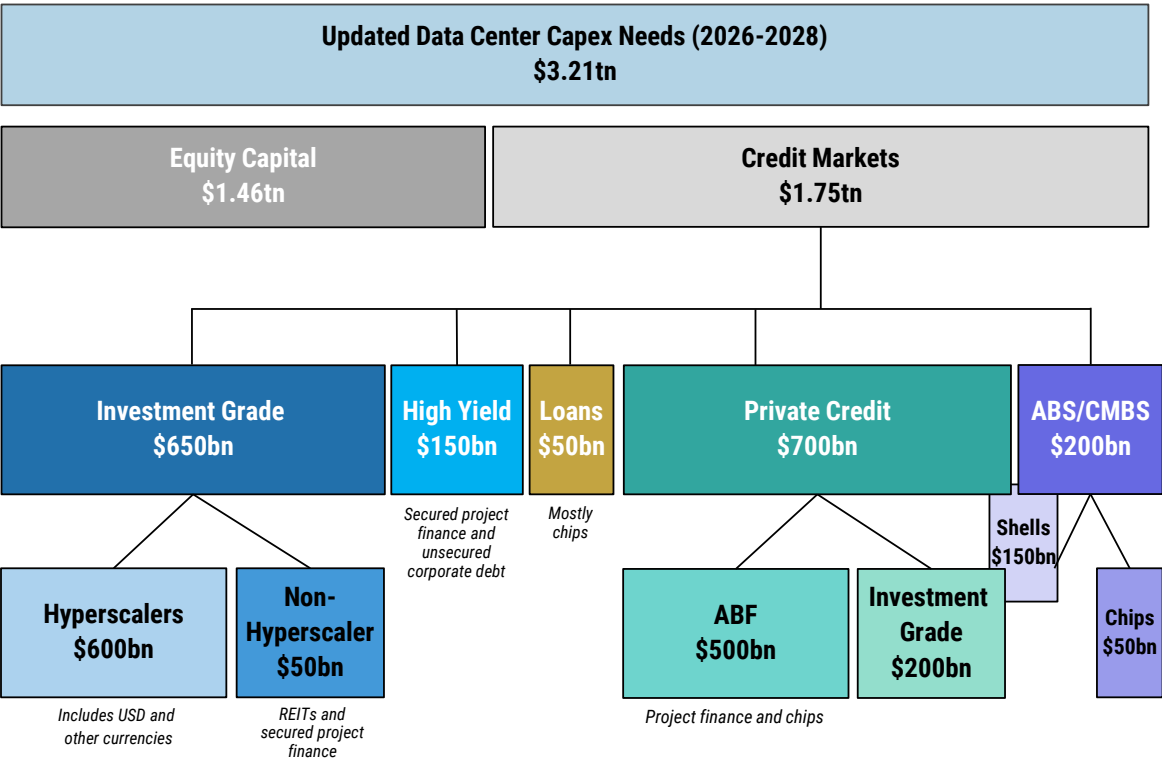
Are We Concerned about Financing or Credit?

Our View: No. Despite ~\$450bn of AI-related debt issuance globally year-to-date, we think there's still significant capacity to absorb additional issuance, especially from high-quality hyperscalers. Rather than constraining issuance, the market adjustment is likely to come through wider spreads.

In addition, we see the 4 main builders (AMZN/GOOGL/MSFT/META) generating accelerating OCF, set to reach \$980bn/\$1.2tn in '27/'28... compared to only \$290bn of debt needed to be raised over that period.

Stock Implications: Further appetite in credit markets gives hyperscalers more leeway to finance data center buildouts in the near to medium term.

Financing AI Infrastructure Capex: Growing Role of Credit Markets



Source: Morgan Stanley Research.

Morgan Stanley

MORGAN STANLEY RESEARCH

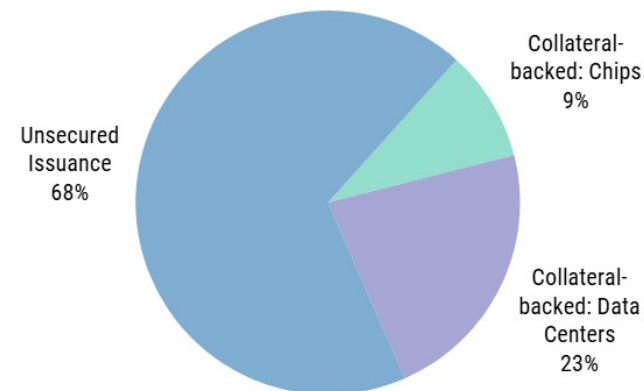
AI-Related Debt Issuance YTD ~450bn, Already More Than Double FY2025 Issuance

- With AI-related debt at ~7% of the IG index, we think there's still significant capacity to absorb more issuance. Hyperscaler credit quality remains strong, suggesting capacity for additional debt issuance with market adjustment likely to come through wider spreads.
- Credit market channels have been broader and deeper than we anticipated, spanning public and private markets blurring the lines between the two along with remarkable structural innovation.

AI-related debt issuance

Issuance (USD bn)	YTD-26	FY25A
HQ Hyperscalers	132	67
Oracle	25	26
Hyperscaler Total	157	93
Other AI Total	114	64
Public IG Total	271	157
4a2 Priv. Plac.	35	0
Total HY	40	15
Secd Lev. Loans	10	2
Total US Corporate Credit	356	174
Total US Securitized Credit	18	27
Corporate Credit non-USD	69	15
Securitized Credit non-USD	2	1
Total non-USD	70	16
Total Global Credit Markets	445	217

Majority of AI-related debt is unsecured corporate issuance



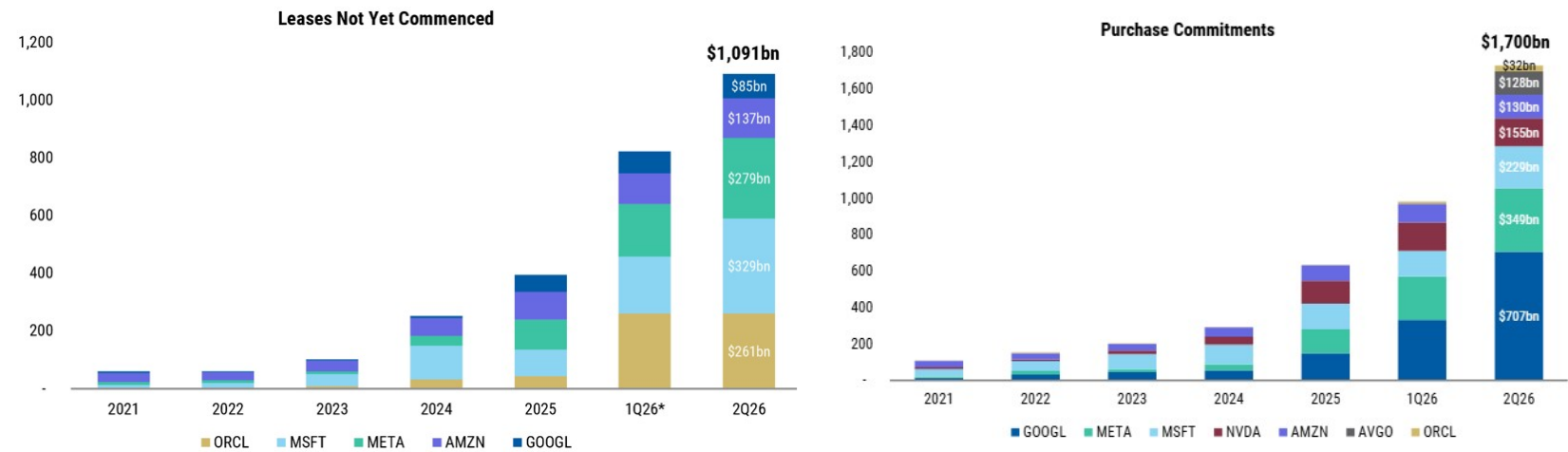
Source: Bloomberg, Pitchbook | LCD, CreditFlow, TREPP, Morgan Stanley Research estimates. Note: YTD 2026 data includes deals priced until 8/19/26. 4a2 Priv. Plac. includes only the AVGO chip financing, deals like Sopaipilla and Beignet are included in "Other AI Total". Figures are rounded to the nearest \$bn; totals are calculated using unrounded figures and may differ slightly from the sum of displayed values.

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Growing Hyperscaler Off-Balance-Sheet Leverage and Purchase Commitments

- Off-balance-sheet leverage is rising as hyperscalers enter into long-dated leases for data center shells and commit to larger and longer-term purchases of compute, chips, equipment, and power.
- Off-balance-sheet commitments will impact leverage, and disclosures are limited, but they're included by rating agencies in their adjusted debt calculations. To the extent those increase, there is less issuance capacity at current rating levels.



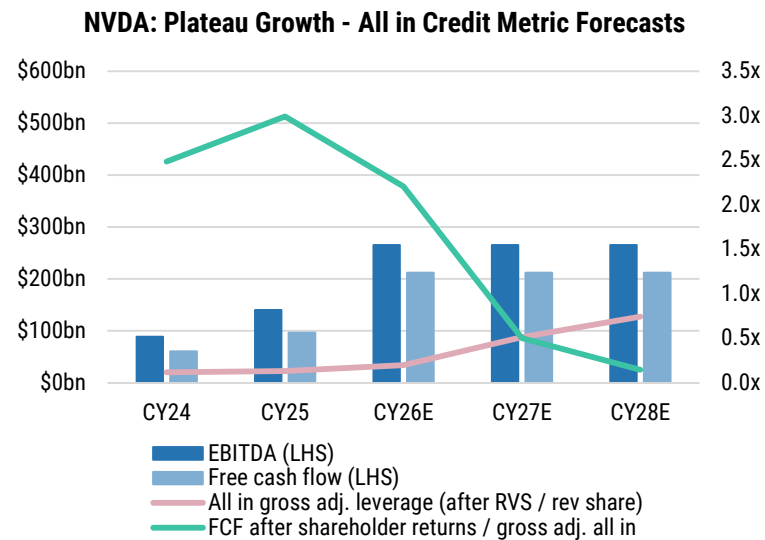
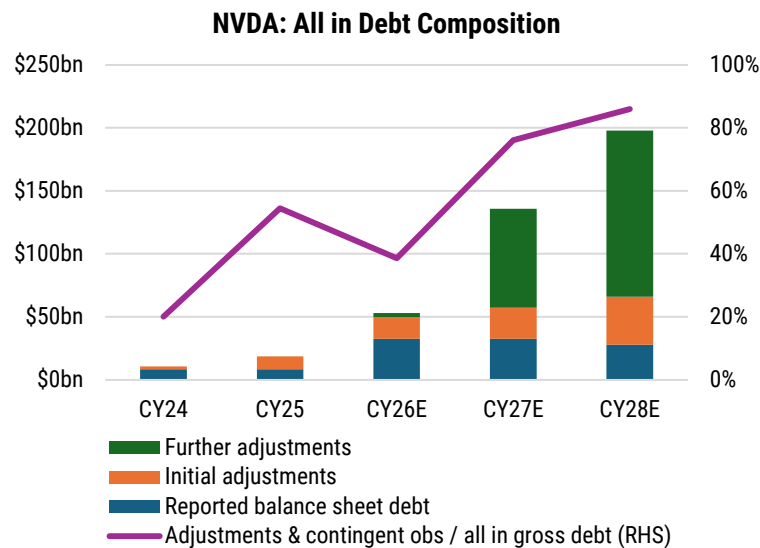
Source: Company data, Morgan Stanley Research estimates

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NVDA's 'Balance Sheet-as-a-Service' – Recent Credit Initiation with Neutral View

- Exceptional growth is turning balance-sheet strength into a strategic AI financing tool – and introduces new risks.
- We forecast CY28-end all-in credit exposure scales to ~\$200bn, with ~\$170bn of that away from funded debt (see left).
- But credit metrics look strong as that contingent exposure scales, even if headline growth moderates (shown on right).
- Our model and relative value are supportive, but we think the tail remains too early-stage, opaque, and sizable to step in.



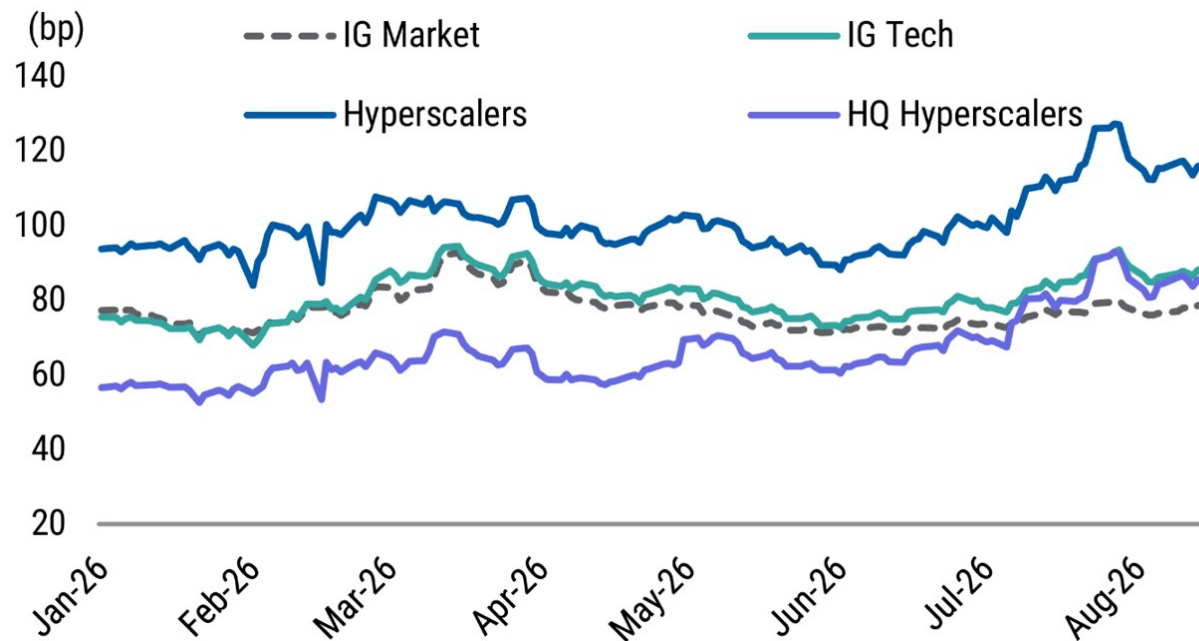
Source: Company filings, Morgan Stanley Equity Research (Moore model dated 5/21/26), Moody's, S&P, Bloomberg, Morgan Stanley Credit Research

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MORGAN STANLEY RESEARCH

Wider Spreads Are the Release Valve to Accommodate Supply

- High-Quality Hyperscaler Spreads Are Now 7bp Wider than the IG Index.
- We also don't think HQ hyperscalers are sensitive to the move we saw in spreads given their expected returns; however, lower-quality names could be more impacted.



Source: Company data, Morgan Stanley Research estimates

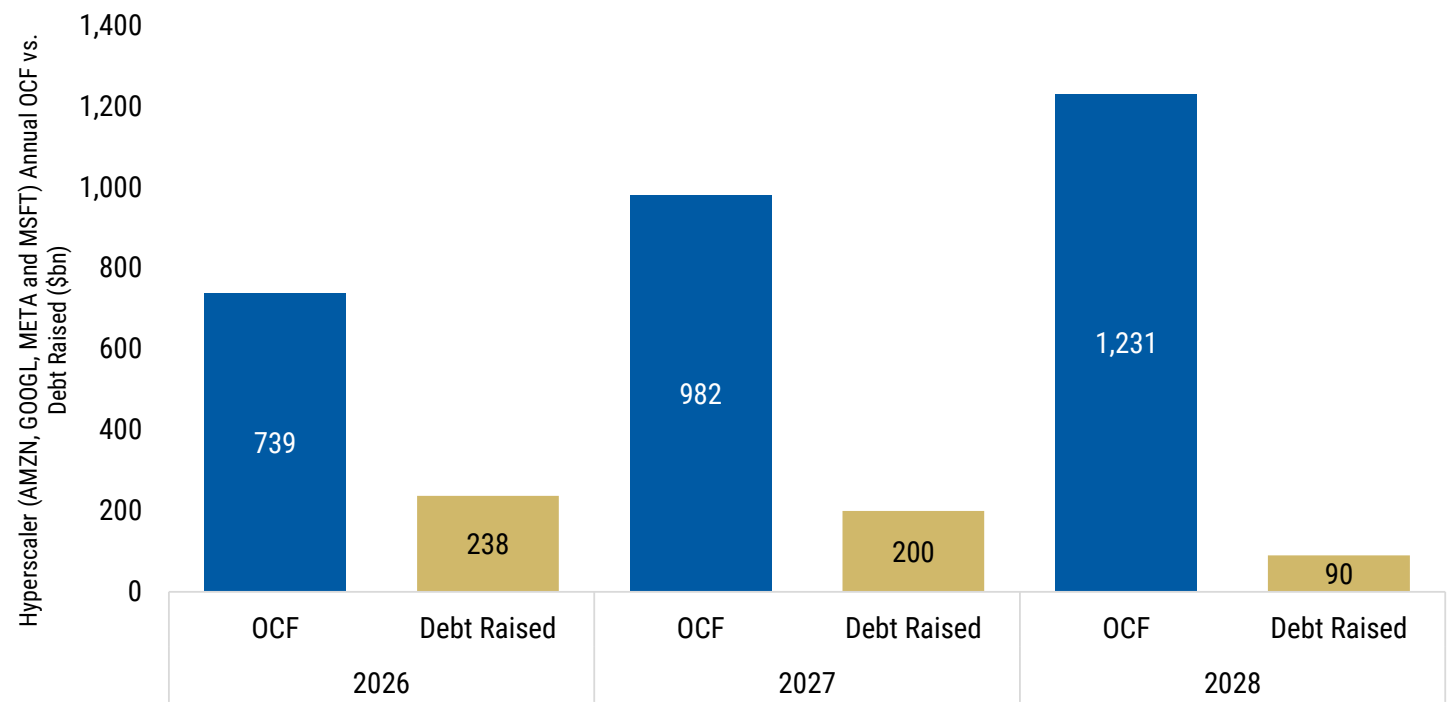
57

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Hyperscaler Operating Cash Flow Still Heavily Outweighs Modeled New Debt Raised...

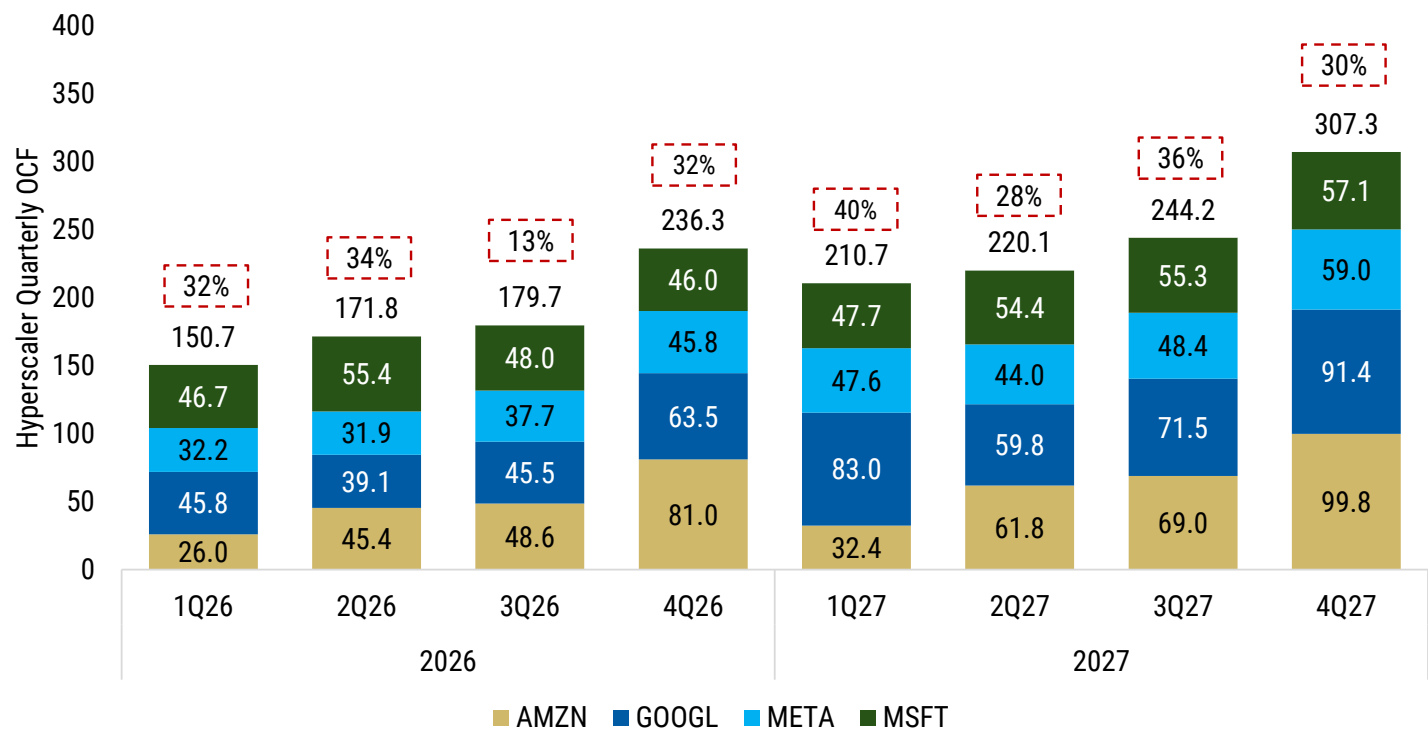
- We see hyperscaler operating cash flow significantly outpacing the growth of new debt, as AMZN, GOOGL, META, and MSFT begin to monetize newly built capacity and generate over \$1tn of OCF in 2028.



Source: Company data, Morgan Stanley Research estimates

... As YoY Hyperscaler OCF Growth Accelerates

- We see annual OCF accelerating to 33% amongst the hyperscalers, as the quarterly pace of growth ramps substantially in 2027.



Source: Company data, Morgan Stanley Research estimates

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Regulatory Concerns and Power Bottlenecks

Our View: As state government scrutiny of data centers intensifies, bottlenecks around time to power, interconnection, and labor will also become more important. We see higher costs and behind-the-meter power solutions becoming more common as the US faces a 38 GW power shortfall.

Stock Implications: Data centers are likely to increasingly adopt onsite generation, benefiting **SEI, INIO, BE** as well as powered shell providers including **CIFR, WULF, HUT, and RIOT**.

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State and Local Pushback Against Data Center Construction Has Grown Louder...

- Opposition largely falls into 3 main categories: (1) concern around potential impacts on power bills; (2) environmental concerns including water use and waste heat; and (3) local quality-of-life concerns around large construction projects including noise, dust, and traffic.
- Data center moratoriums have precedent and are expanding at the state and local level.
- If the local policy efforts are either 1) made permanent and continue to expand, and/or 2) reflected at the federal level, then we think the US risks keeping pace with China on the global AI competitive landscape.
- From a hyperscaler standpoint, data centers are likely to increasingly adopt onsite (behind-the-meter) generation.

State	Bill(s)	Status	Description	Governor	Senate	House	Alignment
Delaware	SB 353	Introduced	Moratorium through Jan. 31, 2027 on applications and permits for data centers able to use 100 MW or more.	Matt Meyer (D)	D	D	Aligned
Georgia	HB 1059	Introduced	Forbids local governments from permitting data centers until Dec. 31, 2028.	Brian Kemp (R)	R	R	Aligned
Maine	LD 307 / HP 207	Vetoed	Creates Maine Data Center Coordination Council and temporarily limits certain data centers over 20 MW until Nov. 2027.	Janet Mills (D)	D	D	Aligned
Maryland	HB 120	Failed	Prohibits construction or permitting of new data centers under stated co-location/generation conditions.	Wes Moore (D)	D	D	Aligned
Michigan	HB 5594 / HB 5595	Introduced	Local governments and state agencies may not permit/approve certain new data centers until Apr. 1, 2027.	Gretchen Whitmer (D)	D	R	Divided
Minnesota	HF 4888 / SF 4298	Failed	Bans new data-center permits until a Public Utilities Commission report, no earlier than July 2027.	Tim Walz (D)	D	NP	Divided
New Hampshire	HB 1265	Failed	Establishes a one-year moratorium on data-center construction.	Kelly Ayotte (R)	R	R	Aligned
New York	A11560 / S10642	Passed Legislature	Places a one-year moratorium on large data-center permits and adds service-classification, energy-efficiency and host-community provisions.	Kathy Hochul (D)	D	D	Aligned
Oklahoma	SB 1488	Failed	Establishes a moratorium on data-center construction until Nov. 2029 and requires a study.	Kevin Stitt (R)	R	R	Aligned
Pennsylvania	SB 1359 / HB 2533	Introduced	Creates a statewide three-year moratorium on hyperscale data centers and impact studies, or authorizes municipal moratoriums.	Josh Shapiro (D)	R	D	Divided
South Carolina	H 5526	Introduced	Blocks state/local acceptance or action on data-center applications until a comprehensive oversight framework is enacted.	Henry McMaster (R)	R	R	Aligned
South Dakota	SB 232	Failed	Imposes a one-year moratorium on construction or expansion of hyperscale data centers.	Larry Rhoden (R)	R	R	Aligned
Vermont	S.205	Introduced	Creates a moratorium on AI data centers until 2030 and requires a report on impacts.	Phil Scott (R)	D	D	Divided
Virginia	HB 1515	Continued	Prohibits final local approval for a new data center until interconnection requests are fulfilled or July 1, 2028.	Abigail Spanberger (D)	D	D	Aligned
Wisconsin	SB 1061 / AB 1099	Failed	Prohibits data-center operation until 14 listed conditions are satisfied.	Tony Evers (D)	R	R	Divided

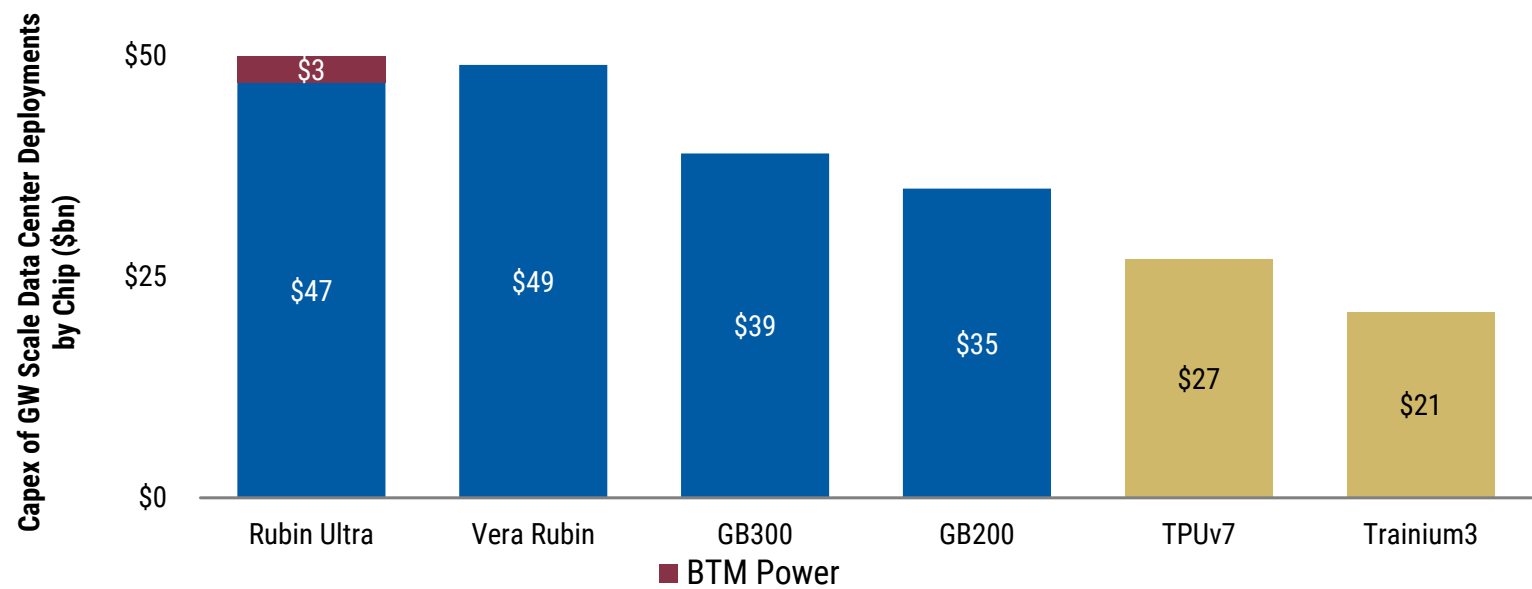
Source: Company data, Morgan Stanley Research estimates

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... Which Should Increase Prevalence of Behind-the-Meter Power Solutions, Consistent with our Bottom-Up Methodology

- As pushback increases, so too will opex requirements for each individual data center as power costs rise.
- Informed by industry datapoints, **we believe behind-the-meter (BTM) power adds an incremental ~\$3bn to our capex/GW estimates.**
- Our bottom-up capex work already accounts for this additional expense starting with the Rubin Ultra generation NVDA chips, which we believe will begin coming online in 2H27 and 2028 at an all-in cost of \$50bn/GW.



Source: Company data, Morgan Stanley Research estimates

Risk Reward – Bloom Energy Corp. (BE.N)

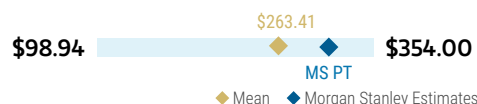
Time-to-Power Constraints Drives Adoption; AI Enabler

PRICE TARGET \$310.00

Derived from an unlevered DCF analysis with 10.8% discount rate, based on a 4.0% risk-free rate, 1.75 Beta and 4.29% equity risk premium.

Consensus Price Target Distribution

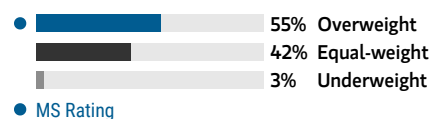
Source: Refinitiv, Morgan Stanley Research



OVERWEIGHT THESIS

We see BE as a key beneficiary of several key themes including: (i) time-to-power for large load customers, (ii) the growing "economic wedge" or value proposition of distributed energy (i.e., fuel cells for C&I customers), (iii) rising grid instability, and (iv) 30% fuel cell tax credit through the 2030s. In the near-term we see a strong growth outlook for its core fuel cell business, particularly for AI power demand, with longer-term tailwinds arising from the OBBB and its hydrogen electrolyzer business, supporting a robust multi-year growth and margin improvement outlook.

Consensus Rating Distribution



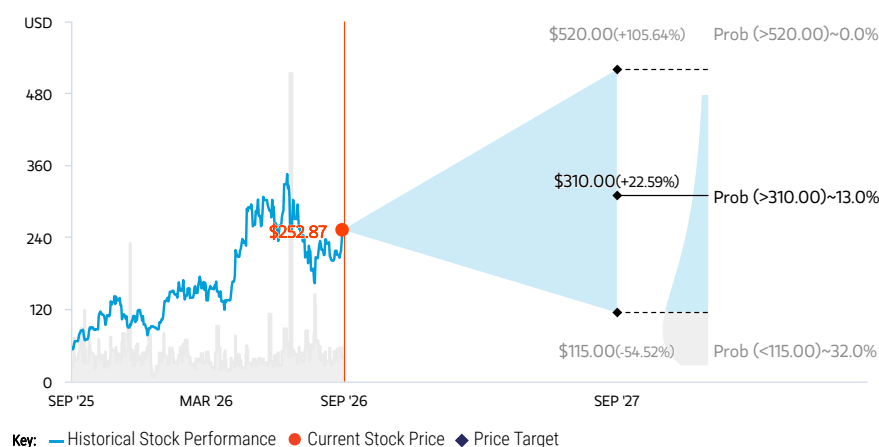
Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Disruption: Positive
Secular Growth: Positive
Renewable Energy: Positive

View descriptions of Risk Rewards Themes [here](#)

RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 4 Sep 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

BULL CASE \$520.00

Broader adoption of SOFCs for BTM power generation

59% revenue CAGR through 2030 and 11% annual revenue growth from 2030-2050 given low market penetration rates and a growing time-to-power constraint across global power markets. Gross margins expand to 45% by 2030, driven by consistent cost-cutting and resilient pricing. Operating margin reaches 38% by 2030 (same as base case).

BASE CASE \$310.00

Time-to-Power Constraint Accelerates Adoption

Continued escalation in utility bills and time-to-power constraints drives 53% revenue CAGR from 2025-2030 and 11% annual revenue growth from 2030-2050. Cost reductions and opex leverage drives expanding operating margins from 10% in 2025 to 35% in 2030.

BEAR CASE

\$115.00

Growth and margins below expectations

Competition and an inability to reach its cost-down targets limits revenue growth (40% CAGR through 2030 vs. 53% in the base case) and margin expansion (33% EBIT margins by 2030 vs. 35% in the base case).

Risk Reward – Bloom Energy Corp. (BE.N)

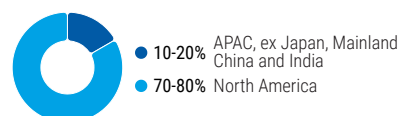
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Volume Growth (%)	49.2	140.2	60.0	68.0
Total Product & Install ASP (\$/kW) (\$)	3,221	3,197	3,085	3,008
Total Installed System Cost-TISC (\$/kW) (\$)	2,043	1,910	1,815	1,724

INVESTMENT DRIVERS

- Increased penetration with existing customers and/or new customer relationships.
- Ability to decrease product and install cost (improving product economics).
- Increased demand for micro-grid solutions that offer 24x7 power.

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	2/5 MOST	3 Month Horizon
-------------	---------------------	-------------	--------------------

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Higher revenue growth through '35, given current low market penetration rates
- Cost reduction initiatives outpace expectations
- Customer adoption increase in light of de-energization in CA.

RISKS TO DOWNSIDE

- Regulators impose charges on C&I customers that use distributed generation.
- Stricter emissions regulation.
- New entrants / technologies emerge with cost-competitive products.
- Cost reductions don't materialize.

OWNERSHIP POSITIONING

Inst. Owners, % Active	56.7%				
HF Sector Long/Short Ratio	1.4x				
HF Sector Net Exposure	8%				

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e

EPS (\$)	2.95	5.04	7.01
		4.89	
Net income (\$, mm)	935	1,706	2,275
		1,581	
EBITDA (\$, mm)	1,170	1,738	2,353
		1,787	

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Innio N.V. (INIO.O)

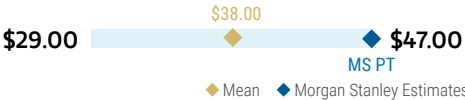
Top Tier Engine OEM with Strong Market Tailwinds

PRICE TARGET \$47.00

In our DCF-based valuation we assume 17% revenue CAGR through 2035, including 19% annual growth in Equipment and 12% annual growth in Service. Adj. EBITDA margins expand from 19% in 2026 to 29.9% by 2035. High FCF conversion (as a % of adj. EBITDA) drives ~\$500bn of FCF in 2026 growing to ~\$1.6bn by 2030. WACC of 9.3%, driven by a 4.25% risk-free rate, 4.60% equity risk premium, 1.25 beta, and 2.5% terminal growth rate.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



OVERWEIGHT THESIS

Innio is well positioned to capture a growing share of onsite and BTM power from data centers as a well established and high quality OEM as overall power demand in the US is set to grow ~3% annually for the next decade. INIO is strategically leaning into this opportunity by expanding its manufacturing footprint ~3x (vs 2025 levels) and therefore the company is set to one of the fastest growing companies in its peer set while also increasing its margin contribution.

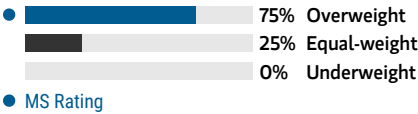
RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: Positive
Pricing Power: Positive
Secular Growth: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

\$76.00

18.7x 2030 EBITDA

We assume a modestly higher 19% revenue CAGR through 2035 (vs. our base case), including 22% annual growth in Equipment and 12% annual growth in Service. Adj. EBITDA margins expand from 19% in 2026 to ~36.4% by 2035. High FCF conversion (as a % of adj. EBITDA) drives ~\$500bn of FCF in 2026 growing to ~\$2.1bn by 2030. WACC of 8.5%, driven by a 4.25% risk-free rate, 4.60% equity risk premium, 1.05 beta, and 2.5% terminal growth rate.

BASE CASE

\$47.00

14.9x 2030 EBITDA

We assume 17% revenue CAGR through 2035, including 19% annual growth in Equipment and 12% annual growth in Service. Adj. EBITDA margins expand from 19% in 2026 to 29.9% by 2035. High FCF conversion (as a % of adj. EBITDA) drives ~\$500bn of FCF in 2026 growing to ~\$1.6bn by 2030. WACC of 9.3%, driven by a 4.25% risk-free rate, 4.60% equity risk premium, 1.25 beta, and 2.5% terminal growth rate.

BEAR CASE

\$19.00

9.2x 2030 EBITDA

We assume a lower 15% revenue CAGR through 2035, including 16% annual growth in Equipment and 12% annual growth in Service. Adj. EBITDA margins peak in 2032 at 24% then decline ~100 bps by 2035. Similar FCF conversion to our base case (as a % of adj. EBITDA) drives ~\$500bn of FCF in 2026 growing to ~\$1.5bn by 2030. WACC of 10.3%, driven by a 4.25% risk-free rate, 4.60% equity risk premium, 1.5 beta, and 2.5% terminal growth rate.

Risk Reward – Innio N.V. (INIO.O)

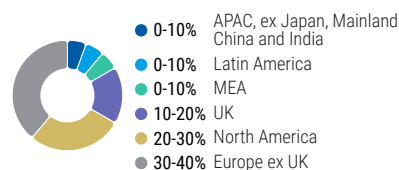
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Power Delivered (MW)	3,381.4	4,610.9	5,855.7	6,534.9
Total Equipment Revenue (\$, mm)	1,365	2,451	3,760	4,400
Service Revenue (\$, mm)	1,271	1,401	1,478	1,579
Total Orders (\$, mm)	3,884	3,998	4,058	4,304

INVESTMENT DRIVERS

- Greater need for onsite power to get data centers online quickly
- Sustained levels of orders driven by growing power demand
- Longer-term higher service opportunity from a larger baseload install base

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

RISKS TO PT/RATING

RISKS TO UPSIDE

- Incremental pricing power for 2029/30 orders
- Greater intensity servicing opportunity near-term
- Expanding capacity beyond 10 GW of nameplate output

RISKS TO DOWNSIDE

- Order cancellation/softer demand from data centers
- Greater than expected capex and opex needed to grow
- Sponsor share overhang

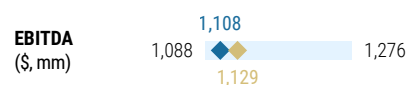
OWNERSHIP POSITIONING

Inst. Owners, % Active	98.7%	
HF Sector Long/Short Ratio	1.4x	
HF Sector Net Exposure	8%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Solaris Energy Infrastructure (SEI.N)

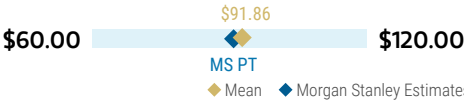
Data Center Power as a Service with Further Expansion Ahead

PRICE TARGET \$90.00

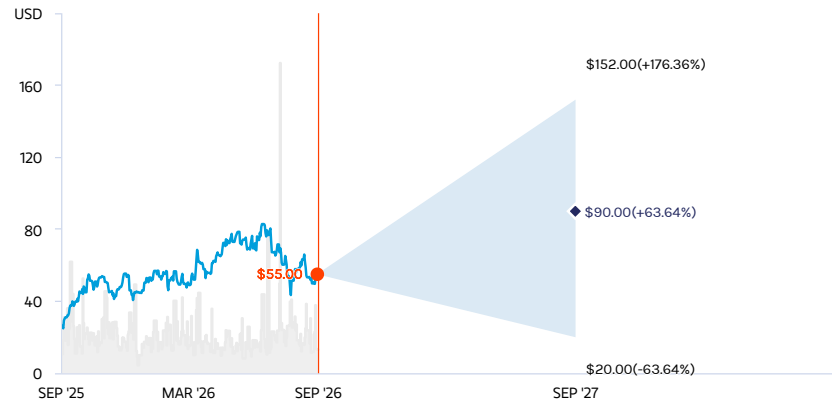
We value SEI's existing assets based on a DCF model through 2050 and no terminal value after that, consistent with the turbines asset life. Our consolidated WACC of 6.8% implies a terminal multiple of 8.6x EV/EBITDA, which is a blend of 10x for Power with long-term contracts, 7.5x for Power with short-term contracts, and 4x for Logistics. We separately add the NPV of 500 MW of incremental capacity procured annually in '29-'35, worth ~\$25/share.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



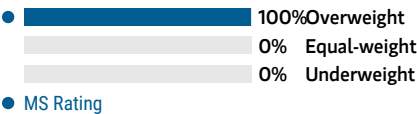
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

Solaris is among the first-movers in providing highly valuable time-to-power solutions to data centers, at competitive costs vs. the traditional grid. With increasing grid constraints, we see an expanding TAM for "de-bottlenecking" solutions that can accelerate the access to power for large loads. We think there is meaningful room for further capacity expansion and more data center contracts going forward.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: Positive
View descriptions of Risk Rewards Themes [here](#)

BULL CASE

\$152.00

Higher capacity growth and enhanced economics

Improvement of micro-grid solutions economics, lower overhaul capex, customers contract SEI's enhanced scope capex, and 1 GW of incremental capacity procured annually in '29-'35.

BASE CASE

\$90.00

Current economics and additional 3.5GW in 7 years

Our base cases assumes \$275/kW overhaul capex every 4 years, \$300-400/kW margin for the most recent hyperscaler contracts and \$330/kW for the JV and short term contracts. Our consolidated WACC of 6.8% implies a terminal multiple of 8.6x EV/EBITDA, which is a blend of 10x for Power with long-term contracts, 7.5x for Power with short-term contracts and 4x for Logistics. We also assume 500 MW of incremental capacity procured annually in '29-'35, worth ~\$25/share.

BEAR CASE

\$20.00

No further growth and economics deterioration

Deterioration of micro-grid solutions economics, smaller portion of the available capacity is contracted, higher overhaul capex, and no incremental capacity procured.

Risk Reward – Solaris Energy Infrastructure (SEI.N)

KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
EBITDA Growth (%)	56.2	66.4	72.4	52.4

INVESTMENT DRIVERS

- Value of time-to-power solutions (e.g. micro-grids)
- Micro-grid deals with data centers
- New capacity procurement

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	2/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- More capacity procured, especially gas turbines
- New LT data center deals
- Increasing scope of solutions offered by SEI
- Worsening of grid bottlenecks, increasing the value of micro-grid solutions

RISKS TO DOWNSIDE

- No new capacity procured
- No new data center deals announced
- Improvement of grid bottlenecks
- Concerns about the terminal value of SEI's gas turbines
- Costumer concentration

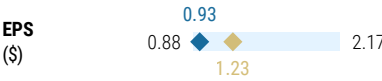
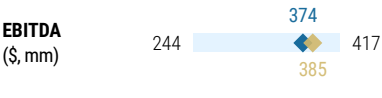
OWNERSHIP POSITIONING

Inst. Owners, % Active	70.5%	
HF Sector Long/Short Ratio	1.7x	
HF Sector Net Exposure	4.2%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure – Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – TeraWulf Inc (WULF.O)

Digital Infrastructure Focus

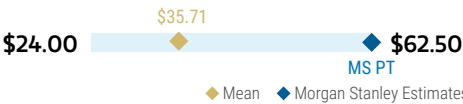
PRICE TARGET \$62.50

TeraWulf has a strong track record of signing agreements with data center customers, and the management team has extensive experience in building a wide range of power infrastructure assets.

- HPC Leases: Discounted NAV
- Other Owned Sites: EV/Watt
- 50% probability of 250 MW Deployed p.a. 2028-2032: Discounted NAV

Consensus Price Target Distribution

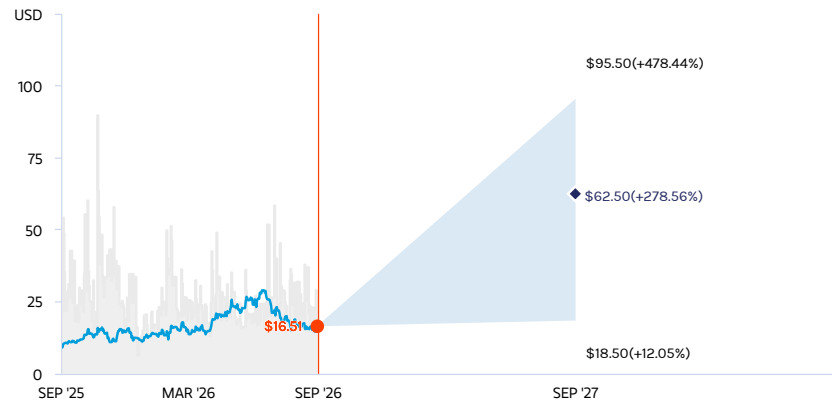
Source: Refinitiv, Morgan Stanley Research



OVERWEIGHT THESIS

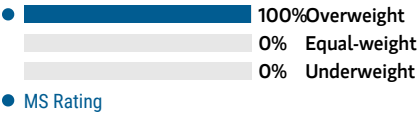
TeraWulf (WULF) remains a strong pick: despite strong stock performance, our base case price target recognizes the opportunity for a tripling (at ~273% upside). Underpinning our conviction is the company's history of successfully building and acquiring power infrastructure and track record of multiple repeat Bitcoin-to-DC conversion transactions with customers. In our view, the company has a viable path to continued, significant annual growth in power and data center capacity, and we include this growth potential in both our base and bull cases.

RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target
Source: Refinitiv, Morgan Stanley Research

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: Positive
View descriptions of Risk Rewards Themes [here](#)

BULL CASE	\$95.50	BASE CASE	\$62.50	BEAR CASE	\$18.50
75% probability of energization		50% probability of energization		0% probability of energization	
In our bull case, we assume the success rate rises to 75%, on the high end of annual project growth of 500 MW/year in 2028-32.		In our base case, we value the existing Bitcoin-to-DC projects based on our "REIT endgame" valuation framework. We project that the company will successfully convert the 2,082 MWs that currently do not have a Bitcoin-to-DC contract, at a present value that equates to ~\$11.50/watt. We also assume that management achieves a 50% success rate in achieving the lower end of its targeted annual DC project growth of 250 MW/year in 2028-32.		WULF downside is limited by the significant value associated with existing contracts. In our bear case we include the value from the existing contracts, plus a very low value assigned to other assets (\$2/watt, much lower than the \$9-13/watt of value created from precedent conversion transactions) and no other Bitcoin-to-DC conversion transactions.	

Risk Reward – TeraWulf Inc (WULF.O)

KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Digital Assets (\$, mm)	151.6	68.1	85.8	50.1
HPC Lease Revenue (\$, mm)	24.1	329.0	829.7	1,389.6
Operating profit (\$, mm)	(186.2)	(221.7)	293.4	540.2
GAAP operating margin (%) (%)	(110.5)	(55.8)	32.0	37.5
Adjusted EBITDA (\$, mm)	(23.1)	239.2	809.7	1,352.2

INVESTMENT DRIVERS

- The magnitude of improvement in AI capabilities will continue to increase at a non-linear rate
- The value creation for both AI Adopters and AI Enablers is high
- The demand for compute is likely to be systematically much higher than the supply

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	4/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- We view a slowdown in AI spend as the largest single risk.
- Execution risks on construction could result in project delays and cost overruns.
- Legislative risk to New York and Maryland sites could inhibit construction.

RISKS TO DOWNSIDE

- AI customer agreements could validate the BTC-to-HPC pivot faster than modeled.
- Additional acquisitions could lift pipeline value beyond the 2,082 MW base case.

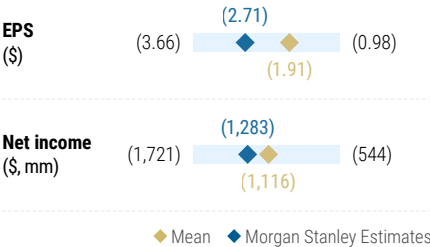
OWNERSHIP POSITIONING

Inst. Owners, % Active	57%	
HF Sector Long/Short Ratio	2.1x	
HF Sector Net Exposure	33.6%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure – Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



Source: Refinitiv, Morgan Stanley Research

Risk Reward – Hut 8 Mining Corp. (HUT.O)

Consistently Strong HPC Leases

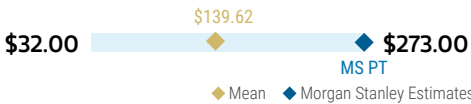
PRICE TARGET \$273.00

We assume a strong cash flow yield on future HPC leases, based on (i) the strength of HUT's currently controlled leases and (ii) precedent of strong deals in the space for said attractive sites.

- HPC leases: Discounted NAV
- Bitcoin Mining Capacity: EV/Watt
- Pipeline Sites (50% Probability of Energization): Discounted NAV

Consensus Price Target Distribution

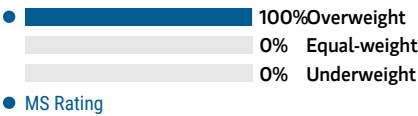
Source: Refinitiv, Morgan Stanley Research



OVERWEIGHT THESIS

HUT remains a cohort leader in contracting HPC leases at stronger-than-expected terms. Our conviction is that these deals for powered shells will come in at a "15/15/15" framework: 15 years, 15% yield on capex, and \$15 of equity value creation. HUT surpassed our expectations each of the three times they've inked a deal. Their robust power pipeline—currently split across bitcoin mining operations, expansions on current sites, and power under exclusivity—provides continued upside on top of their existing lease value.

Consensus Rating Distribution



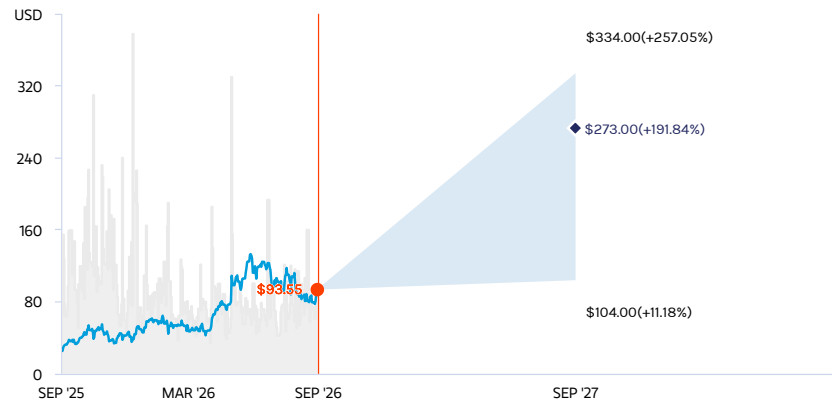
Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: Positive

View descriptions of Risk Rewards Themes [here](#)

RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

BULL CASE

\$334.00

75% chance of electrification

In our bull case, we project that the company will successfully convert the ~2GW of critical IT MWs currently not contracted for HPC operations at a value of \$8.80/watt and increase the value of potentially converted BTC sites to \$15/watt.

BASE CASE

\$273.00

50% chance of electrification

In our base case, we value the existing Bitcoin-to-DC projects based on our "REIT endgame" valuation framework. We project that the company will successfully convert the ~2GW of critical IT MWs currently not contracted for HPC operations at a present value that equates to ~\$5.90/watt and value currently energized MWs (those being used for BTC mining operations) at \$11.70/watt.

BEAR CASE

\$104.00

0% chance of electrification

We reduce expectations on additional signed MWs to zero, increase expectations on the discount rate of signed leases to 12%, and assume the company maintains BTC mining operations.

Risk Reward – Hut 8 Mining Corp. (HUT.O)

KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Total Capacity (EH/s)	26.8	29.9	29.9	29.9
Revenue - Bitcoin mining (\$, mm)	82	59	51	22
Total HPC Revenue (\$, mm)	0	0	138	326
Operating profit (\$, mm)	(435)	(29)	32	106
Operating margin (%) (%)	60.4	51.7	84.3	91.3

INVESTMENT DRIVERS

- The magnitude of improvement in AI capabilities will continue to increase at a non-linear rate
- The value creation for both AI Adopters and AI Enablers is high
- The demand for compute is likely to be systematically much higher than the supply

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	4/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- HUT brings increased power into exclusivity from their ~5GW under diligence.
- Increased relevance of the time-to-power theme increase the strength of lease terms at future sites.

RISKS TO DOWNSIDE

- HUT fails to contract any additional MWs out of its pipeline.
- Execution risk on its current leases cause cancellation or delayed starts.

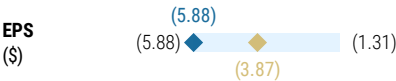
OWNERSHIP POSITIONING

Inst. Owners, % Active	61.1%	
HF Sector Long/Short Ratio	2.1x	
HF Sector Net Exposure	33.6%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure – Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Cipher Mining Inc. (CIFR.O)

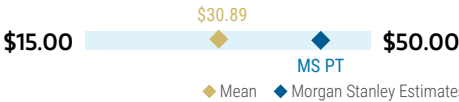
HPC Data Center Conversions

PRICE TARGET \$43.50

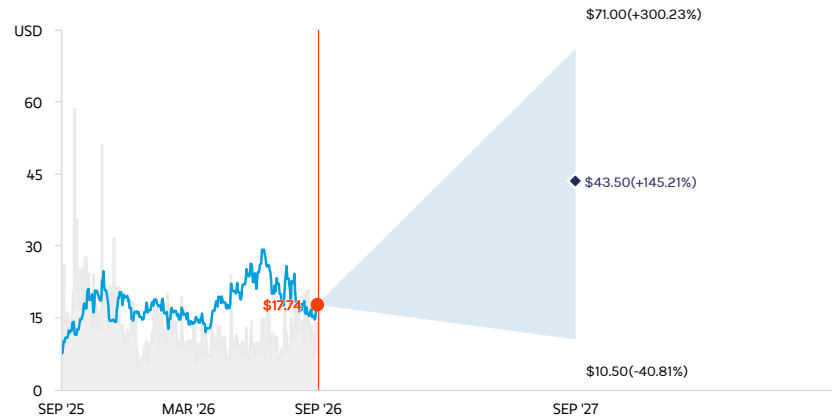
We assume a moderately lower cash flow yield on future Bitcoin-to-DC projects, based on the comparable transaction history in which CIFR's contracted cash flow yields are lower than WULF's.

- HPC leases: Discounted NAV
- Bitcoin Mining Capacity: EV/Watt
- Pipeline Sites (50% Probability of Energization): Discounted NAV

Consensus Price Target Distribution



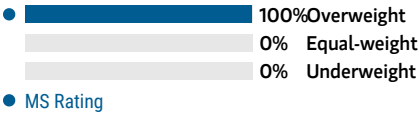
RISK REWARD CHART



OVERWEIGHT THESIS

Our approach is similar to the approach we take with WULF, though we assume a moderately lower cash flow yield on future Bitcoin-to-DC projects, based on the comparable transaction history in which CIFR's contracted cash flow yields are lower than WULF's.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

\$71.00

75% probability of energization

In our bull case, we assume the success rate rises to 75%. We also assume CIFR brings an additional 1GW of power into its pipeline through a behind-the-meter deal.

BASE CASE

\$43.50

50% probability of energization

In our base case, we value the existing Bitcoin-to-DC projects based on our "REIT endgame" valuation framework. We project that the company will successfully convert the other sites that currently do not have a Bitcoin-to-DC contract, at a present value that equates to ~\$10.70/watt. We also assume that management achieves a 50% success rate in executing DC transactions for the company's pipeline of development assets.

BEAR CASE

\$10.50

0% probability of energization

In our bear case we assume the existing contracts, plus a very low value assigned to other assets (\$2/watt, much lower than the ~\$9-18/watt of value created from precedent conversion transactions) and no other Bitcoin-to-DC conversion transactions.

Risk Reward – CIPHER Mining Inc. (CIFR.O)

KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Total Capacity (EH/s)	15.6	11.6	11.6	11.6
Revenue - Bitcoin mining (\$, mm)	219	118	123	81
Total HPC Revenue (\$, mm)	0	89	629	791
Operating profit (\$, mm)	(422)	(287)	145	224
Operating margin (%) (%)	(188.2)	(140.4)	19.4	25.7

INVESTMENT DRIVERS

- The magnitude of improvement in AI capabilities will continue to increase at a non-linear rate
- The value creation for both AI Adopters and AI Enablers is high
- The demand for compute is likely to be systematically much higher than the supply

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	5/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Lease economics could fall short of \$/watt assumptions.
- Delays/cost overruns may pressure campus returns.
- Failure to monetize HPC pipeline would limit valuation upside.

RISKS TO DOWNSIDE

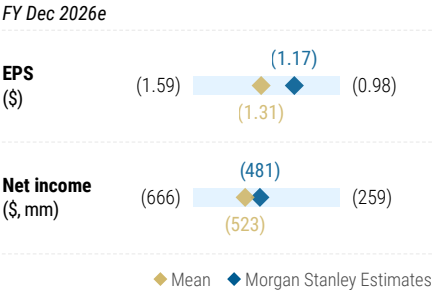
- Future hyperscaler leases could move CIFR toward consistent \$13-18/watt value creation.
- Black Pearl conversion and IT-load gains could lift pipeline value.
- Faster monetization of the 2.6GW HPC pipeline could raise valuation upside.

OWNERSHIP POSITIONING

Inst. Owners, % Active	59.8%	
HF Sector Long/Short Ratio	2.1x	
HF Sector Net Exposure	33.6%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure – Short Exposure).

MS ESTIMATES VS. CONSENSUS



Risk Reward – Riot Platforms Inc. (RIOT.O)

BTC Miner Pivoting to HPC Operations

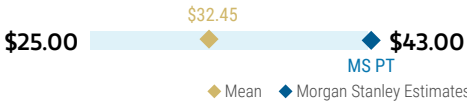
PRICE TARGET \$43.00

We assume a strong cash flow yield on future HPC leases, based on (i) the strength of RIOT's currently controlled sites and (ii) precedent of strong deals in the space for said attractive sites.

- HPC leases: Discounted NAV
- Bitcoin Mining Capacity: EV/Watt
- Pipeline Sites (75% Probability of Energization): Discounted NAV

Consensus Price Target Distribution

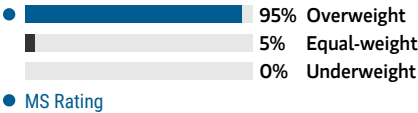
Source: Refinitiv, Morgan Stanley Research



OVERWEIGHT THESIS

We maintain our approach in valuing Powered Shell Providers as a function of their leases and pipeline MWs. We assume a strong cash flow yield on future lease signings given the strength of RIOT's signed Rockdale contracts and precedent from other companies in the cohort.

Consensus Rating Distribution



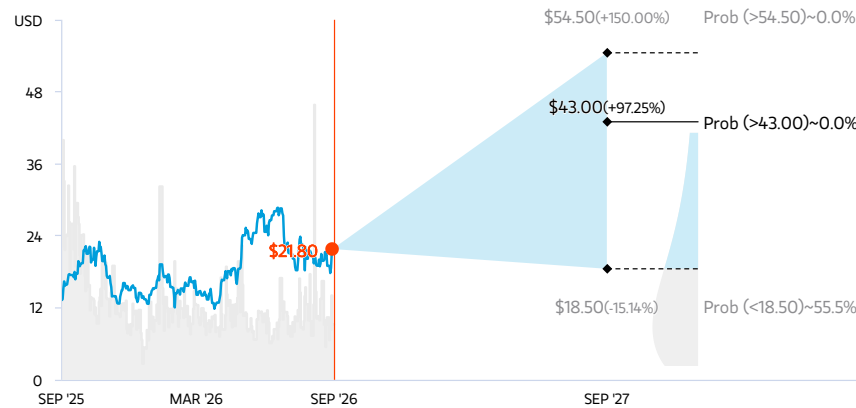
Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: Positive

View descriptions of Risk Rewards Themes [here](#)

RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 4 Sep 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

BULL CASE	\$54.50	BASE CASE	\$36.00	BEAR CASE	\$18.50
100% probability of energization		75% probability of energization		25% probability of energization	
We assume RIOT contracts all available pipeline MWs across Corsicana, Rockdale, and Kentucky. We also reduce our expectations of PUE and increase our expected NOI yield to ~17.6% to reflect strong deal economics, raising the value of pipeline MWs to ~\$12.25/watt.		In our base case, we expect RIOT to continue to sign deals at their Rockdale (700MW) and Corsicana (1GW) sites, thus the 75% change of energization for their pipeline MWs. This conviction is underpinned by the recent LOI signed by a prospective tenant for the entire Corsicana site—we view Corsicana's current grid interconnect in a Tier 1 location as a catalyst for a strong lease in the coming months. We value the ~1.5GWs of critical power in RIOT's portfolio at a \$8.8/watt.		In our bear case, we expect RIOT to sign only a single additional deal, thus the 25% change of energization for their pipeline MWs. Further, we reduce our expectations on the EV/Watt ratio to ~\$3.0/watt.	

Risk Reward – Riot Platforms Inc. (RIOT.O)

KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Total Capacity (EH/s)	38.5	44.4	44.4	44.4
Revenue - Bitcoin Mining (\$, mm)	132	118	102	44
Total HPC Revenue (\$, mm)	0	40	134	480
Operating Profit (\$, mm)	(684)	(89)	(135)	(310)
Operating Margin (%)	30.8	46.1	33.5	15.9

INVESTMENT DRIVERS

- The magnitude of improvement in AI capabilities will continue to increase at a non-linear rate
- The value creation for both AI Adopters and AI Enablers is high
- The demand for compute is likely to be systematically much higher than the supply

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

4/5 BEST	24 Month Horizon	5/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Future site acquisitions could raise the value of pipeline MWs.
- Time-to-Power considerations may lift deal economics above our expectations.

RISKS TO DOWNSIDE

- RIOT could fail to ink a deal at one of their sites.
- Deal economics for said deal could come in below expectations.

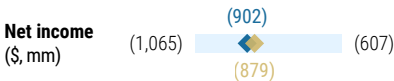
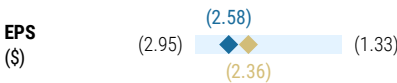
OWNERSHIP POSITIONING

Inst. Owners, % Active	51.5%	
HF Sector Long/Short Ratio	2.1x	
HF Sector Net Exposure	33.6%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure – Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Amazon.com Inc (AMZN.O)

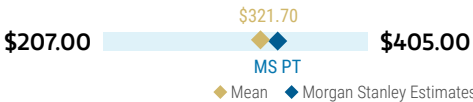
AMZN Risk Reward

PRICE TARGET \$335.00

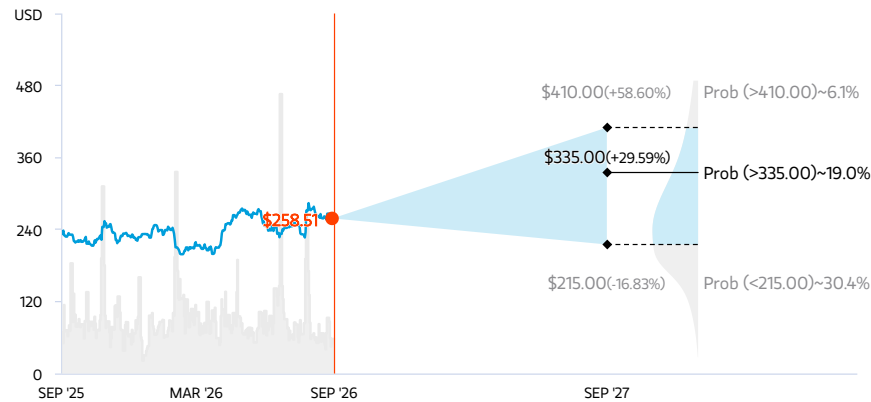
Our \$335 PT is based on applying a ~25X P/E multiple to an average of our \$14 of '27/'28 EPS (roll to mid-year), which implies ~28X '27 EPS and ~22X '28 EPS.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



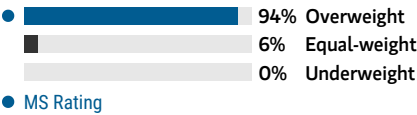
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 4 Sep 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

OVERWEIGHT THESIS

- Amazon's high-margin businesses continue to allow Amazon to drive greater profitability while still continuing to invest (last mile delivery, fulfillment, Prime Now, Fresh, Prime digital content, Alexa/Echo, India, AWS, etc).
- Amazon Prime membership growth drives recurring revenue and positive mix shift.
- Cloud adoption hitting an inflection point.
- Advertising serves as a key area for both further growth potential and profitability flow-through.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

- Secular Growth: Positive
- New Data Era: Positive
- Technology Diffusion: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE \$410.00

~27X an average of \$15 of bull '27/'28 EPS

Our Bull case is derived from higher Revenue and EBIT margin assumptions than in our base case

BASE CASE \$335.00

~25X an average of \$14 of base '27/'28 EPS

Our \$335 PT is based on applying a ~25X P/E multiple to an average of our \$14 of '27/'28 EPS (roll to mid-year), which implies ~28X '27 EPS and ~22X '28 EPS.

BEAR CASE \$215.00

~17X an average of \$12 of bear '27/'28 EPS

Our Bear case is derived from lower Revenue and EBIT margin assumptions than in our base case

Risk Reward – Amazon.com Inc (AMZN.O)

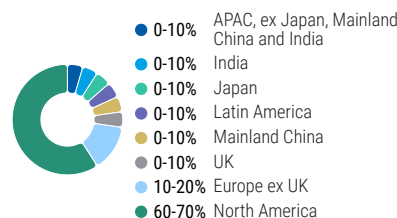
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Revenue (\$, mm)	716,924	833,352	973,667	1,134,182
Gross Profit (\$, mm)	361,287	433,328	534,679	656,405
GAAP EBIT (\$, mm)	79,975	116,753	166,089	227,300
Paid Unit Growth (y/y) (%)	10.8	12.9	9.6	9.1
AWS Revenue Growth (%)	19.7	37.4	40.8	36.8

INVESTMENT DRIVERS

- Amazon's high-margin businesses continue to allow Amazon to drive greater profitability while still continuing to invest
- Cloud is in a multi-decade secular adoption cycle
- Amazon is gaining share in eCommerce/retail and share of consumers' wallets

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

3/5 BEST	24 Month Horizon	2/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Faster than expected AWS rev growth and stable margin expansion
- Faster than expected Retail rev growth, stable to expanding 1P Merch Margins
- Greater than expected shipping and fulfillment leverage

RISKS TO DOWNSIDE

- Investments step up and continue for longer than expected
- Merch margins worse than expected
- AWS revenue decelerates and/or margins decline

OWNERSHIP POSITIONING

Inst. Owners, % Active	55.8%	
HF Sector Long/Short Ratio	1.7x	
HF Sector Net Exposure	10.6%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (\$, mm) ◆ **833,352**
Note: There are not sufficient brokers supplying consensus data for this metric

EBITDA (\$, mm) ◆ **224,292**
Note: There are not sufficient brokers supplying consensus data for this metric

Net income (\$, mm) ◆ **146,557**
Note: There are not sufficient brokers supplying consensus data for this metric

EPS (\$) ◆ **13.43**
Note: There are not sufficient brokers supplying consensus data for this metric

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Microsoft (MSFT.O)

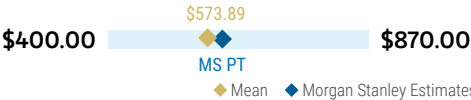
Durability of Earnings Growth & AI Leadership Not Priced-In

PRICE TARGET \$600.00

25x Base Case FY28e EPS of \$24.06. 25x PE is a slight premium to large cap software peers, 1.2x PEG is a discount to MSFT historical PEG and large-cap software peers.

Consensus Price Target Distribution

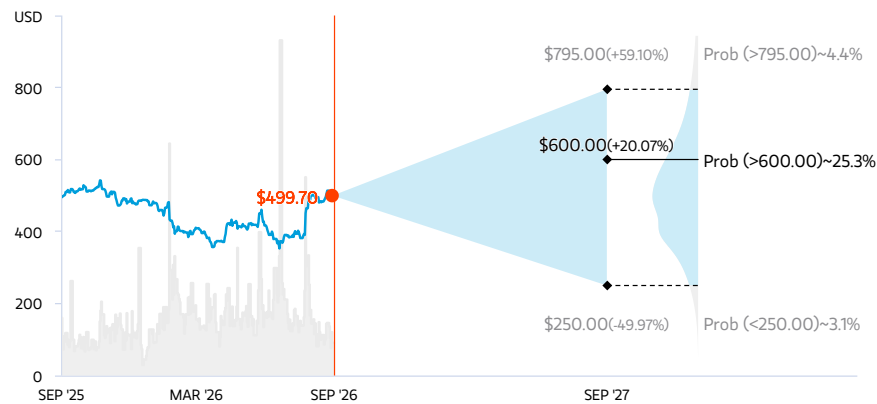
Source: Refinitiv, Morgan Stanley Research



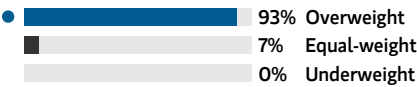
OVERWEIGHT THESIS

- Greater than expected capacity-driven Azure growth inflection and Copilot monetization supports high-teens rev growth.
- Meaningful gross margin % headwinds are offset by operating leverage, yielding 20%+ EBIT growth and >20% EPS growth
- Shares trading at ~20x FY28 GAAP EPS too cheap. Overweight

RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



Consensus Rating Distribution



MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

- New Data Era: Positive
- Pricing Power: Positive
- Secular Growth: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE \$795.00

~29x Bull Case FY28e EPS: \$27.64

Azure, O365 & Robust Contribution from AI Drive Top-Line Growth. Acceleration in Azure growth, adoption of higher priced M365 Commercial SKUs and continued seat growth, and robust adoption of Azure AI services and Microsoft AI Copilots across business lines drive low-20s revenue CAGR in the coming years. Operating margins expand to ~49.0% by FY28 and FY28e EPS is \$27.64. ~29x PE represents a premium to large cap software peers, but below MSFT historical avg PEG at ~1.2x given >20% EPS CAGR.

BASE CASE \$600.00

~25x Base Case FY28e EPS of \$24.06

Azure, M365 & Ramping Contribution from AI Drive Top-Line Growth. Durability of Azure growth, adoption of higher priced M365 Commercial SKUs and continued seat growth, and ramping adoption of Azure AI services & Microsoft AI Copilots across business lines drive high-teens rev CAGR ahead. Operating margins expand beyond ~46.7% by FY28 and FY28e EPS is \$24.06. 25x PE is a premium to large cap software peers, given strong positioning and execution. 1.2x PEG is a discount to MSFT historical PEG.

BEAR CASE \$250.00

~11x Bear Case FY28e EPS: \$21.84

Macro, Scale & Limited AI Adoption Drives Top-Line Growth Deceleration. Azure growth continues deceleration given scale, while M365 reaches penetration and adoption of Azure AI services & Microsoft AI Copilots across business lines remains limited. This drives mid-teens rev CAGR. Operating margins only expand to ~45.0% by FY28 from gross margin pressure and FY28 EPS is \$21.84. ~12x PE is a discount to large cap software peers and a discount on a PEG.

Risk Reward – Microsoft (MSFT.O)

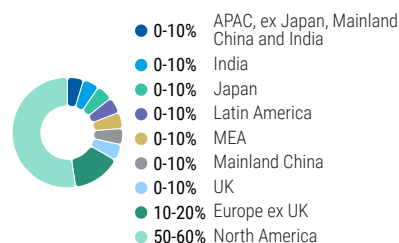
KEY EARNINGS INPUTS

Drivers	Jun 2026	Jun 2027e	Jun 2028e	Jun 2029e
Azure and Other Cloud Services Growth (%)	40.7	44.9	43.2	41.2
M365 Commercial Cloud Growth (%)	16.4	16.6	17.6	17.7
Gross Margins (%)	67.9	65.7	64.4	63.4
Operating Margins (%)	46.8	46.5	46.7	47.2
GAAP EPS Growth (%)	31.5	9.6	22.4	25.1

INVESTMENT DRIVERS

- Sustainability of commercial cloud growth, Copilot growth
- Improving Azure growth, cloud momentum, improving cloud margins

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

3/5 BEST	24 Month Horizon	1/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Cloud adoption accelerates, with Azure as convincing winner
- AI leadership results in substantial revenue contribution over-time
- Operational efficiencies leading to greater than anticipated economies of scale and margin expansion

RISKS TO DOWNSIDE

- Weak macro impacting IT spending
- On-premises cannibalization by Cloud
- Increased investments hurt margin expansion
- AI adoption proves limited

OWNERSHIP POSITIONING

Inst. Owners, % Active	52.4%	
HF Sector Long/Short Ratio	2.1x	
HF Sector Net Exposure	33.6%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Jun 2027e

Sales / Revenue (\$,mm) ◆ 393,588
Note: There are not sufficient brokers supplying consensus data for this metric

EBITDA (\$,mm) ◆ 235,048
Note: There are not sufficient brokers supplying consensus data for this metric

Net income (\$,mm) ◆ 145,544
Note: There are not sufficient brokers supplying consensus data for this metric

EPS (\$) ◆ 19.66
Note: There are not sufficient brokers supplying consensus data for this metric

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – NVIDIA Corp. (NVDA.O)

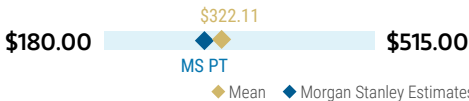
OW as large language model enthusiasm is transforming cloud capex

PRICE TARGET \$300.00

~20x our MW CY27 EPS estimate of \$15.01, in-line with the broader market and a discount to compute semis peers (AMD/AVGO/INTC) as high marketshare and gross margins leave limited levers for multiple expansion in the near term.

Consensus Price Target Distribution

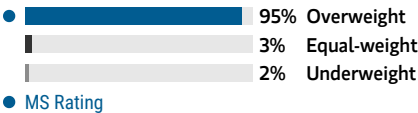
Source: Refinitiv, Morgan Stanley Research



OVERWEIGHT THESIS

- Blackwell remains the premiere solution for gen-AI workloads, where compute demand continues to outstrip supply
- We see continued upward pressure to estimates as demand strength continues, with Rubin expected to maintain Nvidia's performance leadership position

Consensus Rating Distribution



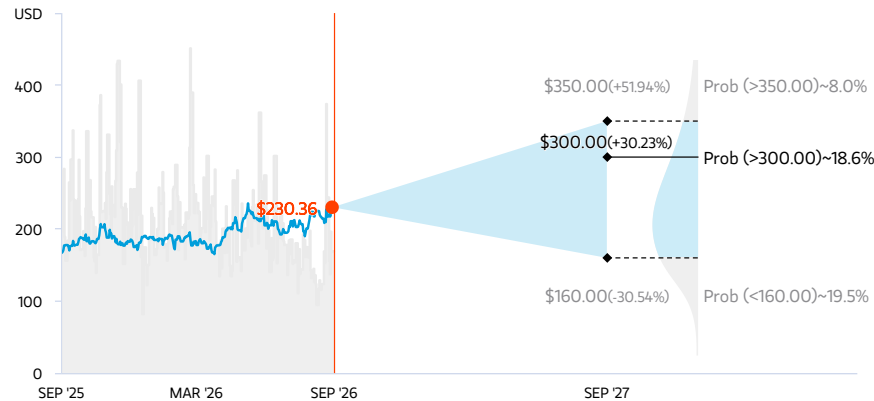
Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

- New Data Era: Positive
- Pricing Power: Positive
- Secular Growth: Positive

View descriptions of Risk Rewards Themes [here](#)

RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 4 Sep 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

BULL CASE \$350.00

~22x bull case MW CY27 EPS of \$16

Bull case has DC revenues surprising to the upside as supply becomes more available and other products gain steam (CPUs, Groq)

- Higher margin data center and AI-focused software and services growth accelerates
- Revenue sharing agreements add billions in 100% margin revenue
- GPU based AI PC gains traction, widely increasing the client TAM

BASE CASE \$300.00

~20x our MW CY27 EPS of \$15.01

~20x PE in-line with the broader market and a discount to compute semis peers (AMD/AVGO/INTC) as high marketshare and gross margins leave limited levers for multiple expansion in the near term..

- Revenue grows by 88% in 2026 and 70% in 2027

BEAR CASE \$160.00

~13x bear case MW CY27 EPS of \$13

Two key debates both go the wrong direction, causing investors to question future prospects for growth

- Growth in DC slows substantially as supply catches up to demand faster than anticipated
- AI development costs come down materially, a strong competitor enters the market to take market share, or customers begin insourcing custom hardware solutions
- Greater than expected impact from tariff headwinds and export controls

Risk Reward – NVIDIA Corp. (NVDA.O)

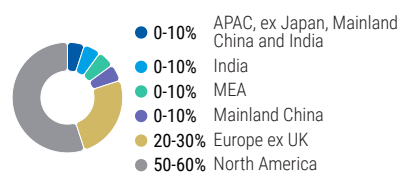
KEY EARNINGS INPUTS

Drivers	Dec 2026	Dec 2027e	Dec 2028e	Dec 2029e
GAAP Revenue (\$, mm)	215,938	405,968	690,168	970,222
MW Gross Margin (%)	71.2	73.7	72.5	72.7
MW EPS (\$)	4.60	9.18	15.01	22.00
Inventory (\$, mm)	21,403	44,841	65,460	79,121
DOI	123.3	151.1	123.9	68.3

INVESTMENT DRIVERS

- Growth in AI capex from customers
- Next gen GPUs continue to outpace the competition
- Systems approach allows for higher monetization over time
- New drivers emerge for Nvidia such as AI PCs, autonomous vehicles, robotics, and software

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

4/5 BEST	24 Month Horizon	4/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Growth in inference and training propel data center revenue
- Gaming sales accelerate as GPU based AI PCs gain traction
- Nvidia can recapture lost revenue in China

RISKS TO DOWNSIDE

- AI end markets don't materialize as expected, customers sharply reduce GPU purchases
- AMD reemerges as a viable GPU competitor
- Cloud customers outside of Google are able to develop competitive custom hardware

OWNERSHIP POSITIONING

Inst. Owners, % Active	51.2%	
HF Sector Long/Short Ratio	2.1x	
HF Sector Net Exposure	33.6%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e

Sales / Revenue (\$, mm) **405,968**
Note: There are not sufficient brokers supplying consensus data for this metric

EBITDA (\$, mm) **269,866**
Note: There are not sufficient brokers supplying consensus data for this metric

Net income (\$, mm) **223,210**
Note: There are not sufficient brokers supplying consensus data for this metric

EPS (\$) **9.21**
Note: There are not sufficient brokers supplying consensus data for this metric

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

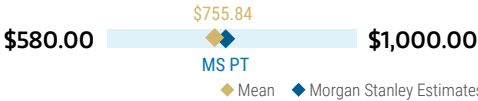
Risk Reward – Meta Platforms Inc (META.O) Top Pick

The Age of Efficiency

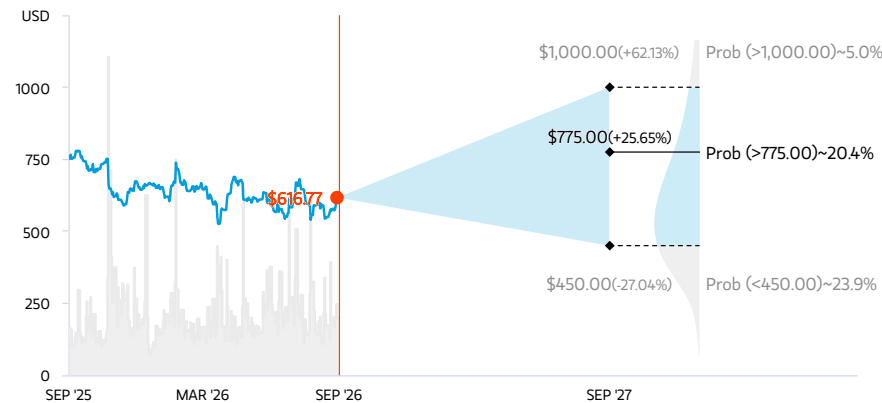
PRICE TARGET **\$775.00**

Our \$775 PT is based on applying a ~23X P/E multiple to the average of our \$34/\$35 EPS in '27/'28.

Consensus Price Target Distribution



RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



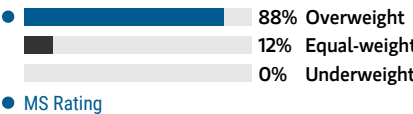
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 4 Sep 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

OVERWEIGHT THESIS

- **A Structural Pivot to Focus on Multi-Year Efficiency, Productivity and Leaner Operations...** We think META's "year of efficiency" is more than just a 365 day change...but rather a structural and cultural pivot to operate leaner and with a greater focus on investor returns...even through investment.
- **Importantly, Revenue and Engagement Trends Are Also Improving Across META's Platform...** our industry and company conversations speak to progress driving incremental Reels engagement and monetization...and traction improving ad measurement/attribution in the post IDFA world.
- **3 Underappreciated "Call Options"** including 1) further AI driven upside, 2) subscription adoption and 3) click-to-message.

Consensus Rating Distribution



Risk Reward Themes

- New Data Era: *Positive*
- Self-help: *Positive*
- Technology Diffusion: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE **\$1,000.00**

Implied ~28x blended '27/'28 P/E

AI investments drive increased engagement and lead to new products/services, which supports outsized ad monetization and revenue growth. Execution on newer offerings (like click to message, subscriptions, and Meta Business AI) could also contribute to higher than expected growth. META could also drive further efficiency gains and be more successful in closing the Reels monetization gap.

BASE CASE **\$775.00**

Implied ~23X base case blended '27/'28 P/E

Ad revenue grows ~27% in '26 as META's AI investments drive incremental Reels engagement/monetization and support improved performance and ad measurement/attribution across the Family of Apps. We also see META continuing to focus on efficiency and explore new ways to monetize META's extensive data center capacity.

BEAR CASE **\$450.00**

Implied ~15x blended '27/'28 P/E

Declines in engagement and/or evidence of Reels monetization ramping more slowly than expected could lead to lower growth and greater uncertainty. We also watch for any macro uncertainty, regulation that limits META's ability to target ads, Higher capital intensity from mis-execution around the data center build widens losses with minimal ROIC. Any additional opex and capex could weigh on operating income growth and FCF.

Risk Reward – Meta Platforms Inc (META.O)

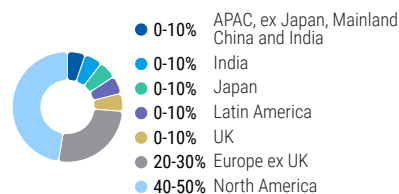
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Constant Currency % Growth (%)	22.5	26.6	22.2	19.9
Operating income (GAAP) (\$, mm)	83,276	89,133	103,424	111,210
Consolidated Total Daily Active Users (mm)	2,239.9	2,294.5	2,344.5	2,389.2

INVESTMENT DRIVERS

- We are positive on META's structural pivot towards efficiency and improving revenue, engagement and Reels. We continue to monitor META's progress on call options (AI, subscriptions, click-to-message) and further investment in AI/data centers.

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

3/5 BEST	24 Month Horizon	2/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Faster than expected Reels monetization & higher engagement growth
- Increased benefit from AI investment
- Revenue driven by subscriptions and/or click-to-message ads
- Further efficiency improvements drive outsized FCF growth

RISKS TO DOWNSIDE

- Macro pressures/weaker consumer spend
- Regulation impacts ability to target ads
- Reality labs losses widen further
- Mis-execution risk could result in higher LT capital intensity

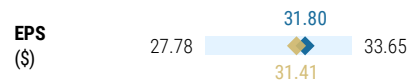
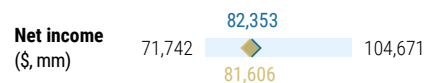
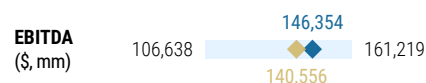
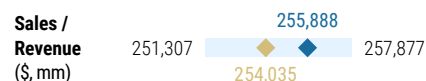
OWNERSHIP POSITIONING

Inst. Owners, % Active	55.2%	
HF Sector Long/Short Ratio	2.4x	
HF Sector Net Exposure	11.4%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure – Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Alphabet Inc. (GOOGL.O)

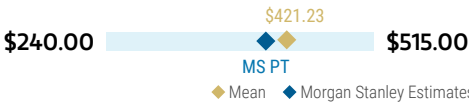
Alphabet Inc.

PRICE TARGET \$400.00

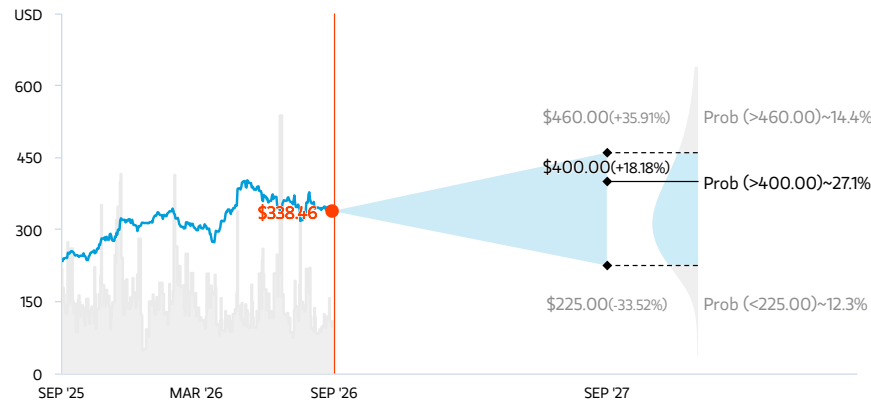
Our \$400 PT is based on applying a ~24X P/E multiple to the average of our \$15/\$18 EPS in '27/'28, and implies paying ~1.6X PEG, a ~35% premium to GOOGL's peer median.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



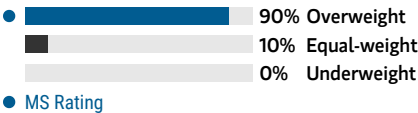
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 4 Sep 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

OVERWEIGHT THESIS

- Continued AI-driven platform-level innovation on Search, YouTube, Cloud, and with other offerings improve confidence in the durability of long term growth.
- Continued expense discipline leads to operating leverage and upward revisions on EPS and FCF estimates.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: Positive
New Data Era: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE \$460.00

~26X Avg. '27/'28 Bull Case EPS of ~\$16/~\$20

Faster innovation and Search/GCP acceleration lead to multiple expansion and higher earnings power. Search revenue accelerates as AI-driven innovation delivers meaningful upside to engagement. New AI offerings prove incremental to the top and bottom line and don't cannibalize core Search. YouTube and Cloud become even bigger contributors to top-line growth, and operate at a higher margin than in our base case.

BASE CASE \$400.00

~24X Avg. '27/'28 Base Case EPS of ~\$15/~\$18

Assumes pragmatic search revenue growth in '26, continued incremental platform monetization and Google Cloud acceleration in '26. As GOOGL continues platform-level innovation on Search and other categories, we assume pragmatic revenue/EBITDA growth. GOOGL introduces new AI offerings which increase confidence in the durability of long term growth and calm fears around competition. Google Cloud accelerates driven by new partnerships and growing backlog.

BEAR CASE \$225.00

~15X Avg. '27/'28 Bear Case EPS of ~\$14/~\$16

Global ad growth slows further and margins face pressure. Assumes slower search advertising growth, and that ad spending slows further. Expense discipline fails to materialize leading to lower than expected margin expansion and adj. EBITDA. New AI products create greater than expected margin pressure due to lower monetization rates and increased compute intensity.

Risk Reward – Alphabet Inc. (GOOGL.O)

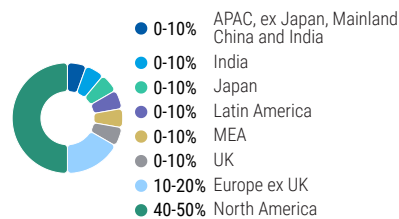
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Total operating income (GAAP) (\$, mm)	129,039	171,465	229,478	284,409
GAAP Operating Income (Loss) (\$, mm)	137,054	178,940	236,678	290,959

INVESTMENT DRIVERS

- Search advertising spend continues to gain share of global advertising budgets.
- Mobile search advertising continues to take share of online budgets.
- Investments in video driving longer-term monetization at YouTube.
- Moderation of expense growth.

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

4/5 BEST	24 Month Horizon	3/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- New products generate higher than expected top line contribution.
- Capital returns through greater share buybacks.
- Hiring and/or spend per headcount is lower than expected.

RISKS TO DOWNSIDE

- High exposure to SMB and travel could pressure ad revenue in a recession
- Improved disclosure around the Google and Other Alphabet segments may not decrease the overall investment activity of the business.

OWNERSHIP POSITIONING

Inst. Owners, % Active	56.5%	
HF Sector Long/Short Ratio	2.4x	
HF Sector Net Exposure	11.4%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (\$, mm)	487,900	500,611	519,818
		498,100	

EBITDA (\$, mm)	223,773	236,736	237,009
		232,993	

Net income (\$, mm)	243,916	250,941	268,656
		253,783	

EPS (\$)	19.70	20.29	21.92
		20.60	

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
5. View explanation of HERS methodology - [ESG_HERS_External_Link.pdf](#)

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(as of August 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1555	42%	450	48%	29%	767	44%
Equal-weight/Hold	1577	43%	395	42%	25%	770	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	559	15%	93	10%	17%	216	12%
Total	3,694		939			1754	

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INDUSTRY COVERAGE: Internet

COMPANY (TICKER)	RATING (AS OF)	PRICE* (09/04/2026)
Brian Nowak, CFA		
Airbnb Inc (ABNB.O)	U (12/06/2022)	\$181.94
Alphabet Inc. (GOOGL.O)	O (08/11/2015)	\$338.46
Amazon.com Inc (AMZN.O)	O (04/24/2015)	\$258.51
Booking Holdings Inc (BKNG.O)	O (02/23/2026)	\$193.29
DoorDash Inc (DASH.O)	O (02/21/2024)	\$211.73
Expedia Inc. (EXPE.O)	E (01/09/2019)	\$298.04
Instacart (CART.O)	E (01/29/2024)	\$51.07
Lyft Inc (LYFT.O)	E (10/24/2019)	\$16.72
Meta Platforms Inc (META.O)	O (03/20/2023)	\$616.77
Pinterest Inc (PINS.N)	O (07/20/2025)	\$20.40
Reddit Inc (RDDT.N)	O (12/08/2024)	\$154.46
Snap Inc. (SNAP.N)	E (07/22/2024)	\$5.47
Uber Technologies Inc (UBER.N)	++	\$75.76
Matthew Cost		
AppLovin Corp (APP.O)	O (04/10/2025)	\$320.56
Compass, Inc. (COMP.N)	E (01/12/2026)	\$11.12
Criteo SA (CRO.O)	E (01/26/2016)	\$17.62
DoubleVerify Holdings Inc (DV.N)	E (06/25/2024)	\$13.36
Liftoff Mobile Inc. (LFTO.O)	E (06/29/2026)	\$18.74
MNTN Inc (MNTN.N)	E (06/16/2025)	\$12.95
Opendoor Technologies Inc (OPEN.O)	E (07/24/2023)	\$3.15
Playtika Holding Corp (PLTK.O)	E (11/27/2022)	\$2.20
Roblox Corporation (RBLX.N)	O (11/04/2024)	\$43.31
Take-Two Interactive Software (TTWO.O)	O (02/01/2018)	\$214.69
Trade Desk Inc (TTD.O)	E (09/10/2025)	\$14.43
Unity Software Inc (U.N)	O (09/02/2024)	\$41.66
Webtoon Entertainment Inc (WBTN.O)	E (07/22/2024)	\$9.97
Yelp Inc (YELP.N)	U (01/10/2019)	\$21.66
Zillow Group Inc (Z.O)	E (04/18/2018)	\$34.59
Nathan Feather		
Bumble Inc. (BMBL.O)	E (03/08/2021)	\$2.90
Chewy Inc (CHWY.N)	O (10/31/2023)	\$23.66
Duolingo Inc (DUOL.O)	E (02/27/2026)	\$154.46
eBay Inc (EBAY.O)	O (04/18/2024)	\$103.41
Etsy Inc (ETSY.N)	E (07/20/2025)	\$76.51
FIGS, Inc. (FIGS.N)	E (02/29/2024)	\$14.42
Grindr Inc. (GRND.N)	O (07/01/2026)	\$15.25
Match Group Inc (MTCH.O)	E (04/18/2024)	\$41.86
Peloton Interactive, Inc. (PTON.O)	E (03/14/2022)	\$5.40
Revolve Group Inc (RVLV.N)	E (10/20/2024)	\$21.35
WW International Inc (WW.O)	E (08/01/2025)	\$18.02

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* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Software

COMPANY (TICKER)	RATING (AS OF)	PRICE* (09/04/2026)
Adam Wood		

Adobe Inc. (ADBE.O)	U (07/21/2026)	\$266.51
Atlassian Corporation PLC (TEAM.O)	O (01/13/2020)	\$189.58
Intuit (INTU.O)	E (07/21/2026)	\$332.70
Microsoft (MSFT.O)	O (07/21/2026)	\$499.70
Salesforce, Inc. (CRM.N)	E (07/21/2026)	\$259.23
Samsara Inc (IOT.N)	E (03/23/2023)	\$40.20
ServiceNow Inc (NOW.N)	O (09/24/2025)	\$141.26
Shopify Inc (SHOP.O)	O (04/19/2024)	\$145.09
Workday Inc (WDAY.O)	U (07/21/2026)	\$195.79
Chris Quintero		
BILL Holdings Inc (BILL.N)	E (06/10/2025)	\$49.16
Blackline Inc (BL.O)	E (07/21/2026)	\$32.71
Box Inc (BOX.N)	E (05/21/2024)	\$35.19
CCC Intelligent Solutions Holdings Inc (CCC.O)	O (11/13/2024)	\$7.51
Descartes Systems Group Inc (DSGX.O)	O (01/15/2026)	\$78.94
DocuSign Inc (DOCU.O)	E (01/16/2024)	\$68.41
Manhattan Associates Inc. (MANH.O)	E (10/21/2025)	\$213.77
Navan Inc (NAVN.O)	O (11/24/2025)	\$27.78
ServiceTitan Inc (TTAN.O)	O (01/20/2026)	\$87.92
SPS Commerce Inc (SPSC.O)	U (07/21/2026)	\$83.00
Vertex Inc. (VERX.O)	E (07/21/2026)	\$13.77
Via Transportation Inc (VIA.N)	O (01/20/2026)	\$28.13
Elizabeth Porter, CFA		
Amplitude Inc. (AMPL.O)	O (01/15/2026)	\$13.02
Asana Inc (ASAN.N)	U (05/20/2025)	\$8.81
Autodesk (ADSK.O)	O (08/23/2024)	\$217.90
Figma Inc (FIG.N)	E (08/25/2025)	\$24.12
Five9 Inc (FIVN.O)	E (10/10/2022)	\$33.27
Freshworks Inc (FRSH.O)	E (10/18/2021)	\$12.73
GoDaddy Inc (GDDY.N)	E (07/19/2021)	\$101.31
HubSpot, Inc. (HUBS.N)	O (03/21/2023)	\$247.60
Klaviyo, Inc (KVYO.N)	O (04/29/2026)	\$18.61
Liveramp Holdings Inc (RAMP.N)	E (01/13/2025)	\$37.75
monday.com Ltd (MNDY.O)	O (08/12/2025)	\$91.07
NICE Ltd. (NICE.O)	E (07/21/2026)	\$105.50
RingCentral Inc (RNG.N)	E (08/08/2023)	\$73.39
Sprinklr Inc (CXM.N)	E (07/19/2021)	\$5.91
Sprout Social Inc (SPT.O)	E (11/17/2020)	\$11.36
Toast, Inc. (TOST.N)	O (12/16/2021)	\$33.95
Twilio Inc (TWLO.N)	O (02/24/2025)	\$232.98
Wix.Com Ltd (WIX.O)	E (07/21/2026)	\$76.21
Zeta Global Holdings Corp (ZETA.N)	E (08/01/2024)	\$31.35
Zoom Communications (ZM.O)	E (10/11/2022)	\$101.33
ZoomInfo Technologies Inc (GTM.O)	E (02/01/2024)	\$4.13
Meryl R Thomas, CFA		
Lightspeed Commerce Inc. (LSPD.N)	E (02/18/2021)	\$10.51
Meta A Marshall		
Check Point Software Technologies Ltd. (CHKP.O)	E (10/16/2023)	\$137.05
CrowdStrike Holdings Inc (CRWD.O)	O (03/10/2026)	\$213.10
Fortinet Inc. (FTNT.O)	E (07/21/2026)	\$156.29
Gen Digital Inc. (GEN.O)	E (06/07/2024)	\$30.65
Netskope, Inc. (NTSK.O)	O (10/13/2025)	\$13.94
Okta, Inc. (OKTA.O)	O (12/02/2024)	\$170.60
Palo Alto Networks Inc (PANW.O)	O (10/10/2017)	\$333.26

Qualys Inc (QLYS.O)	U (02/09/2021)	\$171.65
Rapid7 Inc (RPD.O)	U (07/21/2026)	\$11.00
SailPoint Inc (SAIL.O)	O (09/02/2025)	\$18.82
SentinelOne, Inc. (S.N)	E (12/02/2024)	\$19.88
Tenable Holdings Inc (TENB.O)	E (12/02/2024)	\$34.25
Varonis Systems, Inc. (VRNS.O)	E (01/26/2026)	\$46.09
Zscaler Inc (ZS.O)	E (04/22/2026)	\$169.80
Ryan Lountzis		
CoreWeave (CRWV.O)	E (04/22/2025)	\$89.36
Nebius Group NV (NBIS.O)	E (01/15/2026)	\$226.39
Sanjit K Singh		
Akamai Technologies, Inc. (AKAM.O)	O (01/12/2026)	\$105.22
Appian Corp (APPN.O)	E (04/30/2026)	\$38.37
C3.ai (AI.N)	U (01/04/2021)	\$10.46
Cloudflare Inc (NET.N)	O (12/02/2024)	\$278.92
Datadog, Inc. (DDOG.O)	O (01/12/2026)	\$212.93
DigitalOcean Holdings Inc (DOCN.N)	O (01/16/2025)	\$112.47
Dynatrace Inc (DT.N)	O (08/25/2026)	\$51.90
Elastic NV (ESTC.N)	E (07/21/2026)	\$91.81
GitLab Inc (GTLB.O)	E (01/12/2026)	\$49.83
JFrog Ltd. (FROG.O)	E (07/21/2026)	\$87.60
MongoDB Inc (MDB.O)	O (04/12/2023)	\$368.74
Oracle Corporation (ORCL.N)	E (01/15/2019)	\$158.78
PagerDuty, Inc. (PD.N)	U (07/21/2026)	\$13.62
Palantir Technologies Inc. (PLTR.O)	E (02/04/2025)	\$174.33
Snowflake Inc. (SNOW.N)	O (06/24/2025)	\$337.18
UiPath Inc (PATH.N)	E (09/07/2022)	\$15.19

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* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (09/04/2026)
Joseph Moore		
Advanced Micro Devices (AMD.O)	E (06/09/2024)	\$477.57
Aeva Technologies Inc (AEVA.O)	E (07/19/2021)	\$15.69
Allegro Microsystems Inc (ALGM.O)	O (02/13/2026)	\$36.61
Ambarella Inc (AMBA.O)	O (03/29/2016)	\$62.89
Amkor Technology Inc (AMKR.O)	E (11/08/2023)	\$47.77
Analog Devices Inc. (ADI.O)	O (11/16/2023)	\$362.25
Astera Labs Inc (ALAB.O)	O (05/11/2025)	\$310.40
Broadcom Inc. (AVGO.O)	O (06/09/2024)	\$357.90
Cerebras Systems (CBRS.O)	O (06/08/2026)	\$210.05
GlobalFoundries Inc (GFS.O)	E (10/28/2024)	\$45.21
Intel Corporation (INTC.O)	E (02/22/2023)	\$95.80
IonQ Inc (IONQ.N)	E (04/25/2023)	\$39.52
Marvell Technology Group Ltd (MRVL.O)	E (09/14/2015)	\$223.55
Microchip Technology Inc. (MCHP.O)	E (07/10/2024)	\$74.17
Micron Technology Inc. (MU.O)	O (10/06/2025)	\$1,016.59
Navitas Semiconductor Corp (NVT.S.O)	U (04/06/2025)	\$11.80
NVIDIA Corp. (NVDA.O)	O (03/16/2023)	\$230.36
NXP Semiconductor NV (NXPI.O)	O (02/11/2025)	\$227.84
ON Semiconductor Corp. (ON.O)	++	\$74.38
Qorvo Inc (QRVO.O)	E (10/28/2025)	\$102.84
Qualcomm Inc. (QCOM.O)	E (06/24/2026)	\$168.74

Quantinuum (QNT.O)	E (06/29/2026)	\$49.65
SanDisk Corporation. (SNDK.O)	O (03/03/2025)	\$1,740.00
Semtech Corp. (SMTC.O)	E (04/06/2025)	\$147.89
Silicon Laboratories Inc. (SLAB.O)	E (01/19/2021)	\$220.58
Skyworks Solutions Inc (SWKS.O)	E (11/28/2018)	\$74.02
Texas Instruments (TXN.O)	U (04/13/2020)	\$258.44
Wolfspeed, INC (WOLF.N)	NR (04/06/2025)	\$28.35
Lee Simpson		
Arm Holdings plc (ARM.O)	E (04/07/2026)	\$252.09
Cadence Design Systems Inc (CDNS.O)	O (02/14/2024)	\$292.70
Synopsys Inc. (SNPS.O)	E (02/27/2026)	\$393.84

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